# Set Theory: (Week 1- Lecture1)

## Video segment 1:

**Set theory** is a branch of mathematical logic that studies sets, which informally are collections of objects. Although any type of object can be collected into a set, set theory is applied most often to objects that are relevant to mathematics. There are some set expressions that is given below:

The following notations is more common in Discrete Mathematics.

- The empty set, denoted ∅, is the set that has no element.
- N := {1, 2, 3, 4…….}, the set of Natural numbers;
- W := {0, 1, 2, . . .}, the set of whole numbers
- Z := {0, 1, −1, 2, −2, . . .}, the set of Integers;
- Q := { p q : p, q ∈ Z, q 6= 0}, the set of Rational numbers;
- R := the set of Real numbers; and
- C := the set of Complex numbers. We start with the basic set theory.

A set is typically expressed by curly braces, { } enclosing its elements. If A is a set and a is an element of it, we write a ∈ A. The fact that a is not an element of A is written as a 6∈ A. For instance, if A is the set {1, 4, 9, 2}, then 1 ∈ A, 4 ∈ A, 2 ∈ A and 9 ∈ A. But 7 6∈ A, π 6∈ A, the English word 'four' is not in A, etc.

- The set A = {1, 2, 3, 4} in the predicate notation can be written as i. A = {x ∈ N : 1 ≤x ≤ 4} where, x is natural number.
- The set B= {$2^1$, $2^2$, $2^3$, $2^4$} in the predicate notation can be written as i. A = {x ∈ N : x=$2^k$}
  Where, k ∈ N is natural number and 1 ≤k ≤ 4.

### Set Intervals:

### Closed Set Intervals:



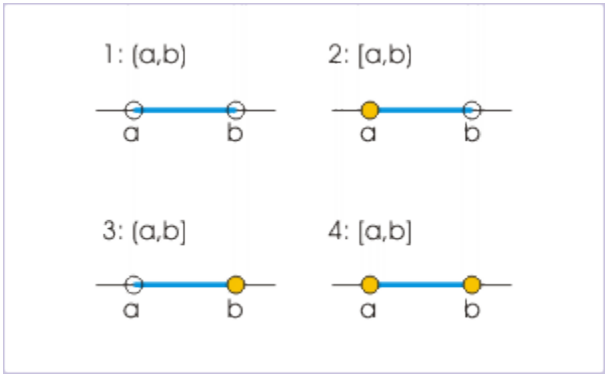

*closed interval* [*a*, *b*]

A closed interval is an interval that includes all of its limit points. If the endpoints of the interval are finite numbers a and b, then the interval {x : a ≤ x ≤ b} is denoted [a, b].

## Open Set Intervals:

An open interval does not include its endpoints and is indicated with parentheses. In case of (a,b), the interval will represent as {x : a < x < b)

To indicate that only one endpoint of an interval is included in that set, both symbols will be used. For example, the interval of numbers between 1 and 5, including 1 but excluding 5, is written as [1,5). The interval will represent as {x : 1≤ x < 5)




## Example of Open and closed ended intervals:Closed Set intervals:

Example 1: [1,6]

Answer: The expression is {x : 1≤ x ≤ 2} .




Example 2: [-5,3]

Answer: The expression is {x : -5≤ x ≤ 3}



## Open Set intervals:

Example 3: (-5,3)

Answer: The expression is {x : -5< x < 3}




Example 4: (2,6]

Answer: The expression is {x : 2< x ≤ 6}




Example 5: $(-\infty,3) \cup [4, \infty)$

Answer: The expression is {x : x < 3 and x ≥4 }

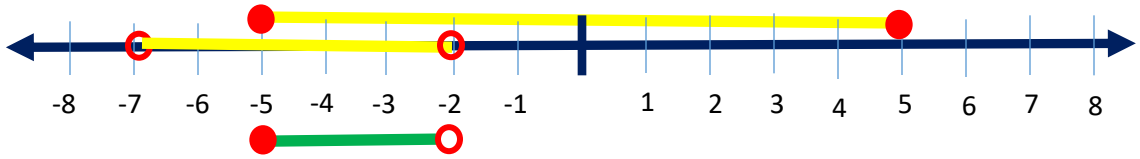



Example 5: $(-\infty,3) \cap [4, \infty)$

Answer: The answer will be empty set.

Example 5: (-7,-2) ∩ [-5, 5]

Answer: The expression is {x : -7<x <- 2 ∩ -5≤ x ≤ 5 } or {x : -5≤ x < -2 }




-5≤ x < -2 or [-5,-2)

## Video segment 2:

### Empty Set:

The set S that contains no element is called the empty set or the null set and is denoted by { } or ∅. A set that has only one element is called a singleton set.

### Universal Set:

A **Universal Set** is the set of all elements under consideration, denoted by U. All other sets are subsets of the universal set.

Let , A={1,3,5}

B={3,6}

C={7,8,9}

Let U be the universal set Then, the complement Universal set will be U={1, 2, 3, 5, 6, 7, 8, 9}

### Equal Set:

Two sets A and B can be equal only if each element of set A is also the element of the set B. Also if two sets are the subsets of each other, they are said to be equal. This is represented by:

$A = B$

$A \subset B$ and $B \subset A \Leftrightarrow A = B$

If the condition discussed above is not met, then the sets are said to be unequal. This is represented by

$A \neq B$

If P = {1, 3, 9, 5, −7} and Q = {5, −7, 3, 1, 9,}, then P = Q. It is also noted that no matter how many times an element is repeated in the set, it is only counted once. Also, the order doesn't matter for the elements in a set. So, to rephrase in terms of cardinal number, we can say that

## SubSet:

Suppose X is the set such that whenever $x \in X$, then $x \in Y$ as well. Here, X is said to be a subset of the set Y , and is denoted by $X \subseteq Y$ .

Example: the set {1, 2, 3, 4, 5}

A **subset** of this is {1, 2, 3}. Another subset is {3, 4} or even another is {1}, etc.

But {1, 6} is **not** a subset, since it has an element (6) which is not in the parent set.

## Common Symbols Used in Set Theory

*In the examples C = {1, 2, 3, 4} and D = {3, 4, 5}*

| Symbol | Meaning | Example |
|---|---|---|
| { } | Set: a collection of elements | {1, 2, 3, 4} |
| A ∪ B | Union: in A or B (or both) | C ∪ D = {1, 2, 3, 4, 5} |
| A ∩ B | Intersection: in both A and B | C ∩ D = {3, 4} |
| A ⊆ B | Subset: every element of A is in B. | {3, 4, 5} ⊆ D |
| A ⊂ B | Proper Subset: every element of A is in B, but B has more elements. | {3, 5} ⊂ D |
| A ⊄ B | Not a Subset: A is not a subset of B | {1, 6} ⊄ C |
| A ⊇ B | Superset: A has same elements as B, or more | {1, 2, 3} ⊇ {1, 2, 3} |
| A ⊃ B | Proper Superset: A has B's elements and more | {1, 2, 3, 4} ⊃ {1, 2, 3} |
| A ⊅ B | Not a Superset: A is not a superset of B | {1, 2, 6} ⊅ {1, 9} |
| $A^c$ | Complement: elements not in A | $D^c$ = {1, 2, 6, 7}<br>When $\mathbb{U}$ = {1, 2, 3, 4, 5, 6, 7} |
| A − B | Difference: in A but not in B | {1, 2, 3, 4} − {3, 4} = {1, 2} |
| *a* ∈ A | Element of: *a* is in A | 3 ∈ {1, 2, 3, 4} |
| *b* ∉ A | Not element of: *b* is not in A | 6 ∉ {1, 2, 3, 4} |
| ∅ | Empty set = {} | {1, 2} ∩ {3, 4} = ∅ |
| $\mathbb{U}$ | Universal Set: set of all possible values (in the area of interest) | |
| | | |
| **P**(A) | Power Set: all subsets of A | P({1, 2}) = { {}, {1}, {2}, {1, 2} } |
| A = B | Equality: both sets have the same members | {3, 4, 5} = {5, 3, 4} |
| A×B | Cartesian Product (set of ordered pairs from A and B) | {1, 2} × {3, 4} = {(1, 3), (1, 4), (2, 3), (2, 4)} |
| \|A\| | Cardinality: the number of elements of set A | \|{3, 4}\| = 2 |
| | | |
| \| | Such that | { *n* \| *n* > 0 } = {1, 2, 3,...} |
| : | Such that | { *n* : *n* > 0 } = {1, 2, 3,...} |
| ∀ | For All | ∀x>1, $x^2$>x |
| ∃ | There Exists | ∃ x \| $x^2$>x |
| ∴ | Therefore | *a*=*b* ∴ *b*=*a* |
| | | |
| $\mathbb{N}$ | Natural Numbers | {1, 2, 3,...} or {0, 1, 2, 3,...} |
| $\mathbb{Z}$ | Integers | {..., −3, −2, −1, 0, 1, 2, 3, ...} |
| $\mathbb{Q}$ | Rational Numbers | |

## Venn Diagram:

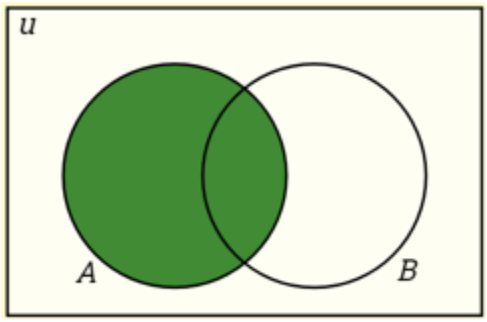
A Venn Diagram is a pictorial representation of the relationships between sets.

We can represent sets using **Venn diagrams.** In a Venn diagram, the sets are represented by

shapes; usually circles or ovals. The elements of a set are labeled within the circle. In the figure A and B set is represented in a different relationship.

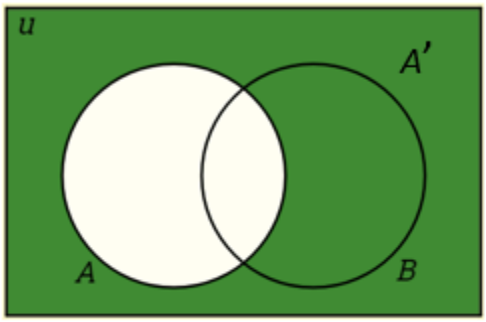
The following diagrams show the set operations and Venn Diagrams for Complement of a Set, Disjoint Sets, Subsets, Intersection and Union of Sets.

The set of all elements being considered is called the **Universal Set (U)** and is represented by a rectangle. Here, Colored portion represent the element of Set A.



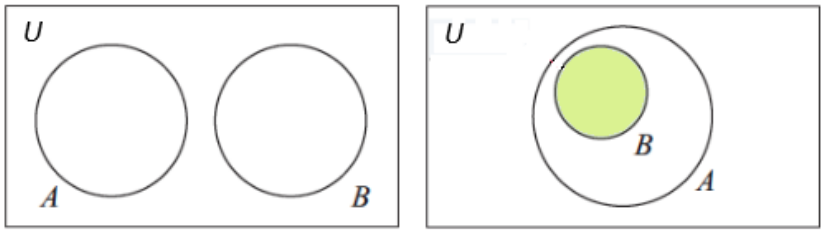

Set *A*

The **complement of A, A'**, is the set of elements in U but not in A. A' ={x| x ∈ U and x ∉ A}. Colored section is A'.




*A'* the complement of *A*

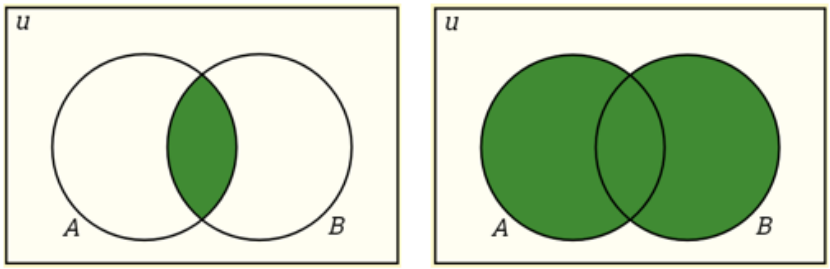
Set A and B are **disjoint sets** if they do not share any common elements.
B is a **proper subset** of A. This means B is a subset of A, but B ≠ A.




*A* and *B* are disjoint sets

*B* is proper subset of *A* $B \subset A$

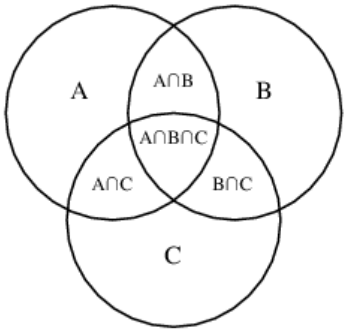
The **intersection of A and B** is the set of elements in both set A and set B. A ∩ B = {x| x ∈ A and x ∈ B}
The **union of A and B** is the set of elements in set A or set B. A ∪ B = {x| x ∈ A or x ∈ B}
A ∩ ∅ = ∅
A ∪ ∅ = A




Both A and B
*A* intersect *B*
$A \cap B$

Either A or B
*A* union *B*
$A \cup B$

The order-three diagram consists of three symmetrically placed mutually intersecting circles comprising a total of eight regions. The regions labeled $A$, $B$, and $C$ consist of members which are only in one set and no others, the three regions labelled $A \cap B$, $A \cap C$, and $B \cap C$ consist of members which are in two sets but not the third, the region $A \cap B \cap C$ consists of members which are simultaneously in all three, and no regions occupied represents $\emptyset$




## Video segment 3:

Cardinality of set:

Total number of elements of a set is known as cardinality or size of the set. If a set A={2,4,5,8,10} then the cardinality of set A is 5. Which is denoted as |A| or n(A). So, |A|=5.

## Finite set

Finite sets are the sets having a finite/countable number of members. Finite sets are also known as **countable sets** as they can be counted. The process will run out of elements to list if the elements of this set have a finite number of members.

**Examples of finite sets:**

P={ 0, 3, 6, 9, …, 99}

Q={ a : a is an integer, 1<a<10}

A set of all English Alphabets (because it is countable).

**Another example of a Finite set:**

A set of months in a year.
**M =** {January, February, March, April, May, June, July, August, September, October, November, December}

n (M) = 12

It is a finite set because the number of elements is countable.

### Cardinality of Finite Set

If 'a' represents the number of elements of Set A, then the cardinality of a finite set is **n(A)=a**.

So, the cardinality of the above set is 26, because the number of elements(alphabets) is 26.

Hence, n (A) = 26.

Similarly, for a set containing the months in a year will have a cardinality of 12.

So, this way we can list all the elements of any finite set and list them in the *curly braces* or in *Roster form.*

### Properties of Finite sets

The following finite set conditions are always finite.

- A subset of Finite set.
- The union of two finite sets
- The power set of a finite set

## Infinite set

If a set is not finite, it is called an **infinite set** because the number of elements in that set is not countable and also we cannot represent it in Roster form. Thus, infinite sets are also known as **uncountable sets**.

So, to represent the elements of an Infinite set are represented by 3 dots (ellipse) to represent the infinity of that set.

## Examples of Infinite Sets

- A set of all whole numbers.W= {0, 1, 2, 3, 4,…}
- A set of all points on a line.
- A set of all triangles.
- The set of leaves on a tree
- The set of hair on the head
- The set of integers.

## Cardinality of Infinite Sets

The cardinality of a set is n (A) = x, where x is the number of elements of a set A. The cardinality of an infinite set is n (A)= infinite as the number of elements is unlimited in it.

## Properties of Infinite Sets

- The union of two infinite sets is infinite
- The power set of an infinite set is infinite
- The superset of an infinite set is also infinite

The power set of an infinite set is unlimited because its elements are also unlimited.

## Power Set:

In set theory, the power set (or powerset) of a Set A is defined as the set of all subsets of the Set A including the Set itself and the null or empty set. It is denoted by P(A). Basically, this set is the combination of all subsets including null set, of a given set.

If the given set has n elements, then its Power Set will contains $2^n$ elements.

Here, Let us say Set A = { a, b, c }

Number of elements: 3

Therefore, the subsets of the set are:

{ } which is the null or the empty set

{ a }

{ b }

{ c }

{ a, b }

{ b, c }

{ c, d }

{ a, b, c }

**The power set P(A) = { { } , { a }, { b }, { c }, { a, b }, { b, c }, { c, d }, { a, b, c } }**

Now, the Power Set should have $2^3 = 8$ elements.

The number of elements of a set is written as |A|, If A has n elements then it can be written as

$|P(A)| = 2^n$

## Properties

- It is much larger than the original set.
- The powerset of a countable finite set is uncountable.
- For a set of natural numbers, we can do one-to-one mapping of the resulted set, P(S), with the real numbers.
- P(s) of set S, if operated with the union of sets, the intersection of sets and complement of sets, denotes the example of Boolean Algebra.

## Power Set of a Empty Set

An empty set has zero element. Therefore, powerset of a empty set{ }, can be mentioned as;

- A set containing a null set.
- It contains zero or null elements.
- Empty set is the only subset.

## Cartesian Products:

If A and B are two non-empty sets, then their Cartesian product A × B is the set of all ordered pair of elements from A and B.

$$A \times B = \{(x, y) : x \in A, y \in B\}$$

Suppose, if A and B are two non-empty sets, then the Cartesian product of two sets, A and set B is the set of all ordered pairs (a, b) such that a ∈A and b∈B which is denoted as A × B.

## For Example;

**1.** If A = {7, 8} and B = {2, 4, 6}, find A × B.

**Solution:**

A × B = {(7, 2); (7, 4); (7, 6); (8, 2); (8, 4); (8, 6)}

The 6 ordered pairs thus formed can represent the position of points in a plane, if a and B are subsets of a set of real numbers.

**2.** If A × B = {(p, x); (p, y); (q, x); (q, y)}, find A and B.

**Solution:**

A is a set of all first entries in ordered pairs in A × B.

B is a set of all second entries in ordered pairs in A × B.

Thus A = {p, q} and B = {x, y}

**3.** If A and B are two sets, and A × B consists of 6 elements: If three elements of A × B are (2, 5) (3, 7) (4, 7) find A × B.

**Solution:**

Since, (2, 5) (3, 7) and (4, 7) are elements of A × B.

So, we can say that 2, 3, 4 are the elements of A and 5, 7 are the elements of B.

So, A = {2, 3, 4} and B = {5, 7}

Now, A × B = {(2, 5); (2, 7); (3, 5); (3, 7); (4, 5); (4, 7)}

Thus, A × B contain six ordered pairs.

## Video segment 4:

Set Operations:

## Union:

The **union** of sets ***A*** and ***B***, denoted by $A \cup B$ **,** is the set defined as

$$A \cup B = \{ x \mid x \in A \vee x \in B \}$$

**Example 1:** If $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ , then $A \cup B = \{1, 2, 3, 4, 5\}$ .

**Example 2:** If $A = \{1, 2, 3\}$ and $B = \{1, 2, 4, 5\}$ , then $A \cup B = \{1, 2, 3, 4, 5\}$ .

Note that elements are not repeated in a set.

## Intersection:

The **intersection** of sets ***A*** and ***B***, denoted by $A \cap B$ **,** is the set defined as

$$A \cap B = \{ x \mid x \in A \wedge x \in B \}$$

**Example 3:** If $A = \{1, 2, 3\}$ and $B = \{1, 2, 4, 5\}$ , then $A \cap B = \{1, 2\}$ .

**Example 4:** If $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ , then $A \cap B = \emptyset$ .

## Difference:

The **difference** of sets ***A*** from ***B*** , denoted by $A - B$ **,** is the set defined as

$$A - B = \{ x \mid x \in A \wedge x \notin B \}$$

**Example 5:** If $A = \{1, 2, 3\}$ and $B = \{1, 2, 4, 5\}$ , then $A - B = \{3\}$ .
**Example 6:** If $A = \{1, 2, 3\}$ and $B = \{4, 5\}$ , then $A - B = \{1, 2, 3\}$ .

Note that in general $A - B \neq B - A$

## Complement:

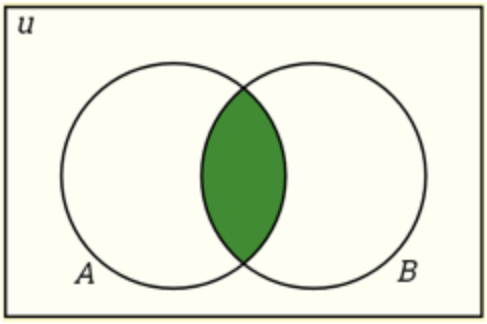
For a set ***A***, the difference $U - A$ **,** where ***U*** is the universe, is called the **complement** of ***A*** and it is denoted by $\overline{A}$ .

Thus $\overline{A}$ is the set of everything that is not in ***A***.

## Joint set:

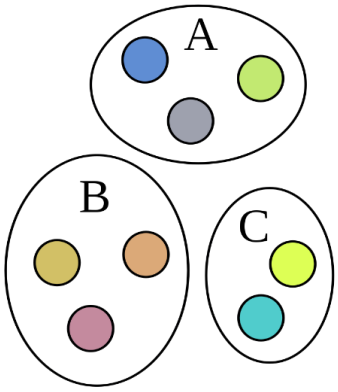
**If at least one element of two or more sets are common then those are called by joint set.**

A={1,2,6}A={1,2,6} and B={6,12,42}B={6,12,42}. There is a common element ‘6’, hence these sets are overlapping sets.




**Disjoint set** are those sets whose intersection with each other results in a null set. In Set theory, sometimes we notice that there are no common elements in two sets or we can state that the intersection of the sets is an empty set or null set. This type of set is called a **disjoint set**. For example, if we have X = {a, b, c} and Y = {d, e, f}, then we can say that the given two sets are disjoint since there are no common elements in these two sets X and Y.

Here is a disjoint set collection is represented.



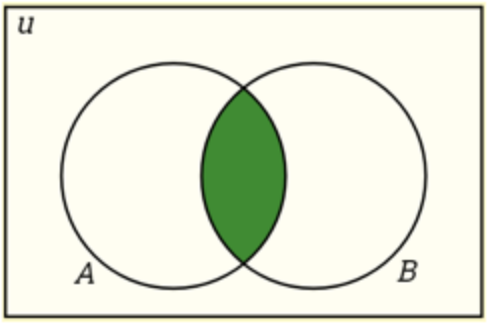

## Inclusion and Exclusion Principle:

The inclusion-exclusion principle can be expressed as follows:To compute the size of a union of multiple sets, it is necessary to sum the sizes of these sets **separately**, and then subtract the sizes of all **pairwise** intersections of the sets, then add back the size of the intersections of **triples** of the sets, subtract the size of **quadruples** of the sets, and so on, up to the intersection of **all** sets.

For two set In combinatorics, a branch of mathematics, the **inclusion–exclusion principle** is a counting technique which generalizes the familiar method of obtaining the number of elements in the union of two finite sets; symbolically expressed as

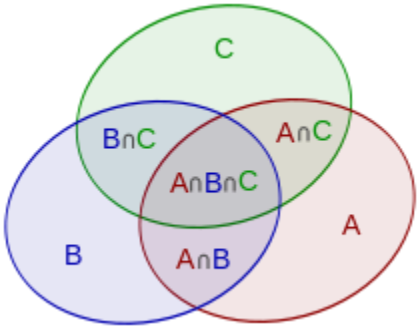
### The formulation using Venn diagrams

Let the diagram show three sets A, B




$$S(A \cup B) = S(A) + S(B) - S(A \cap B)$$

Let the diagram show three sets A, B and C:




Then the area of their union $A \cup B \cup C A \cup B \cup C$ is equal to the sum of the areas A, Band C less double-covered areas $A \cap B A \cap B$, $A \cap C A \cap C$, $B \cap C B \cap C$, but with the addition of the area covered by three sets $A \cap B \cap C A \cap B \cap C$:

$$S(A \cup B \cup C) = S(A) + S(B) + S(C) - S(A \cap B) - S(A \cap C) - S(B \cap C) + S(A \cap B \cap C)$$

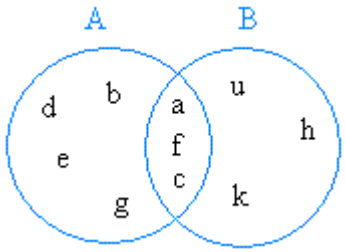
## Video Segment 5:

### Set Difference Example:

Given set A and set B the set difference of set B from set A is the set of all element in A, but not in B.

We can write A − B

**Example #1.**




Take a close look at the figure above. Elements in A **only** are b, d, e,g.

Therefore, A − B = { b, d, e, g}

Notice that although elements a, f, c are in A, we did not include them in A − B because we must not take anything in set B.

Sometimes, instead of looking at a the Venn Diagrams, it may be easier to write down the elements of both sets

Then, we show in bold the elements that are in A, but not in B

A = {**b**

, **d**, **e**, **g**, a, f, c}

B = { k, h, u, a, f, c}

**Example #2.**

Find B − A

Notice that this time you are looking for anything you see in B **only**

Elements that are in B only are shown in bold below

Let A = {1 orange, 1 pinapple, 1 banana, 1 apple}

Let B = {1 orange, **1 apricot**, 1 pinapple, 1 banana, **1 mango**, 1 apple, **1 kiwifruit** }

B − A = {1 apricot, 1 mango, 1 kiwifruit}

**Example #3.**

Find A − B

B = { 1, 2, 4, 6}

A = {1, 2, 4, 6, **7**, **8**, **9** }

What I see in A that are not in B are 7, 8, and 9

A − B = { 7, 8, 9}

**Example #4.**

Find B − A

A = { x / x is a number bigger than 6 and smaller than 10}

B = { x / x is a positive number smaller than 15}

A = {7, 8, 9} and B = { **1**, **2**, **3**, **4**, **5**, **6**, 7, 8, 9, **10**, **11**, **12**, **13**, **14**}

Everything you see in bold above are in B only.

B − A = {1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14}

**Complement Example:**

**Examples :**

1) If A = { 1, 2, 3, 4} and U = { 1, 2, 3, 4, 5, 6, 7, 8} then find A complement ( A').

**Solution :**
A = { 1, 2, 3, 4} and Universal set = U = { 1, 2, 3, 4, 5, 6, 7, 8}

Complement of set A contains the elements present in universal set but not in set A.

Elements are 5, 6, 7, 8.

∴ A complement = A' = { 5, 6, 7, 8}.

2) If B = { x | x is a book on Algebra in your library} . Find B'.

**Solution :** B' = { x | x is a book in your library and x ∉ B }

3) If A = { 1, 2, 3, 4, 5 } and U = N , then find A'.

**Solution :**
A = { 1, 2, 3, 4, 5 }

U = N

⇒ U = { 1, 2, 3, 4, 5, 6, 7, 8, 9,10,… }

A' = { 6, 7, 8, 9, 10, … }

4) If A = { x | x is a multiple of 3, x ∉ N }. Find A'.

**Solution :**
As a convention, x ∉ N in the bracket indicates N is the universal set.

N = U = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10,11, … }

A = { x | x is a multiple of 3, x ∉ N }

A = { 3, 6, 9, 12, 15, … }

So, A' = { 1, 2, 4, 5, 7, 8, 10,11, … }

## De Morgan's Law proof: Follow the book.

## Generalized Union and intersection:

As we saw earlier, union, intersection and Cartesian product of sets are associative For example

$$(A \cup B) \cup C = A \cup (B \cup C)$$

To denote either of these we often use $A \cup B \cup C$.

This can be generalized for the union of any finite number of sets as $A_1 \cup A_2 \cup .... \cup A_n$, which we write as

$\cup_{i=1}^{n} A_i$

This **generalized union** of sets can be rigorously defined as follows:

**Definition** ($\cup_{i=1}^{n} A_i$) :

**Basis Clause:** For $n = 1$, $\cup_{i=1}^{n} A_i = A_1$.

**Inductive Clause:** $\cup_{i=1}^{n+1} A_i = (\cup_{i=1}^{n} A_i) \cup A_{n+1}$

Similarly the **generalized intersection** $\cap_{i=1}^{n} A_i$ and **generalized Cartesian product** $\times_{i=1}^{n} A_i$ can be defined.

Based on these definitions, De Morgan's law on set union and intersection can also be generalized as follows:

**Theorem (Generalized De Morgan)**

$\overline{\cup_{i=1}^{n} A_i} = \cap_{i=1}^{n} \overline{A_i}$, and

$\overline{\cap_{i=1}^{n} A_i} = \cup_{i=1}^{n} \overline{A_i}$

# CONCISE NOTE OF LECTURE 2- FUNCTION

## Relations vs. Functions

A relation is just a relationship between x- and y-coordinates that means two variables or two equations.

Example 1: $y^2=x$

Example 2: If we consider two set like, A= {1,5} and B={3,4}

Then the product of the two sets are A×B= {(1,3), (1,4) (5,3) (5,4)} and the order should be (x,y). But if we say that x>y then the product answer should be A×B= { (5,3) (5,4)}. So relation depends on the given condition.

In the set of everything that is a relation, there's a smaller subset which we call functions.

A function tries to define these relationships. It tries to give the relationship a mathematical form. An equation is a mathematical way of looking at the relationship between concepts or items. These concepts or items are represented by what are called variables.

Example: y=x+3

A variable represents a concept or an item whose magnitude can be represented by a number, i.e. measured quantitatively. Variables are called variables because they vary, i.e. they can have a variety of values. Thus a variable can be considered as a quantity which assumes a variety of values in a particular problem. Many items in economics can take on different values. Mathematics usually uses letters from the end of the alphabet to represent variables.

Example1: Set variable A={2, 6, 8, 10} so the value of a set is 2 , 6, 8 and 10 which is changeable.

Example2: Equation Variable Y=X+3, Here Y and X is two variable and Y is changeable for the all distinct value of X.

Domain:

The “domain” of a function or relation is:

- the set of all values for which it can be evaluated
- the set of  allowable “input” values
- the values along the horizontal axis for which a point can be plotted along the vertical axis

Example 1:

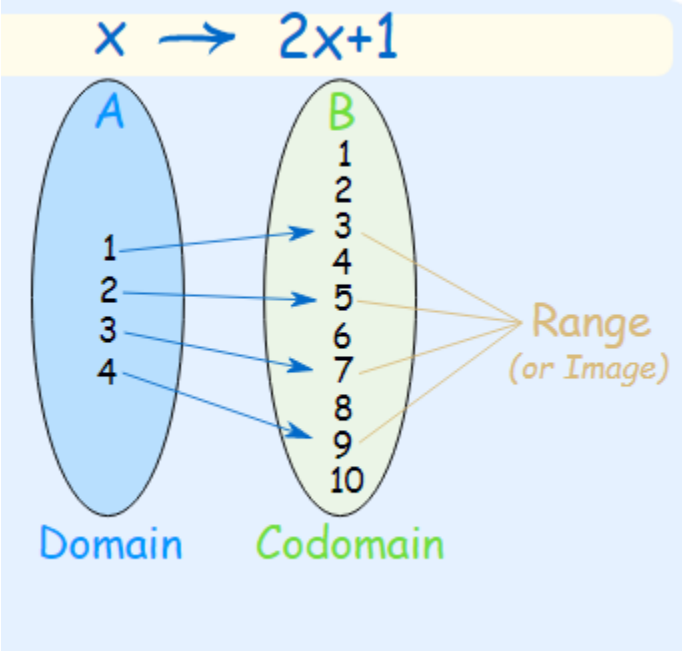
• The set "A" is the Domain,

• The set "B" is the Codomain,

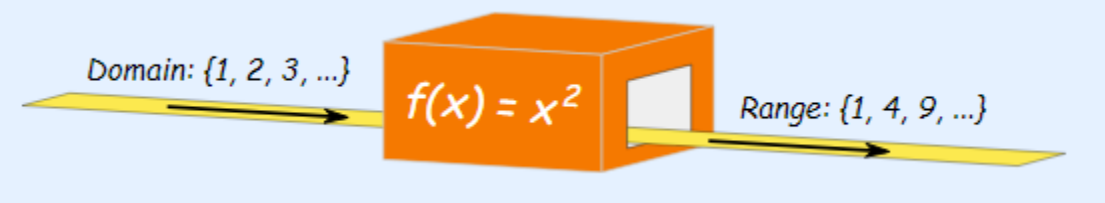
• And the set of elements that get pointed to in B (the actual values produced by the function) are the Range, also called the Image.

And we have:

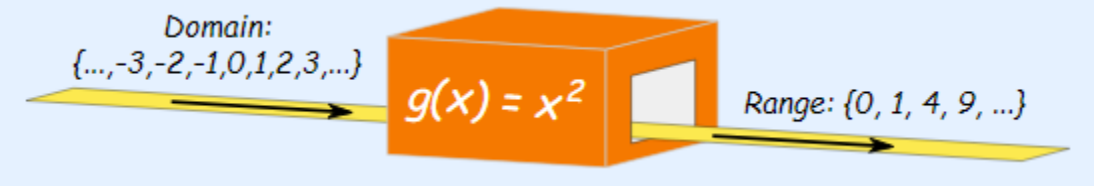
- Domain: {1, 2, 3, 4}
- Codomain: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
- Range: {3, 5, 7, 9}




Example 2: a simple function like $f(x) = x^2$ can have the domain (what goes in) of just the counting numbers {1,2,3,...}, and the range will then be the set {1,4,9,...}




And another function $g(x) = x^2$ can have the domain of integers {...,-3,-2,-1,0,1,2,3,...}, in which case the range is the set {0,1,4,9,...}




Range

The "range" of a function or relation is:

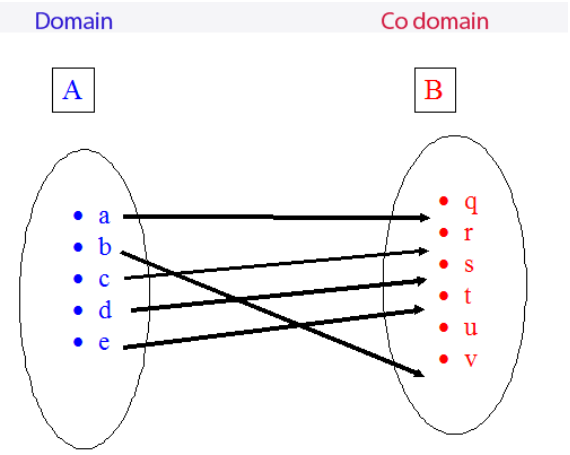
- the set of all values that it can produce
- its "output" set of values
- the set of values along the vertical axis for which a point can be plotted on its graph

Codomain

The "codomain" of a function or relation is a set of values that includes the Range as described above, but may also include additional values beyond those in the range.

Codomains can be useful when:

- You need to restrict the output of a function. For example, by specifying a codomain to be “the set of positive Real numbers”, you are instructing any who use the function to ignore any negative values it produces.
- The Range might be difficult to specify exactly, but a larger set of numbers that includes the entire Range can be specified. For example, a codomain could specify the set of all positive Real numbers, even though the function does not generate all possible positive Real numbers.
- More examples are given below:
  Example1:




*Domain* $A\{a,b,c,d,e\}$

*Codomain* $B\{q, r, s, t, u, v\}$

*Range* $\{q, r, s, t, v\}$

Example:2

$f(x) = x^2 + 2,\ x \in \{-3,-2,-1,0,1,2,3\}$

*a*) State the domain

*b*) State the range

a) The domain is $\{-3,-2,-1,0,1,2,3\}$

b) $f(x)=x^2+2$

$f(-3)=(-3)^2+2=9+2=11$

$f(-2)=(-2)^2+2=4+2=6$

$f(-1)=(-1)^2+2=1+2=3$

$f(0)=(0)^2+2=0+2=2$

$f(1)=1^2+2=1+2=3$

$f(2)=2^2+2=4+2=6$

$f(3)=3^2+2=9+2=11$

The range is $\{2,3,6,11\}$

Example 3:

The function $f(x)=\cos x^\circ$

has domain $\{x:\ 0\le x\le 360, x\in\mathbb{R}\}$

*a*) State the range

*b*) If $f(a)=1$, find $a$

a) The range is $\{f(x): -1\le f(x)\le 1,\ x\in\mathbb{R}\}$

Example 4:

The function $h(x) = x^2 + 1$

has domain $x \subset \mathbb{Z}$ and range $\{1,2,5,10\}$

List all possible elements of the domain.

$h(x) = x^2 + 1$ so set $h(x) = range = x^2 + 1$

$h(x) = x^2 + 1$

$1 = x^2 + 1$

$0 = x^2$

$x = \pm\sqrt{0}$

$x = 0$

$h(x) = x^2 + 1$

$2 = x^2 + 1$

$1 = x^2$

$x = \pm\sqrt{1}$

$x = \pm 1$

$h(x) = x^2 + 1$

$5 = x^2 + 1$

$4 = x^2$

$x = \pm\sqrt{4}$

$x = \pm 2$

$h(x) = x^2 + 1$

$10 = x^2 + 1$

$9 = x^2$

$x = \pm\sqrt{9}$

$x = \pm 3$ Domain $\{-3,-2,-1,0,1,2,3\}$

How to Find the Domain of a Rational Function: Examples with Solutions

**Example 1**

Find the domain of the function f defined by

$$f(x) = \frac{1}{x-2}$$

f(x) can take real values if the denominator of f(x) is NOT ZERO because division by zero is not allowed in mathematics

$x - 2 \neq 0$

Solve the above inequality for to obtain the domain: $x \neq 2$
Which in interval form may be written as follows
$(-\infty , 2) \cup (2 , +\infty)$

## Example 2

Find the domain of the function f defined by

$$f(x) = \frac{x+3}{x^2+7}$$

**Solution to Example 2**

For f(x) to have real values, the denominator must be different from zero. Hence

$x^2 + 7 \neq 0$

Expression $x^2 + 7$ is always positive (square added to a positive number). Hence the domain of f is given by the interval

$(-\infty , +\infty)$

## Example 3

Find the domain of the function f defined by:

$$f(x) = \frac{2x+9}{2x^2+x-15}$$

**Solution to Example 3**

For f(x) given above to be real, its denominator must be different from zero. Let us first find the roots of the denominator by solving the equation

$2 x^2 + x - 15 = 0$

The roots are

- 3 and 5 / 2

The denominator $2x^2 + x - 15$ is not equal to zeros for all real values except - 3 and 5 / 2.

Hence the domain of the given function is given by

$(-\infty , - 3) \cup (- 3 , 5 / 2) \cup (5 / 2 , +\infty)$

## Example 4

Find the domain of the function f given by:

$$f(x) = \frac{2}{2x - 6} - \frac{x}{4x + 7}$$

**Solution to Example 4**

For f(x) to be real, both denominators 2x - 6 and - 4x + 7 must not be equal to zero. Let us find the values of x that make the two denominators equal to zero

2x - 6 = 0 gives x = 3

4x + 7 gives x = - 7 / 4

f(x) is real for all real values except 3 and - 7/4. The domain of the above function is given by

$(-\infty , -7/4) \cup (-7/4 , 3) \cup (3 , +\infty)$

Examples on How to Find the Domain of Square Root Functions with Solutions

Example 1
Find the domain of function f defined by

$f(x) = \sqrt{(x - 1)}$

Solution to Example 1

- For f(x) to have real values, the radicand (expression under the radical) of the square root function must be positive or equal to 0. Hence
  $x - 1 \geq 0$
- The solution set to the above inequality is the domain of f(x) and is given by:
  $x \geq 1$
  or in interval form $[1 , +\infty)$

Example 2
Find the domain of function f defined by

$f(x) = \sqrt{ [ (x - 2)(x + 3) ] }$

Solution to Example 2

- For f(x) to have real values, the radicand $(x - 2)(x + 3)$ must be positive. Hence
  $(x - 2)(x + 3) \geq 0$
- Solve the above inequality to obtain the solution set, which is also the domain, in interval form as follows:
  $(-\infty , -3] \cup [2 , + \infty)$

Example 3
Find the domain of function f defined by:

$f(x) = \sqrt{ [ x^2 + 2 x - 1 ] }$

Solution to Example 3

- For $\sqrt{[\ x^2 + 2x - 1\ ]}$ to be real, the radicand must be positive or equal to 0. Hence the inequality
  $x^2 + 2x - 1 \geq 0$
- The solution set of the above inequality, which is also the domain, is given in interval form as follows:
  $(-\infty , -1-\sqrt{2}] \cup [-1+ \sqrt{2} , + \infty)$
- The domain of the given function is given by the interval $(-\infty , -1-\sqrt{2}] \cup [-1+ \sqrt{2} , + \infty)$.

Example 4
Find the domain of function f defined by:

$f(x) = \sqrt{[\ (2x - 1)/(x + 3)\ ]}$

Solution to Example 4

- The domain of this function is the set of all values of x such that $(2x - 1)/(x + 3) \geq 0$ which is an inequality to solve. The solution set of the above inequality which is also the domain is given by
  $(-\infty , -3) \cup [\ 1 / 2 , +\infty)$

Domain and Range of Logarithmic Functions

**EXAMPLE 1: IDENTIFYING THE DOMAIN OF A LOGARITHMIC SHIFT**

What is the domain of $f(x) = \log_2(x + 3)$?

**SOLUTION**

The logarithmic function is defined only when the input is positive, so this function is defined when $x + 3 > 0$. Solving this inequality,

$$\begin{cases} x + 3 > 0 & \text{The input must be positive.} \\ x > -3 & \text{Subtract 3.} \end{cases}$$

The domain of $f(x) = \log_2(x + 3)$ is $(-3, \infty)$.

**EXAMPLE 2: IDENTIFYING THE DOMAIN OF A LOGARITHMIC SHIFT AND REFLECTION**

What is the domain of $f(x) = \log(5 - 2x)$?

**SOLUTION**

The logarithmic function is defined only when the input is positive, so this function is defined when $5 - 2x > 0$. Solving this inequality,

$$\begin{cases} 5 - 2x > 0 & \text{The input must be positive.} \\ -2x > -5 & \text{Subtract 5.} \\ x < \frac{5}{2} & \text{Divide by } -2 \text{ and switch the inequality.} \end{cases}$$

The domain of $f(x) = \log(5 - 2x)$ is $\left(-\infty, \frac{5}{2}\right)$.

When we call relation as a function?

| Example 1 | | |
|---|---|---|
| Problem | Is the relation given by the set of ordered pairs below a function?<br>$\{(-3, -6),(-2, -1),(1, 0),(1, 5),(2, 0)\}$ | |
| | $x$ \| $y$<br>$-3$ \| $-6$<br>$-2$ \| $-1$<br>$1$ \| $0$<br>$1$ \| $5$<br>$2$ \| $0$ | Organizing the ordered pairs in a table can help.<br>By definition, the inputs in a function have only one output.<br><br>The input 1 has two outputs: 0 and 5. |
| *Answer* | The relation is not a function. | |

**Example 2**

**Problem** Is the relation given by the set of ordered pairs below a function?
$\{(-3, 4),(-2, 4),(-1, 4),(2, 4),(3, 4)\}$

| $x$ | $y$ |
|---|---|
| −3 | 4 |
| −2 | 4 |
| −1 | 4 |
| 2 | 4 |
| 3 | 4 |

You could reorganize the information by creating a table.

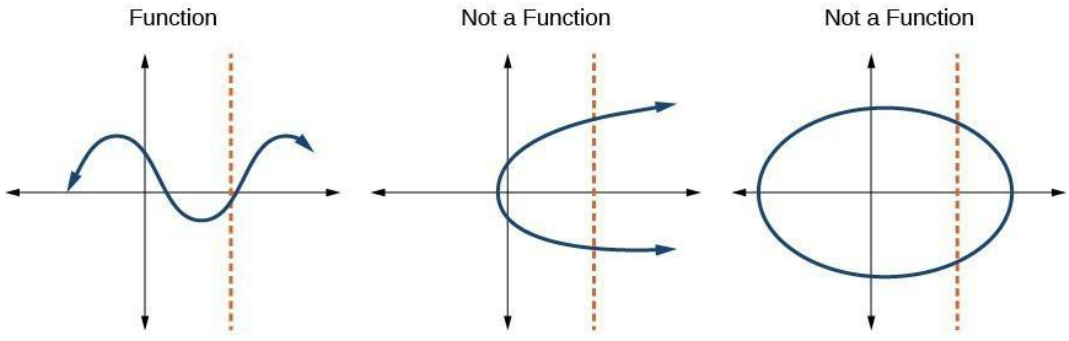
Each input has only one output.

Each input has only one output, and the fact that it is the same output (4) does not matter.

*Answer* This relation is a function.

Identify Function from a graph:

The vertical line test can be used to determine whether a graph represents a function. If we can draw any vertical line that intersects a graph more than once, then the graph does *not* define a function because a function has only one output value for each input value.



Principles of Mathematics 12

# Permutations and Combinations

MATH

# Lesson 1

## Permutations

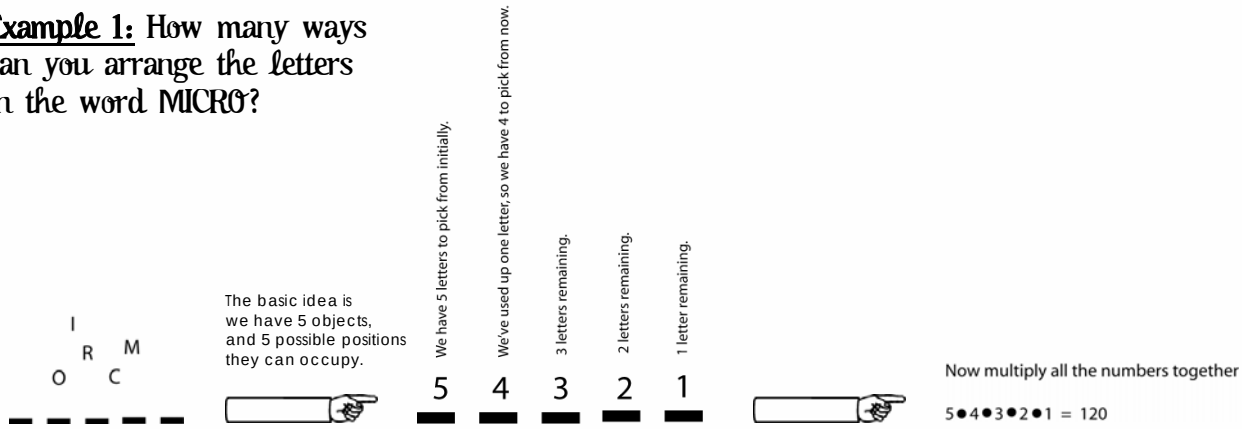


# Permutations and Combinations

## Lesson 1, Part One: The Fundamental Counting Principle

The Fundamental Counting Principle: This is an easy way to determine how many ways you can arrange items. The following examples illustrate how to use it:

**Example 1:** How many ways can you arrange the letters in the word MICRO?




**Example 2:** How many ways can 8 different albums be arranged?

We *could* approach this question in the same way as the last one by using the spaces and multiplying all the numbers, but there is a shorter way.
The factorial function on your calculator will perform this calculation for you!

$$8! = 8 \bullet 7 \bullet 6 \ldots . 2 \bullet 1$$
$$8! = \mathbf{40320}$$

**TI-83 Info**
*You can get the factorial function using:*
**Math → Prb → !**

## Questions:

**1)** How many ways can the letters in the word PENCIL be arranged?

**2)** If there are four different types of cookies, how many ways can you eat all of them?

**3)** If three albums are placed in a multi-disc stereo, how many ways can the albums be played?

**4)** How many ways can you arrange all the letters in the alphabet?

**5)** How many ways can you arrange the numbers 24 through 28 (inclusive)?

**Answers:**
**1)** 6! = 720
**2)** 4! = 24
**3)** 3! = 6
**4)** $26! = 4.03 \bullet 10^{26}$
**5)** 5! = 120

# Permutations and Combinations

## Lesson 1, Part Two: Repetitions Not Allowed

**Repetitions Not Allowed:** In many cases, some of the items we want to arrange are identical. For example, in the word TOOTH, if we exchange the places of the two O's, we still get TOOTH. Because of this, we have to get rid of extraneous cases by dividing out repetitions.

**Example 1:** How many ways can you arrange the letters in the word THESE?

Do this as a fraction. Factorial the total number of letters and put this on top. Factorial the repeated letters and put them on the bottom.

$\frac{5!}{2!}$ — $5!$: There are 5 letters altogether; $2!$: There are 2 E's

$$\frac{5!}{2!} = \frac{120}{2} = \mathbf{60}$$

**Example 2:** How many ways can you arrange the letters in the word REFERENCE?

$\frac{9!}{2!\bullet 4!}$ — $9!$: There are 9 letters altogether; $2!\bullet 4!$: There are 2 R's and 4 E's

$$\frac{9!}{2!\bullet 4!} = \frac{362880}{2 \bullet 24} = \mathbf{7560}$$

**TI-83 Info**
*Make sure you put the denominator in brackets or you'll get the wrong answer!*

## Questions:

1) How many ways can the letters in the word SASKATOON be arranged?

2) How many ways can the letters in the word MISSISSIPPI be arranged?

3) How many ways can the letters in the word MATHEMATICS be arranged?

4) If there are eight cookies (4 chocolate chip, 2 oatmeal, and 2 chocolate) in how many different orders can you eat all of them?

5) If a multiple choice test has 10 questions, of which one is answered *A*, 4 are answered *B*, 3 are answered *C*, and 2 are answered *D*, how many answer sheets are possible?

**Answers:**

1) $\frac{9!}{2!\bullet 2!\bullet 2!} = 45360$

2) $\frac{11!}{4!\bullet 4!\bullet 2!} = 34650$

3) $\frac{11!}{2!\bullet 2!\bullet 2!} = 4989600$

4) $\frac{8!}{4!\bullet 2!\bullet 2!} = 420$

5) $\frac{10!}{4!\bullet 3!\bullet 2!} = 12600$

# Permutations and Combinations

## Lesson 1, Part Three: Repetitions Are Allowed

**Repetitions Are Allowed:** Sometimes we are interested in arrangements allowing the use of items more than once.

**Example 1:** There are 9 switches on a fuse box. How many different arrangements are there?

2 _ 2 _ 2 _ 2 _ 2 _ 2 _ 2 _ 2 _ 2

Each switch has two possible positions, on or off.
Placing a 2 in each of the 9 positions, we have $2^9$ = **512**.

**Example 2:** How many 3 letter words can be created, if repetitions are allowed?

26 _ 26 _ 26

There are 26 letters to choose from, and we are allowed to have repetitions.
There are $26^3$ = **17576** possible three letter words.

### Questions:

**1)** If there are 4 light switches on an electrical panel, how many different orders of on/off are there?

**2)** How many 5 letter words can be formed, if repetitions are allowed?

**3)** How many three digit numbers can be formed? (Zero can't be the first digit)

*There are **ten** digits in total, from zero to nine.*

**4)** A coat hanger has four knobs. If you have 6 different colors of paint available, how many different ways can you paint the knobs?

Answers:

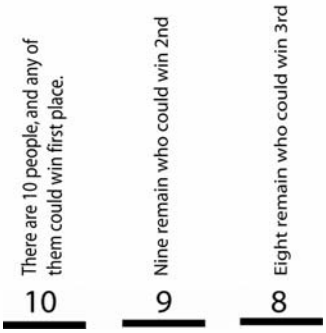
**1)** $2^4 = 16$
**2)** $26^5 = 11881376$
**3)** $9 \cdot 10 \cdot 10 = 900$
**4)** $6^4 = 1296$

# Permutations and Combinations

## Lesson 1, Part Four: Arranging A Subset

Arranging a subset of items: Sometimes you will be given a bunch of objects, and you want to arrange only a few of them:

**Example 1:** There are 10 people in a competition. How many ways can the top three be ordered?




Since only 3 positions can be filled, we have 3 spaces. Multiplying, we get 720.

A fast way to do questions like this is to use the $_nP_r$ feature on your calculator.

n is the total number of items.
r is the number of items you want to order.

For Example 1, you would type:
10 → Math → PRB → $_nP_r$ → enter → 3

**Example 2:** There are 12 movies playing at a theater, in how many ways can you see two of them consecutively?

12 11 OR $_{12}P_2$

You could use the spaces, but let's try this question with the permutation feature. There are 12 movies, and you want to see 2, so type $_{12}P_2$ into your calculator, and you'll get **132**.

**Example 3:** How many 4 letter words can be created if repetitions are not allowerd?

26 25 24 23 OR $_{26}P_4$

The answer is: $_{26}P_4$ = **358800**

## Questions:

**1)** How many three letter words can be made from the letters of the word KEYBOARD

**2)** If there are 35 songs and you want to make a mix CD with 17 songs, how many different ways could you arrange them?

**3)** There are six different colored balls in a box, and you pull them out one at a time. How many different ways can you pull out four balls?

**4)** A committee is to be formed with a president, a vice-president, and a treasurer. There are 10 people to be selected from. How many different committees are possible?

**5)** A baseball league has 13 teams, and each team plays each other twice; once at home, and once away. How many games are scheduled?

Answers:

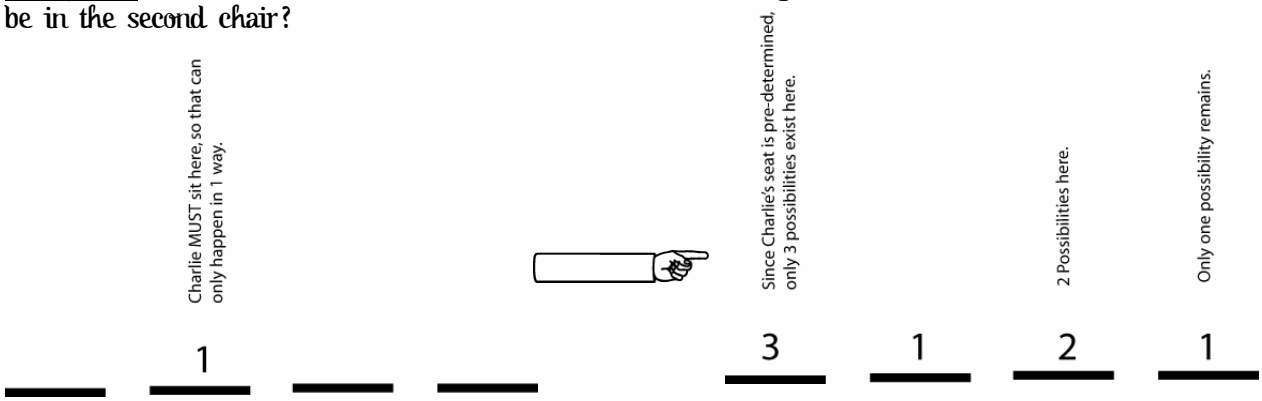
**1)** $_8P_3$ =336
**2)** $_{35}P_{17} = 1.6 \cdot 10^{24}$
**3)** $_6P_4$ = 360
**4)** $_{10}P_3$ = 720
**5)** $_{13}P_2$ = 156

# Permutations and Combinations

## Lesson 1, Part Five: Specific Positions

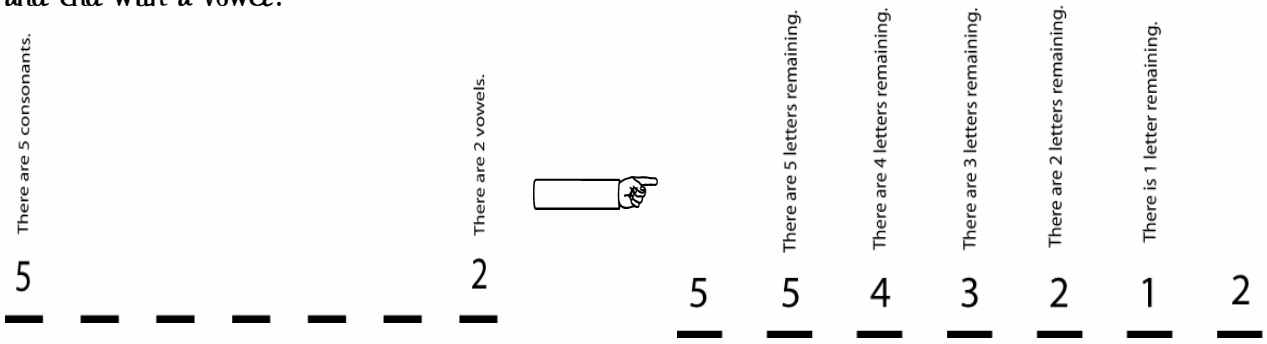
Specific Positions: Frequently when arranging items, a particular position must be occupied by a particular item. The easiest way to approach these questions is by analyzing how many possible ways each space can be filled.

**Example 1:** How many ways can Adam, Beth, Charlie, and Doug be seated in a row if Charlie must be in the second chair?



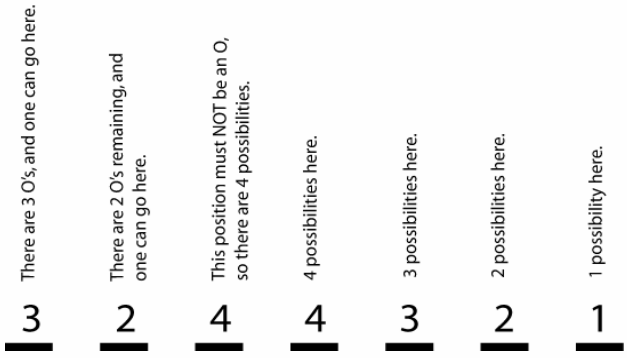

The answer is **6**.

**Example 2:** How many ways can you order the letters of KITCHEN if it must start with a consonant and end with a vowel?




The answer is **1200**.

**Example 3:** How many ways can you order the letters of TORONTO if it begins with exactly two O's?




***Exactly Two O's*** *means the first 2 letters must be O, and the third must NOT be an O.*

*If the question simply stated* ***two O's****, then the third letter could also be an O, since that case wasn't excluded.*

Don't forget repetitions! The answer from the left will be the numerator with repetitions divided out. $\frac{576}{3! \cdot 2!} = \mathbf{48}$

## Questions:

1) Six Pure Math 30 students (Brittany, Geoffrey, Jonathan, Kyle, Laura, and Stephanie) are going to stand in a line: How many ways can they stand if:



**a)** Stephanie must be in the third position?

**b)** Geoffrey must be second and Laura third?

**c)** Kyle can't be on either end of the line?

**d)** Boys and girls alternate, with a boy starting the line?

**e)** The first three positions are boys, the last three are girls?

**f)** A girl must be on both ends?

**g)** The row starts with two boys?

**h)** The row starts with exactly two boys?

**i)** Brittany must be in the second position, and a boy must be in the third?

2) How many ways can you order the letters from the word TREES if:

**a)** A vowel must be at the beginning?

**b)** It must start with a consonant and end with a vowel?

**c)** The R must be in the middle?

**d)** It begins with an E?

**e)** It begins with exactly one E?

**f)** Consonants & vowels alternate?

## Answers:

**1)** a) If Stephanie must be in the third position, place a one there to reserve her spot. You can then place the remaining 5 students in any position. $5 \quad 4 \quad 1 \quad 3 \quad 2 \quad 1 = 120$

b) Place a 1 in the second position to reserve Geoffrey's spot, and place a 1 in the third position to reserve Laura's spot. Place the remaining students in the other positions. $4 \quad 1 \quad 1 \quad 3 \quad 2 \quad 1 = 24$

**c)** Since Kyle can't be on either end, 5 students could be placed on one end, then 4 at the other end. Now that 2 students are used up, there are 4 that can fill out the middle. $5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 4 = 480$

d) Three boys can go first, then three girls second. Two boys remain, then two girls. Then one boy and one girl remain. $3 \quad 3 \quad 2 \quad 2 \quad 1 \quad 1 = 36$

e) Three boys can go first, then place the girls in the next three spots. $3 \quad 2 \quad 1 \quad 3 \quad 2 \quad 1 = 36$

f) Three girls could be placed on one end, then 2 girls at the other end. There are four students left to fill out the middle. $3 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 = 144$

g) Three boys could go first, then 2 boys second. Once those positions are filled, four people remain for the rest of the line. $3 \quad 2 \quad 4 \quad 3 \quad 2 \quad 1 = 144$

h) Three boys could go first, then two boys second. The third position can't be a boy, so there are three girls that could go here. Then, three people remain to fill out the line. $3 \quad 2 \quad 3 \quad 3 \quad 2 \quad 1 = 108$

i) Place a 1 in the second position to reserve Brittany's spot, then 3 boys could go in the third position. Now fill out the rest of the line with the four remaining people. $4 \quad 1 \quad 3 \quad 3 \quad 2 \quad 1 = 72$

2) Note that since there are 2 E's, all answers MUST be divided by 2! to eliminate repetitions.

a) There are two vowels that can go first, then four letters remain to fill out the other positions. $2 \quad 4 \quad 3 \quad 2 \quad 1 = 48$ *(Answer = 48 / 2! =* ***24****)*

b) Three consonants could go first, and two vowels could go last. There are three letters to fill out the remaining positions. $3 \quad 3 \quad 2 \quad 1 \quad 2 = 36$ *(Answer = 36 / 2! =* ***18****)*

c) Place a 1 in the middle spot to reserve the R's spot. Then fill out the rest of the spaces with the remaining 4 letters. $4 \quad 3 \quad 1 \quad 2 \quad 1 = 24$ *(Answer = 24 / 2! =* ***12****)*

d) Two E's could go in the first spot, then fill the remaining spaces with the 4 remaining letters. $2 \quad 4 \quad 3 \quad 2 \quad 1 = 48$ *(Answer = 48 / 2! =* ***24****)*

e) Two E's could go in the first spot, but the next letter must NOT be an E, so there are 3 letters that can go here. Fill out the three last spaces with the 3 remaining letters. $2 \quad 3 \quad 3 \quad 2 \quad 1 = 36$ *(Answer = 36 / 2! =* ***18****)*

f) Three consonants could go first, then 2 vowels, and so on. $3 \quad 2 \quad 2 \quad 1 \quad 1 = 12$ *(Answer = 12 / 2! =* ***6****)*

## Lesson 1, Part Six: Adding Permutations

**More than one case (Adding):** Given a set of items, it is possible to form multiple groups by ordering any 1 item from the set, any 2 items from the set, and so on. If you want the total arrangements from multiple groups, you have to *ADD* the results of each case.

**Example 1:** How many words (of any number of letters) can be formed from CANS

| If you are ordering only one letter, any of the 4 can go here. | | If ordering two letters, any 4 can go here. | Any 3 can go here. | | If ordering three letters, any 4 can go here. | Any 3 can go here. | Any 2 can go here. | | If ordering four letters, any 4 can go here. | Any 3 can go here. | Any 2 can go here. | Any 1 can go here. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | + | 4 | 3 | + | 4 | 3 | 2 | + | 4 | 3 | 2 | 1 |

Since we are allowed to have any number of letters in a word, we can have a 1 letter word, a 2 letter word, a 3 letter word, and a 4 letter word.
We can't have more than 4 letters in a word, since there aren't enough letters for that!

We could also write this using permutations: ${}_4P_1 + {}_4P_2 + {}_4P_3 + {}_4P_4 = 64$

The answer is **64**

**Example 2:** How many four digit positive numbers less than 4670 can be formed using the digits 1, 3, 4, 5, 8, 9 if repetitions are not allowed?

We must separate this question into different cases. Numbers in the 4000's have extra restrictions.

**Case 1 - Numbers in the 4000's:** There is only one possibility for the first digit {4}. The next digit has three possibilities. {1, 3, 5}. There are 4 possibilities for the next digit since any remaining number can be used, and 3 possibilities for the last digit.

$$1 \quad 3 \quad 4 \quad 3 = 36$$

**Case 2 - Numbers in the 1000's and 3000's:** There are two possibilities for the first digit {1, 3}. Anything goes for the remaining digits, so there are 5, then 4, then 3 possibilities.

$$2 \quad 5 \quad 4 \quad 3 = 120$$

Add the results together: 36 + 120 = **156**

## Questions:

1) How many one-letter, two-letter, or three-letter words can be formed from the word PENCIL?

2) How many 3-digit, 4-digit, or 5-digit numbers can be made using the digits of 46723819?

3) How many numbers between 999 and 9999 are divisible by 5 and have no repeated digits?

### Answers:

**1)** ${}_6P_1 + {}_6P_2 + {}_6P_3 =$ **156**

**2)** ${}_8P_3 + {}_8P_4 + {}_8P_5 =$ **8736**

**3)** There are two cases: The first case has five as the last digit, the second case has zero as the last digit. Remember the first digit can't be zero!

$$\underset{\text{Must Not Be Zero}}{8} \quad 8 \quad 7 \quad \underset{\{5\}}{1} = 448$$

$$9 \quad 8 \quad 7 \quad \underset{\{0\}}{1} = 504$$

Add the results to get the total: **952**

# Permutations and Combinations

## Lesson 1, Part Seven: Items Always Together

**Always Together:** Frequently, certain items must always be kept together. To do these questions, you must treat the joined items as if they were only one object.

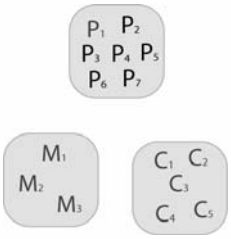
**Example 1:** How many arrangements of the word ACTIVE are there if C & E must always be together?




There are 5 groups in total, and they can be arranged in 5! ways.
The letters EC can be arranged in 2! ways.

The total arrangements are 5! x 2! = **240**

**Example 2:** How many ways can 3 math books, 5 chemistry books, and 7 physics books be arranged on a shelf if the books of each subject must be kept together?




There are three groups, which can be arranged in 3! ways.
The physics books can be arranged in 7! ways.
The math books can be arranged in 3! ways.
The chemistry books can be arranged in 5! ways.

The total arrangements are 3! x 7! x 3! x 5! = **21772800**

## Questions:

1) How many ways can you order the letters in KEYBOARD if K and Y must always be kept together?

2) How many ways can the letters in OBTUSE be ordered if all the vowels must be kept together?

3) How many ways can 4 rock, 5 pop, & 6 classical albums be ordered if all albums of the same genre must be kept together?

### Answers:

1) 

EBOARD 7! • 2! = 10080

2) 

BTS 4! • 3! = 144

3) 

3! • 4! • 5! • 6! = 12441600

# Permutations and Combinations

## Lesson 1, Part Eight: Items Never Together

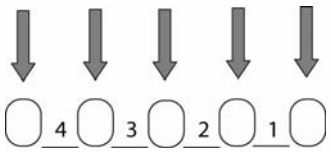
**Never Together:** If certain items must be kept apart, you will need to figure out how many possible positions the separate items can occupy.

**Example 1:** How many arrangements of the word ACTIVE are there if C & E must never be together?

**Method 1:**

4 3 2 1

First fill in the possible positions for the letters ATIV




Next draw empty circles representing the positions C & E can go.

You can place C & E in these spaces ${}_5P_2$ ways.
Get the answer by multiplying:

$4! \times {}_5P_2 = \mathbf{480}$

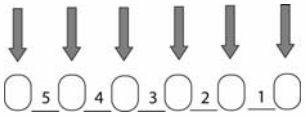
**Method 2:** First determine the number of ways ACTIVE can be arranged if C & E are ALWAYS together. (5! • 2!)
Then subtract that from the total number of possible arrangements without restrictions (6!)
The answer is 6! – (5! • 2!) = 480
***This method does not work if there are more then two items you want to keep separate.

**Example 2:** How many arrangements of the word DAUGHTER are there if none the vowels can ever be together?

First fill in the possible positions for the consonants

5 4 3 2 1

Next draw empty circles representing the positions the vowels can go.




You can place the 3 vowels in the 6 spaces in ${}_6P_3$ ways.

$5! \times {}_6P_3 = \mathbf{14400}$

**Example 3:** In how many ways can the letters from the word EDITOR be arranged if vowels and consonants alternate positions?

First determine the number of arrangements with consonants first in the arrangement:

$\underset{c}{3}\ \underset{v}{3}\ \underset{c}{2}\ \underset{v}{2}\ \underset{c}{1}\ \underset{v}{1} = 36$

Then determine the number of arrangements with vowels coming first:

$\underset{v}{3}\ \underset{c}{3}\ \underset{v}{2}\ \underset{c}{2}\ \underset{v}{1}\ \underset{c}{1} = 36$

Add the results to get **72** possible arrangements.

## Questions:

1) How many ways can you order the letters in QUEST if the vowels must never be together?

2) If 8 boys and 2 girls must stand in line for a picture, how many line-up's will have the girls separated from each other?

3) How many ways can you order the letters in FORTUNES if the vowels must never be together?

4) In how many ways can the letters AEFGOQSU be arranged if vowels and consonants alternate positions?

### Answers:

**1)** $3! \bullet {}_4P_2 = 72$ **or** $5! - (4! \bullet 2!) = 72$

**2)** $8! \bullet {}_9P_2 = 2903040$ **or** $10! - (9! \bullet 2!) = 2903040$

**3)** $5! \bullet {}_6P_3 = 14400$

**4)** Consonants first: $4\times4\times3\times3\times2\times2\times1\times1 = 576$
Vowels first: $4\times4\times3\times3\times2\times2\times1\times1 = 576$
Add results: 576 + 576 = **1152**

## *A group of 5 boys and 3 girls is to be photographed.*

- How many ways can they be arranged in one row?
  8! = 40320 ways

- How many ways can they be arranged with the girls in the front row and the boys in the back?
  Front row = 3! = 6  Back row = 5! = 120
  They can be arranged in: 6.120 = 720 ways.

- What if a boy is to sit in the end chairs? (All the chairs in one row)
  There exists 5 choices for seating a boy in the left seat, then 4 choices right end.
  Left (8-2 =) 6 people may be seated with no restriction:
  5.(6! (4)) = 14400 ways  b _ _ _ _ _ _ b
  5 6 5 4 3 2 1 4

- How many ways can this be done if boys sit side by side & girls side by side?
  Boys-Girls or Girls-Boys $\Rightarrow$ Total numbers = (5! 3!) + (3! 5!) = 1440

- In how many ways can be seated on a bench if only 4 seats are available?
  Number of arrangements of 8 people taken 4 at a time = P(8,4) = 8.7.6.5 = 1680

- How many ways can be arranged if the girls occupy the even places? (All the chairs in one row)
  The boys can be seated in P(5,5) = 5! = 120 ways, and the girls in P(3,3) = 3! = 6 ways.
  Number of arrangements = (120)(6) = 720

- In how many ways 2 particular people must not sit next to each other?
  Consider 2 particular people as 1 person. Then there are 7 people altogether and they can be arranged in 6! = 720 ways. But the 2 people can be arranged in 2! ways. Therefore, the number of ways to arranging 8 people with 2 particular people together is = 6! 2! = 1440.
  The total number of ways in which 8 people can be seated so that the 2 particular people do not sit together

  = (# of 1 seated anywhere) - (# 2 particular people seated together)
  = 7! – 1440 = 3600 ways.

- How many possibilities for 2 boys and 3 girls.
  C(5,2) C(3,2) = 10 .(3) = 30 possibilities.

- To select 3 how many ways for any mixture of boys and girls  $C(8,3) = \frac{8.7.6}{3.2.1} = 56$ ways

- How many to select 3 with majority boys

  All boys and no girl: C(5,3).C(3,0) = 10.(1) = 10
  2 boys and 1 girl:  C(5,2).C(3,1) = 10.(3) = 30
  $\Rightarrow$ the # of selecting majority boys = 10 + 30 = 40 ways.

**12 Boys & 10 Girls. 7 are chosen.**

- What is the probability that at least 2 Girls are chosen?

$$1 - [\Pr(\text{no Girls}) + \Pr(1\ G)] = 1 - \frac{C(12,7)+C(12,6)C(10,1)}{C(22,7)} \approx .94$$

- What is the probability no boys are chosen? $\Pr(\text{no boys}) = \Pr(\text{all Girls}) = \frac{C(10,7)}{C(22,7)} = \frac{5}{7106}$

- What is the probability that 1st three are boys? $\Pr = \frac{12.11.10}{22.21.20} = \frac{1}{7}$

- What is the probability that more boys than girls are chosen?

$$\Pr(B > G) = \frac{C(12,4).C(10,3)+C(12,5).C(10,2)+C(12,6).C(10,1)+C(12,7)}{C(22,7)} \approx .616$$

## *Poker hand consists of 5 cards selected from a deck of 52 cards.*

- How many different poker hands are there?
  C(52,5) = 2598960 hands  ({1,2,3,4,5} = {5,3,4,2,1})

- How many different poker hands consist entirely of aces and kings?
  Number of aces + kings = 8;  C(8,5) = 56 hands.

- How many different poker hands consist entirely of clubs?  # of clubs = 13,  C(13,5) = 1287 hands.

- How many consist of 3 aces and 2 kings?  C(4,3).C(4,2) = 4.6 = 24

- How many different poker hands consist entirely of red cards?
  # red cards = 26;  C(26,5) = 65780 hands.

- How many combinations have cards from exactly 2 suits?
  a)Consider one from the 1st suit, then there are C(4,1) = 4, and left 4 for the other suit then there are C(3,1) = 3. Therefore there are 4.C(13,1) . 3C(13,4) = 111540 ways.
  b) Consider 2 from the 1st suit, then there are C(4,1) = 4, and left 3 for the other suit then there are C(3,1) = 3. Therefore there are 4.C(13,2) . 3C(13,3) = 267696 ways
  c)Total = 111540 + 267696 = 379236 ways

- How many ways all the cards from the same suit?
  Select a suit, there are C(4,1) = 4 ways to do this. For each selection of a suit there are C(13,5) = 1287.
  Final = 4. C(13,5) = 5148 ways.

- How many ways 3 from one suit and 2 from another?
  Select a suit, there are C(4,1) = 4 ways to do this. The other suit is C(3,1) = 3 (since 3 suits left to choose from). First 3 from 1 suit there are 4.C(13,3) = 286 ways, and 2 from another 3.C(13,2) = 78.
  Total = 4.C(13,3) . 3C(13,2) = 22308 ways.

- How many ways 2 aces, 2 cards of another denomination, and 1 card of a 3rd denomination.
  - For 2 aces = C(4,2) = 6
  - 2 cards of another denomination are C(4,2) = 6 ways, there are 12 ways for the 2nd denomination.
    Therefore, there are 12.(6) = 72 ways
  - 3rd denomination the are 11 ways, 1 card $\Rightarrow$ 11.C(4,1) = 44
    The outcomes: 6.(72).(44) = 19008 hands.

- How many hands are in 2 cards of 1 denomination, 2 cards of another different denomination, and 1 card of a 3rd denomination.

  Select 2 cards of 1 denomination = C(13,2) = 78 ways.
  Select 2 of one denomination, there are C(4,2) = 6
  Then select 2 of the other = C(4,2) = 6
  Select the 3rd denomination, there are 11.C(4,1) = 44
  # of poker hands = 78.6.6.44 = 123552 hands.

Let the event A: card is a spade, and B is the face card.

- ***What is the probability of A?*** Pr (A={card is a spade}) = $\frac{\#\text{spades}}{\#\text{cards}} = \frac{13}{52} = \frac{1}{4}$

- ***What is the probability of B?*** Pr (B={face card}) = $\frac{\#\text{faces}}{\#\text{cards}} = \frac{12}{52} = \frac{3}{13}$

- ***What is the probability of $A \cap B$?*** Pr ($A \cap B$) = $\frac{\#\text{spade face cards}}{\#\text{cards}} = \frac{3}{52}$

* 13 cards are dealt from a deck of 52 cards

*a)* ***What is the probability that the ace of spades is one of the 13 cards***?

Pr (ace of spades) = $\frac{13}{52} = 1/4$

*b)* ***Suppose 1 of the 13 cards is chosen at random and not found to be the ace spades. What is the probability that none of the 13 cards is the ace of spades***?

$$\Pr(\text{none}|\ 1 \text{ is not}) = \frac{\Pr(\text{none}) \text{x } \Pr(\text{random 1 is not found} | \text{none})}{\Pr(\text{none}) \text{x} \Pr(\text{1 is not found} | \text{none}) + \Pr(\text{1 is}) \text{x} \Pr(\text{Ace is not} | \text{ace is})} = \frac{\frac{3}{4}\text{x}1}{\frac{3}{4}\text{x}1 + (\frac{1}{4}\text{x}\frac{12}{13})} = \frac{13}{17}$$

***c) Repeat previous experiment 10 times (replacing the card), and the ace of spades is not seen. What is the probability that the ace of spades actually is one of the 13 cards?***

$$\Pr(\text{aces is one}|\ 10 \text{ is not}) = \frac{\Pr(1\ is) x \Pr(10 \text{ are not} | \text{is})}{\Pr(1 \text{ is}) \text{x} \Pr(10 \text{ are not} | \text{is}) + \Pr(0 \text{ is}) \text{x} \Pr(10 \text{ are not}|\ 0 \text{ is})}$$

$$= \frac{\frac{1}{4} x \left(\frac{12}{13}\right)^{10}}{\frac{1}{4} x \left(\frac{12}{13}\right)^{10} + (\frac{3}{4} x 1)} \approx .13$$

## *COIN*

- 3 coins to be tossed.
  - ***What is the probability at least one head appears***?

    $\Pr(\{1,2,3\}) = \Pr(1) + \Pr(2) + \Pr(3) = \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8}$

  - ***What is the probability all heads or all tails***?

    $\Pr(\text{all heads or all tails}) = \Pr(0) + \Pr(3) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$

- 1 coin, ***what is the probability that heads is twice as likely to appear as tails***?

  Let $\Pr(T) = p \Rightarrow \Pr(H) = 2p \Rightarrow p + 2p = 1 \Rightarrow p = 1/3$ $\quad \Pr(T) = 1/3 \Rightarrow \Pr(H) = 2/3$

- 1 coin is tossed 10 times?

  *a)* ***How many different outcomes are possible***? $2^{10} = 1024$ outcomes.

  *b)* ***How many different outcomes have exactly 4 heads***? $C(10,4) = 210$ outcomes

  *c)* ***How many different outcomes at the most 2 heads?*** $C(10,0) + C(10,1) + C(10,2) = 56$

  *d)* ***How many different outcomes at least 3 heads?***

  All outcomes – (at most 2 H) = 1024 – 56 = 968

  *e)* ***What is the probability of obtaining exactly 4 heads?***

  $$P(4H) = \frac{C(10,4)}{2^{10}} = \frac{210}{1024} \approx .205$$

- ***Tosses 3 times, what is the conditional probability that the outcome is HHH given that at least 2H occurs?***

  $\Pr(\{HHH\} \mid \text{at least 2H}) = \frac{\#\{HHH\} \cap \{\text{least 2H}\}}{\#\{\text{least 2H}\}} = \frac{1}{4}$

  at least 2 H = {THH,HTH,HHT,HHH}

## *DICE*

**- Toss 2 dice** come up 7 or 11 (you win $7), for any (loose $2).

Determine the player's mathematical expectation?

36 possible pairs with 2 dice

8 possible for 7 or 11 Pr(7 or 11) = $\frac{8}{36}$ and Pr (not (7 or11)) = 1- $\frac{8}{36}$ = $\frac{28}{36}$

E = 7x $\frac{8}{36}$ + (-2) $\frac{28}{36}$ = 0

Events E= 1st die is a 3, F = 2nd die is 6. Pr(E) = ? Pr(F) = ? E and F are independent?

Pr(E) = Pr(F) = 1/6 , E $\cap$ F = {(3,6)} $\Rightarrow$ Pr(E|F) = 1/6 = Pr(E)

$\Rightarrow$ Pr(F|E) = 1/6 = Pr(F) $\therefore$ E and F are independent

What is the probability that the 2 dice show the same number?

E = {(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)} Pr(E) = $\frac{6}{36} = \frac{1}{6}$

What is the probability that the number add up to 8?

Pr(=8) = $\frac{5}{36}$ E ={(2,6),(6,2),(3,5),(5,3),(4,4)}

What is the probability that the sum is less than 5?

Pr(2) + Pr(3) + Pr(4) = $\frac{1}{36} + \frac{2}{36} + \frac{3}{36} = \frac{1}{6}$

**- Toss 1 die**

What is the probability that an odd number will appear?

E = {1,3,5} $\Rightarrow$ Pr(E) = 3/6 = ½

What is the probability that the result is odd and greater than 4?

F = {5,6} Pr(F) = $\frac{2}{6} = \frac{1}{3}$ $\Rightarrow$ Pr(E $\cup$ F) = Pr(E) + Pr(F) – Pr(E $\cap$ F) = $\frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3}$

# Practice sheet on Combinatorics

Made by Md.Ishrak Khan(T134)

**Type-1:**

(i) Using each digit only once in each number how many significant odd numbers containing five digits can be formed with the digits 6, 5, 2, 3, 0?

(ii) Using each digit only once in each number how many significant even number containing five digits can be formed with the digits 5, 3, 2, 6, 0?

**Type-2:**

(i) The digits from 0 to 9 is written in the telephone dial. All the telephone numbers in the town of Cox's Bazar contain 5 digits. How many connections of telephones can be made in that town?

(ii) 15 letters in each of 3 rings of a lock are printed. If the lock can be opened by only one permutation of three letters, then find the number of permutations by which the lock cannot be opened.

(iii) How many numbers of three digits can be formed with the digits 1, 2, 3, 4, 5 so that any digit may be repeated any number of times in each numbers? How many of these numbers have the same digit twice or thrice?

**Type-3:**

(i) In how many ways can the letters of 'Equation' be arranged by using all the letters at a time?

(ii) In how many ways can the letters of the word 'TRIANGLE' be arranged so that the vowels do not come together?

(iii) Find the number of arrangements of the letters of the word 'Mathematics'. In how many of these arrangements the vowels remain together?

(iv) In how many ways can the letters of the word 'MILLENNIUM' be arranged? How

many of them begin with 'M' and also end with 'M'?

(v) In how many ways can the letters of the word 'Permutation' be re-arranged so that the positions of vowels are not changed?

**Type-4:**

(i) How many different words taking letters at a time can be formed from the letters of the word Cambridge' so that each word contains all the vowels?

(ii) How many different words consisting of 3 consonants and 2 vowels can be formed from 12 different consonants and 5 different vowels?

(iii) How many different words by taking 1 vowel and 2 consonants from the letters of the word "Permutation" can be formed where the vowel always stays in the middle?

**Type-5:**

(i) How many teams of 11 players can be formed from 14 players of which 6 are experienced bowlers so that each team consists of at least 5 experienced bowers?

(ii) From two groups of players of which 6 are in one group and 8 are in the other group, a cricket team of 11 players is to be formed so that each team consists of at least 4 players from the group of 6. In how many ways can that team be formed?

(iii) From 6 students having Mathematics and 4 students having Physics, a committee of 6 is to be formed so that the majority of members of the committee is taken from the students having Mathematics. In how many ways can the committee be formed?

**Type-6:**

(i) An examinee is required to answer 6 questions out 12 questions. He will have to select exactly 4 questions out of 5 questions from the beginning. In how many ways can he select the questions?

(ii) Question paper on Mathematics has been divided into two groups, each group contains 5 questions. An examinee is required to answer 6 questions taking not more than 4 questions from any group. In how many ways can the examinee select

questions?

**Type-7:**

(i) In how many ways can the letters of the word "Degree" taking 4 letters at a time be selected?

(ii) In how many ways can selections be made by taking 4 letters at a time from the letters of the word "Thesis"?

(iii) How many words can be formed by taking 3 letters at a time from the letters of the word 'America'?

**Type-8:**

(i) How many triangles can be formed by joining the angular points of a polygon containing 17 sides?

(ii) How many triangles can be formed by joining the angular points of a polygon containing 12 sides. Find the number of diagonals of this polygon.

# Practice Sheet
## Combinatorics and Pigeonhole Principle

**Combinatorics:**

1. Find the number of different words that can be formed by rearranging the letters in the following words:
    a. MISSISSIPPI.
    b. NORMAL
    c. ABOMINABLE
    d. MONKEYDLUFFY
    e. FMALCHEMIST
2. Make all arrangement of letters of the word VINSMOKESANJI so that
    a) M is always next to O
    b) M and O are always together
3. Consider a license plate of 8 letters, where the first 4 letters include only alphabets and the last 4 letters include digits only. How many license plates can be made in this way if the alphabet comes from your name and digits come from your Student ID?
4. Find the total number of 3 digit numbers possible from the digits of your Student ID.
5. A box contains 12 black and 8 green marbles. How many ways can 3 black and 2 green marbles be chosen?
6. 8 students on a student council are assigned 8 seats around a U-shaped table.
    a. How many different ways can the students be assigned seats at the table?
    b. How many ways can a president and a vice-president be elected from the 8 students?
7. A Club consists of 20 members, of which 9 are male and 11 are female.Seven members will be selected to form an event-planning committee. How many committees of 4 females and 3 males can be formed?
8. In how many ways can a cricket eleven be chosen out of a batch of 15 players if-
    a)There is no restriction on the selection.
    b) One Particular Player is always chosen.
    c)Two Particular Players are never chosen.
9. How many ways are there to seat 10 people, consisting of 5 couples, in a row of seats (10 seats wide) if
    a. The seats are assigned at random
    b. All couples are to get adjacent seats

10. In how many ways can 5 boys and 4 girls be arranged on a bench if
    a. Boys and girls alternate?
    b. Anne and Jim wish to stay together?
11. In a playground, 3 sisters and 8 other girls are playing together. In a particular game, how many ways can all the girls be seated in a circular order so that the three sisters are not seated together?

**Pigeonhole principle:**

1) Show that in a group of 20 people and friendship is mutual, show that there exist two people who have the same number of friends?
2) The population of the US is 300 million. Every person has written somewhere between 0 and 10 million lines of code. What's the maximum number of people that we can say must have written the same number of lines of code?
3) Three people are running for student government. There are 202 people who vote. What is the minimum number of votes needed for someone to win the election?
4) Briar the cat likes to wear socks on all four of its feet. Briar's sock drawer is filled with yellow, white, and pink socks. Every morning Briar pulls socks out of the drawer one at a time until four matching socks are found. What is the largest number of socks Briar may pull from the drawer before finding a complete set?
5) Tammy notices that whenever she selects 7 whole numbers, there is always a triple a, b, c of numbers in her collection that all differ from each other by a multiple of 3. Tammy conjectures this will always be the case. Prove Tammy's conjecture
6) There are 38 different time periods during which classes at a university can be scheduled. If there are 677 different classes, what is the minimum number of different rooms that will be needed?

Binomial:

A polynomial equation with two terms usually joined by a plus or minus sign is called a binomial. Binomials are used in algebra. Polynomials with one term will be called a monomial and could look like 7x. A polynomial with two terms is called a binomial; it could look like 3x + 9. It is easy to remember binomials as **bi** means 2 and a binomial will have 2 terms.

A classic example is the following: 3x + 4 is a binomial and is also a polynomial, $2a(a+b)^2$ is also a binomial (a and b are the binomial factors).

The above are both binomials.

When multiplying binomials, you'll come across a term called the FOIL method which is often just the method used to multiply binomials.

For instance, to find the product of 2 binomials, you'll add the products of the **F**irst terms, the **O**uter terms, the **I**nner terms, and the **L**ast terms.

When you're asked to square a binomial, it simply means to multiply it by itself. The square of a binomial will be a trinomial. The product of two binomials will be a trinomial.

Example of Multiplying Binomials

(5 + 4x) x (3 + 2x)
(5 + 4x)(3 + 2x)
= (5)(3) + (5)(2x) + (4x)(3) + (4x)(2i)
= 15 + 10x + 12x + 8(x)2 = 15 + 22x + 8(-1)
= 15 + 22x - 8 = (15 - 8) + 22x = 7 + 22x

Once you begin taking algebra in school, you'll be doing a great many computations that require binomials and polynomials.

The formal expression of the Binomial Theorem is as follows:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

A few things should be noticed:

- The number of terms is one more than $n$ (the exponent).
- The power of $a$ starts with $n$ and decreases by 1 each term.
- The power of $b$ starts with 0 and increases by 1 each term.
- The sum of the exponents in each term adds up to $n$.
- The coefficients of the first and last terms are both 1 and they follow Pascal's triangle.

If the expansion is short, such as:

$$\begin{aligned}(x+2)^3 &= x^3 + 2x^2 2^1 + 2x^1 2^2 + 2^3 \\ &= x^3 + 4x^2 + 8x + 8\end{aligned}$$

Multinomial:

The multinomial theorem provides a formula for expanding an expression such as $(x_1 + x_2 + \cdots + x_k)^n$ for integer values of $n$. In particular, the expansion is given by

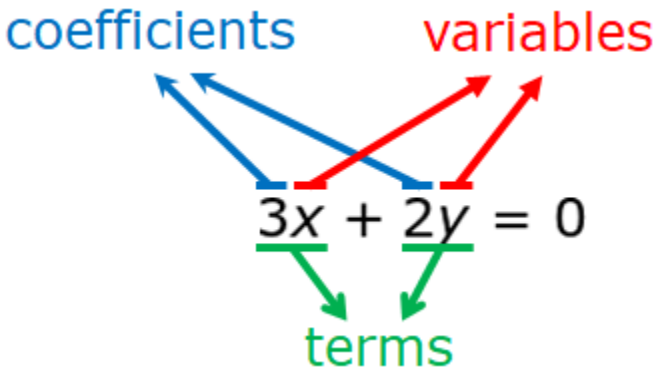
$$(x_1 + x_2 + \cdots + x_k)^n = \sum_{n_1,\, n_2, \ldots,\, n_k \geq 0} \frac{n!}{n_1! n_2! \cdots n_k!}\, x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k},$$

where $n_1 + n_2 + \cdots + n_k = n$ and $n!$ is the factorial notation for $1 \times 2 \times 3 \times \cdots \times n$.

For example, the expansion of $(x_1 + x_2 + x_3)^3$ is $x_1^3 + 3x_1^2x_2 + 3x_1^2x_3 + 3x_1x_2^2 + 3x_1x_3^2 + 6x_1x_2x_3 + x_2^3 + 3x_2^2x_3 + 3x_2x_3^2 + x_3^3$.

**Coefficients:**

In algebra, one of the first things you learn how to do is read algebraic equations. An algebraic equation consists of a number of terms added and/or subtracted together. Each of these terms has two parts to it: variables and coefficients.




***This diagram shows the difference between terms, coefficients, and variables in an equation.***

In statistics, we are introduced to a new type of coefficient known as the multinomial coefficient. In this lesson, you will learn what a multinomial coefficient is and how it is used in statistics.

**Multinomial Coefficient**

The multinomial coefficient gets its name from the multinomial series raised to the *n*th power, as seen below.

$$(x_1 + x_2 + \ldots + x_k)^n$$

In a series like this, the *x*'s represent terms, the *k* represents the number of elements in the series, and *n* is the positive integer power to which the series is raised. How a series like this is expanded is told to us by the **multinomial theorem**.

$$(x_1 + x_2 + \ldots + x_k)^n = \sum \binom{n}{n_1\ n_2\ \ldots\ n_k} x_1^{n_1} x_2^{n_2} \ldots x_k^{n_k}$$

In the multinomial theorem, the sum is taken over $n_1$, $n_2$, ... ,$n_k$ such that $n_1 + n_2 + \ldots + n_k = n$.

The multinomial theorem gives us a sum of multinomial coefficients multiplied by variables. In other words, it represents an expanded series where each term in it has its own associated multinomial coefficient. The **multinomial coefficient** itself is expressed in terms of factorials.

$$\binom{n}{n_1\ n_2\ \dots\ n_k} = \frac{n!}{n_1!\ n_2!\ \dots\ n_k!}$$

**Multinomial Coefficient Example 1**

To better understand how the multinomial coefficient works, let's work though an example together where we find the multinomial coefficient of one term of an expanded multinomial series.

Given the following multinomial series:

$$(a + 2b + 3c)^5$$

What is the coefficient of the $a^3c^2$ term in the expanded series?

In order to solve this problem we'll be working with the multinomial theorem.

$$(x_1 + x_2 + \dots + x_k)^n = \sum \binom{n}{n_1\ n_2\ \dots\ n_k} x_1^{n_1}\ x_2^{n_2}\ \dots x_k^{n_k}$$

Starting by comparing the series in this problem to the left side of the multinomial theorem equation, we can see that $x_1 = a$, $x_2 = 2b$, $x_3 = 3c$, and $n = 5$.

$$(a + 2b + 3c)^5 = \sum \binom{5}{n_1\ n_2\ n_3} a^{n_1} (2b)^{n_2} (3c)^{n_3}$$

Here, $k = 3$ since there are 3 elements ($a$, $b$, $c$) in the series. This is the formula for the full expansion of our series, but we're only looking for the coefficient of one term of the series. To do this, we're going to look at the right side of the equation without the sum, and identify $n_1$, $n_2$, and $n_3$.

These values of $n$ are given by the powers in our given term $a^3c^2$. The $a$ is raised to the third power, which gives us $n_1 = 3$, and $c$ is raised to the second power, giving us $n_3 = 2$.

We see $b$ is absent in this term. This is the equivalent of $b^x = 1$, which occurs when $x = 0$. Therefore, $n_2 = 0$.

With all the values of $n$ found, we now have everything we need to find the coefficient $a^3c^2$.

$$\binom{5}{3\ 0\ 2} a^2 (2b)^0 (3c)^2$$

First, we write the multinomial coefficient out as factorials.

$$\left(\frac{5!}{3!\ 0!\ 2!}\right) a^2 (2b)^0 (3c)^2$$

Then, all that's left to do is work out the basic arithmetic.

$$\left(\frac{120}{6 \times 1 \times 2}\right) a^3 2^0 b^0 3^2 c^2$$

$$\left(\frac{120}{12}\right) 9a^3 c^2$$

$$(10) 9a^3 c^2$$

$$90a^3 c^2$$

With this we've found that the coefficient of $a^3 c^2$ is 90.

# Probability

Random Experiment: An experiment is said to be a random experiment, if it's out-come can't be predicted with certainty.

Example: If a coin is tossed, we can't say whether head or tail will appear. So it is a random experiment.

Sample Space: The sent of all possible out-comes of an experiment is called the sample space. It is denoted by 'S' and its number of elements are n(s).

Example: In throwing a dice, the number that appears at top is any one of 1,2,3,4,5,6. So here:

S ={1,2,3,4,5,6} and n(s) = 6

Similarly in the case of a coin, S={Head,Tail} or {H,T} and n(s)=2.

The elements of the sample space are called sample point or event point.

Event: Every subset of a sample space is an event. It is denoted by 'E'.

Example: In throwing a dice S={1,2,3,4,5,6}, the appearance of an event number will be the event E={2,4,6}.

Clearly E is a sub set of S.

Simple event; An event, consisting of a single sample point is called a simple event.

Example: In throwing a dice, S={1,2,3,4,5,6}, so each of {1},{2},{3},{4},{5} and {6} are simple events.

Compound event: A subset of the sample space, which has more than one element is called a mixed event.

Example: In throwing a dice, the event of appearing of odd numbers is a compound event, because E={1,3,5} which has '3' elements.

Equally likely events: Events are said to be equally likely, if we have no reason to believe that one is more likely to occur than the other.

Example: When a dice is thrown, all the six faces {1,2,3,4,5,6} are equally likely to come up.

Exhaustive events: When every possible outcome of an experiment is considered.

Example: A dice is thrown, cases 1,2,3,4,5,6 form an exhaustive set of events.

*Classical definition of probability:*

If 'S' be the sample space, then the probability of occurrence of an event 'E' is defined as:

$$P(E) = n(E)/N(S) = \frac{\text{number of elements in 'E'}}{\text{number of elements in sample space 'S'}}$$

Example: Find the probability of getting a tail in tossing of a coin.

Solution:

Sample space S = {H,T} and n(s) = 2

Event 'E' = {T} and n(E) = 1

therefore P(E) = n(E)/n(S) = ½

Note: This definition is not true, if
(a) The events are not equally likely.
(b) The possible outcomes are infinite.

Sure event: Let 'S' be a sample space. If E is a subset of or equal to S then E is called a sure event.

Example: In a throw of a dice, S={1,2,3,4,5,6}

Let $E_1$=Event of getting a number less than '7'.

So '$E_1$' is a sure event.

So, we can say, in a sure event n(E) = n(S)

Mutually exclusive or disjoint event: If two or more events can't occur simultaneously, that is no two of them can occur together.

Example: When a coin is tossed, the event of occurrence of a head and the event of occurrence of a tail are mutually exclusive events.

Independent or mutually independent events: Two or more events are said to be independent if occurrence or non-occurrence of any of them does not affect the probability of occurrence or non-occurrence of the other event.

Example: When a coin is tossed twice, the event of occurrence of head in the first throw and the event of occurrence of head in the second throw are independent events.

Difference between mutually exclusive an mutually independent events: Mutually exclusiveness id used when the events are taken from the same experiment, where as independence is used when the events are taken from different experiments.

Complement of an event: Let 'S' be the sample for random experiment, and 'E' be an event, then complement of 'E' is denoted by 'E' is denoted by E'. Here E' occurs, if and only if E' doesn't occur.

**Rudimentary level math example:**

Random Experiment: An experiment is said to be a random experiment, if its out-come can't be predicted with certainly.

Ex.: If a coin is tossed, we can't say, whether head or rail will appear. So it is a random experiment.

Sample Space : The set of all possible out-comes of an experiment is called the sample – space.

I a dice is thrown, the number, that appears at top is any one of 1, 2, 3, 4, 5, 6,

So here :

S = { 1, 2, 3, 4, 5, 6, } and n(s) = 6

Similarly in the case of a coin, s = {H, T} and n (s) = 2.

The elements of the sample of the sample-space are called sample points or event points.

Ex.: if S = {H, T}, than 'H' and 'T' are sample points.

Event: Every subset of a sample space is an event. It is denoted by 'E'.

Ex.: In throwing a dice S = {1, 2, 3, 4, 5, 6,}, the appearance of an even number will be the event E = {2, 4, 6}.

Clearly E $\subset$ S.

Important types of Events:

Simple or elementary event: An event, consisting of a single point is called a simple event.

Ex.: In throwing a dice s = {1, 2, 3, 4, 5, 6} so each of {1}, {2}, {3}, {4}, {5} and {6} is a simple event.

Compound or mixed event: A subset of the sample space which has more than one element is called a mixed event.

Ex.: In throwing a dice, the event of odd numbers appearing is a mixed event, because E = {1, 3, 5}, which has '3' elements.

Equally likely events: Events are said to be equally likely, if we have no reason to believe that one is more likely to occur than the other.

Ex.: When a dice is thrown, all the six-faces {1, 2, 3, 4, 5, 6,} are equally likely to come-up.

Exhaustive events: When every possible outcome of an experiment is considered, the observation is called exhaustive events.

Ex.: When a dice is thrown, cases 1, 2, 3, 4, 5, 5 form an exhaustive set of events.

Mathematical or Classical definition of Probability:

It' 'S' be the sample-space, then the probability of occurrence of an event 'E' is defined as:

P(E) = n (E) / n(S)

= number of elements in 'E' / number of elements in sample space

Ex: Find the probability of getting tails in tossing of a coin.

Sol.: Sample-space S = { H, T} ⇨ n(S) = 2

Event 'E' = {T} ⇨ n (E) = 1

P(E) = n (E) / n (S) = 1 / 2

Note: This definition is not true, if

(a) The events are not equally likely.

(b) The possible out comes are infinite.

Sure-event: Let "S" be a sample – space.

If S ⊆ S, 'S' is an event, called sure event.

Ex.: In a throw of dice, S = {1, 2, 3, 4, 5, 6}

Let $E_1$ = Event of getting a number less than '7'

Clearly each outcome is a number less than '7'.

So '$E_1$' is a sure – event.

We can say, in a sure event n(E) = n(S).

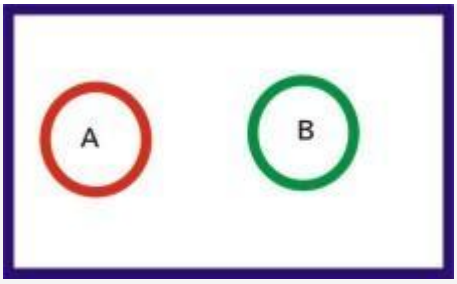
Mutually Exclusive or Disjoint Events: If two or more events can't occur simultaneously, i.e. no two of them can occur together.

So the event 'A' and 'B' are mutually exclusive if

$A \cap B = \phi$, so $P(A \cap B) = 0$

Ex.: When a coin is tossed, the event of occurrence of a head and the event of occurrence of a tail are mutually exclusive events.

Pictorial Representation:




$A \cap B = \phi$

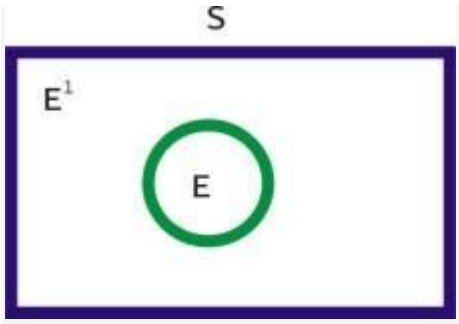
Independent or Mutually independent Events: Two or more events are said to be independent, if occurrence of non-occurrence of any of them does not affect the probability of occurrence or non-occurrence of other event.

Ex.: When a coin is tossed twice, the event of occurrence of head in the first throw and the event of occurrence of head in the second throw and independent events.

Difference between mutually exclusive and mutually Independent events: Mutually exclusiveness is used, when the events are taken from the same experiment, whereas the independence is used, when the events are taken from different experiments.

Complement of an event : let 'S' be the sample space for a random experiment, and 'E' be an event, then complement of 'E' is denoted by 'E'' or E or $E^c$.

Here E' occurs, if and only if 'E' doesn't occur.




Clearly n(E) + n(E') = n(S)

**Theorems related t Probability**

Theorom1 : The probability of an event lies between 'O' and '1'.

i.e. O<= P(E) <= 1.

Proof: Let 'S' be the sample space and 'E' be the event.

Then

O <= n(E) <= n (S)

O / n(S) <= n(E)/ n(S) < = n(S) / n(S)

or O < =P(E) <= 1

The number of elements in 'E' can't be less than 'O' i.e. negative and greater than the number of elements in S.

Theorem 2 : The probability of an impossible event is 'O' i.e. P () = O

Proof: Since has no element, ⇨ n() = O

From definition of Probability:

P() = n () / n(S) = O / n(S)

⇨ P() = O

Theorem 3 : The probability of a sure event is 1. i.e. P(S) = 1. where 'S' is the sure event.

Proof : In sure event n(E) = n(S)

[ Since Number of elements in Event 'E' will be equal to the number of element in sample-space.]

By definition of Probability :

$P(S) = n(S)/n(S) = 1$

$\Rightarrow P(S) = 1$

Theorem 4: If two events 'A' and 'B' are such that $A \subseteq B$, then $P(A) <= P(B)$.

Proof: $A \subseteq B$

$\Rightarrow n(A) <= n(B)$

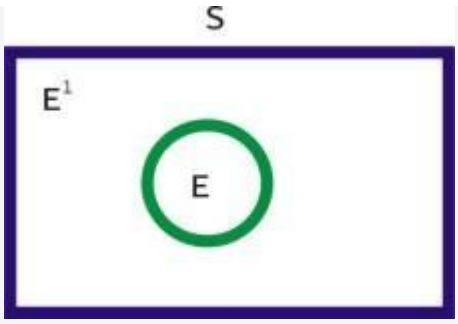
or $n(A)/N(S) <= n(B)/n(S)$

$\Rightarrow P(A) <= P(B)$

Since 'A' is the sub-set of 'B", so from set theory number of elements in 'A' can't be more than number of element in 'B'.

Theorem 5 : If 'E' is any event and $E^1$ be the complement of event 'E', then $P(E) + P(E^1) = 1$.

Proof:




Let 'S' be the sample – space, then

$n(E) + n(E^1) = n(S)$

or $n(E)/n(S) + n(E^1)/n(S) = 1$

or $P(E) + P(E^1) = 1$

Algebra of Events: In a random experiment, let 'S' be the sample – space. Let $A \subseteq S$ and $B \subseteq S$, where 'A' and 'B' are events.

Thus we say that :

(i) $(A \cup B)$, is an event occurs only when at least of 'A' and 'B' occurs.

$\Rightarrow$ $(A \cup B)$ means (A or B).

Ex.: if A = { 2,4,6,} and B = {1, 6}, than the event ‘A’ or ‘B’ occurs, if ‘A’ or ‘B’ or both occur i.e. at least one of ‘A’ and ‘B’ occurs. Clearly ‘A’ or ‘B’ occur, if the outcome is any one of the outcomes 1, 2, 4, 6.

That is A ∪ B. ( From set – theory ).

(ii) (A ∪ B) is an event, that occurs only when each one of ‘A’ and ‘B’ occur ⇨ (A ∩B) means ( A and B).

Ex.: In the above example, if the outcome of an experiment is ‘6’, then events ‘A’ and ‘B’ both occur, because ‘6’ is in both sets. That is A ∩ B.

(iii) A is an event, that occurs only when ‘A’ doesn’t occur – category of problems related to probability :

(1) Category A – When n(E) and n(S) are determined by writing down the elements of ‘E’ and ‘S’.

(2) Category B – When n(E) and n(S) are calculated by the use of concept of permutation and combination.

(3) Category C – Problems based on $P(E) + P(E^1) = 1$

Q1: A coin is tossed successively three times. Find the probability of getting exactly one head or two heads.

Sol.: Let ‘S’ be the sample – space. Then,

S = {HHH, HHT, HTH, THH, TTH, THT, HTT, TTT}

⇨ n (S) = 8

Let ‘E’ be the event of getting exactly one head or two heads.

Then:

E = { HHT, HTH, THH, TTH, THT, HTT }

⇨ n (E) = 6

Therefore:

P(E) = n (E)/ n (S) = 6 / 8 = 3 / 4

Q2: Three coins are tossed. What is the probability of getting (i) all heads, (ii) two heads, (iii) at least one head, (iv) at least two heads?

Sol.: Let ‘S’ be the sample – space. Then

S = { HHH, HHT, HTH, THH, HTT, THT, TTH, TTT }

(i) Let '$E_1$' = Event of getting all heads.

Then $E_1$ = { HHH }

$n(E_1) = 1$

⇨ $P(E_1) = n(E_1) / n(S) = 1 / 8$

(ii) Let $E_2$ = Event of getting '2' heads.

Then:

$E_2$ = { HHT, HTH, THH }

$n(E_2) = 3$

⇨ $P(E_2) = 3 / 8$

(iii) Let $E_3$ = Event of getting at least one head.

Then:

$E_3$ = { HHH, HHT, HTH, THH, HTT, THT, TTH }

$n(E_3) = 7$

⇨ $P(E_3) = 7 / 8$

(iv) Let $E_4$ = Event of getting at least one head.

Then:

$E_4$ = { HHH, HHT, HTH, THH, }

$n(E_4) = 4$

⇨ $P(E_4) = 4/8 = 1/2$

Q3: What is the probability that a number selected from 1, 2, 3, --- 25, is a prime number, when each of the numbers is equally likely to be selected.

Sol.: S = { 1, 2, 3, ---- , 25} ⇨ n(S) = 25

And E = { 2, 3, 5, 7, 11, 13, 17, 19, 23 } ⇨ n(E) = 9

Hence $P(E) = n(E) / n(S) = 9 / 25$

Q4: Two dice are thrown simultaneously. Find the probability of getting :

(i) The same number on both dice,

(ii) An even number as the sum,

(iii) A prime number as the sum,

(iv) A multiple of '3' as the sum,

(v) A total of at least 0,

(vi) A doublet of even numbers,

(vii) A multiple of '2' on one dice and a multiple of '3' on the other dice.

Sol.: Here:

S = { (1,1), (1,2) ------, (1,6), (2,1), (2,2), ---- (2,6), (3,1), (3,2), -----, (3,6), (4,1), (4,2), -------- (4,6), (5,1), (5,2), ----- (5,6), (6,1,), (6,2), ---------- (6,6) }

n(S) = 6 x 6 = 36

(i) Let $E_1$ = Event of getting same number on both side:

⇨ $E_1$ = { (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) }

$n(E_1)$ = 6

$P(E_1) = n(E_1)/n(S) = 6/36 = 1/6$

(ii) Let $E_2$ = Event of getting an even number as the sum.

$E_2$ = { (1,1), (1,3), (1,5), (2,2), (2,4), (2,6), (3,1), (3,3), (3,5), (4,2), (4,4), (4,6), (5,1), (5,5), (6,2), (6,4), (6,6) }

$n(E_2)$ = 18 hence $P(E_2) = n(E_2)/n(S) = 18/36 = 1/2$

(iii) Let $E_3$ = Event of getting a prime number as the sum..

$E_3$ = { (1,1), (1,2), (1,4), (1,6), (2,1), (2,3), (2,5), (3,2), (3,4), (4,1), (4,3), (5,2), (5,6), (6,1), (6,5),}

$n(E_3)$ = 15

$P(E_2) = n(E_3) / n(S) = 15/36 = 5/12$

(iv) Let $E_4$ = Event of getting a multiple of '3' as the sum.

$E_4$ = { (1,2), (1,5), (2,1), (2,4), (3,3), (3,6), (4,2), (4,5), (5,1), (5,4), (6,3), (6,6)}

$n(E_4) = 12$

$P(E_4) = n(E_4)/n(S) = 12/36 = 1/3$

(v) Let $E_5$ = Event of getting a total of at least 10.

$E_5 = \{ (4,6), (5,5), (5,6), (6,4), (6,5), (6,6), \}$

$n(E_5) = 6$

$P(E_5) = n(E_5)/n(S) = 6/36 = 1/6$

(vi) Let $E_6$ = Event of getting a doublet of even numbers.

$E_6 = \{ (2,2), (4,4), (6,6), \}$

$n(E_6) = 3$

$P(E_6) = n(E_6)/ n(S) = 3/36 = 1/12$

(vii) Let $E_7$ = Even of getting a multiple of '2" on one dice and a multiple of '3' on the other dice.

$E_7 = \{ (2,3), (2,6), (4,3), (4,6), (6,3), (3,2), (3,4), (3,6), (6,2), (6,4) \}$

$n(E_7) = 11$

$P(E_7) = n(E_7) / n(S) = 11/36$

Q5.: What is the probability, that a leap year selected at random will contain 53 Sundays?

Sol.: A leap year has 366 days, therefore 52 weeks i.e. 52 Sunday and 2 days.

The remaining 2 days may be any of the following :

(i) Sunday and Monday

(ii) Monday and Tuesday

(iii) Tuesday and Wednesday

(iv) Wednesday and Thursday

(v) Thursday and Friday

(vi) Friday and Saturday

(vii) Saturday and Sunday

For having 53 Sundays in a year, one of the remaining 2 days must be a Sunday.

n(S) = 7

n(E) = 2

P(E) = n(E) / n(S) = 2 / 7

## Intermediary level math example

Problems based on fundamental principal of counting and permutations and combinations :

Q1. A bag contains '6' red, 4 white and 8 blue balls. If three balls are drawn at random, find the probability, that

(i) '1' is red and '2' are white, (ii) '2' are blue and 1 is red, (iii) none is red.

Sol.: We have to select '3' balls, from 18 balls (6+4+8)

⇨ $n(S) = {}^{18}C_3 = 18! / (3! \times 15!) = (18x17x16) / (3x2x1) = 816$

(i) Let $E_1$ = Event of getting '1' ball is red and '2' are white

Total number of ways = $n(E_1) = {}^{6}C_1 \times {}^{4}C_2$

$= 6! / (1! \times 5!) \times 4! / (2! \times 2!)$

$= 6 \times 4 / 2$

$= 36$

$P(E_1) = n(E_1) / n(S) = 36/816 = 3/68$

(ii) Let $E_2$ = Event of getting '2' balls are blue and '1' is red.

= Total no. of ways ⇨ $n(E_2) = {}^{8}C_2 \times {}^{6}C_1$

$= 8! / (2! \times 6!) \times 6! / (1! \times 5!)$

$= (8 \times 7) / 2 \times 6 / 1 = 168$

$P(E_2) = 168 / 816 = 7/34$

(iii) Let $E_3$ = Event of getting '3' non – red balls. So now we have to choose all the three balls from 4 white and 8 blue balls.

Total number of ways :

$n(E_3) = {}^{12}C_3 = 12! / (3! \times 9!) = (12x11x10) / (3x2x1) = 220$

$P(E_3) = n(E_3) / n(S) = 220 / 816 = 55/204$

Q: A box contains 12 bulbs of which '4' are defective. All bulbs took alike. Three bulbs are drawn randomly.

What is the probability that :

(i) all the '3' bulbs are defective?

(ii) At least '2' of the bulbs chosen are defective?

(iii) At most '2' of the bulbs chosen are defective?

Sol.: We have to select '3' bulbs from 12 bulbs.

⇨ $n(S) = {}^{12}C_3 = 12! / (3! \times 9!) = (12x11x10) / (3x2x1) = 220$

(i) Let $E_1$ = All the '3' bulbs are defective.

⇨ All bulbs have been chosen, from '4' defective bulbs.

⇨ $n(E_1) = {}^4C_3 = 4! / (3! \times 1!) = 4$

⇨ $P(E_1) = n(E_1) / n(S) = 4 /220 = 1/55$

(ii) Let $E_2$ = Event drawing at least 2 defective bulbs. So here, we can get '2' defective and 1 non-defective bulbs or 3 defective bulbs.

$n(E_2) = {}^4C_2 \times {}^8C_1 + {}^4C_3$ [Non-defective bulbs = 8]

$= 4! / (2! \times 2!) \times 8! / (1! \times 7!) + 4! / (3! \times 1!)$

$= 4x3 / 2 \times 8/1 + 4/1 = 48+4$

$n(E_2) = 52$

⇨ $P(E_2) = n(E_2) / n(S) = 52/220 = 13/55$

(iii) Let $E_3$ = Event of drawing at most '2' defective bulbs. So here, we can get no defective bulbs or 1 is defective and '2' is non-defective or '2' defective bulbs.

$n(E_3) = {}^8C_3 + {}^4C_1 \times {}^8C_2 + {}^4C_2 \times {}^8C_1$

$= 8? / (3? \times 5?) + 4? / (1? \times 3?) \times 8? / (2? \times 6?) + 4? / (2? \times 2?) \times 8? / (1? \times 7?)$

$= (8x7x6) / (3x2x1) + 4 \times (8x7)/2 + (4 \times 3) / 2 + 8/1$

$= 216$

$P(E_3) = n(E_3) / n(S) = 216 / 220 = 54 / 55$

Q: In a lottery of 50 tickets numbered from ‘1’ to ‘50’ two tickets are drawn simultaneously. Find the probability that:

(i) Both the tickets drawn have prime number on them,

(ii) None of the tickets drawn have a prime number on it.

Sol.: We want to select ‘2’ tickets from 50 tickets.

⇨ $n(S) = {}^{50}C_2 = 50? / (2? \times 48?) = (50x49) / 2 = 1225$

(i) Let $E_1$ = Event that both the tickets have prime numbers Prime numbers between ‘1’ to ‘50’ are :

2,3,5,7,11,13,17,19,23,29,31,37,41,43,47.

Total Numbers = 15.

We have to select ‘2’ numbers from these 15 numbers.

⇨ $n(E_1) = {}^{15}C_2 = 15? / (2? \times 13?) = (15x47) / 2 = 105$

⇨ $P(E_1) = n(E_1) / n(S) = 105/1225 = 21/245$

(ii) Non prime numbers between ‘1’ to ‘50’ = 50-15 = 35

Let $E_2$ = Event that both the tickets have non-prime numbers.

Now we have to select ‘2’ numbers, from ‘35’ numbers.

⇨ $n(E_2) = {}^{35}C_2 = 35? / (2? \times 33?) = (35x34) / 2 = 595$

⇨ $P(E_2) = n(E_2) / n(S) = 595 / 1225 = 17/35$

Q.: A bag contains 30 tickets, numbered from ‘1’ to ‘30’. Five tickets are drawn at random and arranged in ascending order. Find the probability that the third number is 20.

Sol.: Total number of ways of selecting ‘5’ tickets from 30 tickets = ${}^{30}C_5$

⇨ $n(S) = {}^{30}C_5 = 30? / (5? \times 25?) = (30 \times 29 \times 28 \times 27 \times 26) / (5 \times 4 \times 3 \times 2 \times 1)$

$n(S) = 29 \times 27 \times 26 \times 7$

Suppose the ‘5’ tickets are a1, a2,20, a4, a5

They are arranged in ascending order.

⇨ a1, a2 ⊆ {1, 2, 3, ------- , 19} and a4, a5 ⊆ { 21, 22, 23, -----, 30}

We have to select ‘2’ tickets from first ‘19’ tickets and ‘2’ tickets from last 10 tickets.

⇨ $n(E) = {}^{19}C_2 \times {}^{10}C_2$

= 19? / (2? x 17?) = 10? / (2? x 8?) = (19 x 18) / 2 = (10 x 9) / 2

= 19 x 9 x 5 x 9

⇨ P(E) = n(E) / n(S) = (19x9x5x9) / (29x27x26x7) = 285 / 5278

Odds is Favour and Odds against an Event:

Let ‘S” be the sample space and ‘E’ be an event. Let ‘E’ devotes the complement of event ‘E’, then.

(i) Odds in favour of event ‘E’ = $n(E) / n(E^1)$

(ii) Odds in against of an event ‘E’ = $n(E^1) / n(E)$

Note : Odds in favour of ‘E’ = $n(E) / n(E^1)$

= $[n(E) / n(S)] / [n(E^1) / n(S)] = P(E) / P(E^1)$

Similarly odds in against of ‘E’ = $P(E^1) / P(E)$

Ex.: The odds in favours of an event are 3:5 find the probability of the occurrence of this event.

Sol.: Let ‘E’ be an event.

Then odds in favour of E = $n(E) / n(E^1) = 3 / 5$

⇨ $n(E) = 3$ and $n(E^1) = 5$

Total number of out-comes $n(S) = n(E) + n(E^1) = 3+5 = 8$

$P(E) = n(E) / n(S) = 3 / 8$

Q.: If ‘12’ persons are seated at a round table, what is the probability that two particulars persons sit together?

Sol.: We have to arrange 12 persons along a round table.

So if ‘S” be the sample – space, then n(S) = (12-1)? = 11?

n(S) = 11?

Now we have to arrange the persons in away, such that ‘2’ particulars person sit together.

Regarding that 2 persons as one person, we have to arrange 11 persons.

Total no. of ways = (11-1)? = 10? ways.

That ‘2’ persons can be arranged among themselves in 2? ways.

So, total no. of ways, of arranging 12 persons, along a round table, so that two particular person sit together : = 10? x 2?

⇨n(E) = 10? x 2?

⇨ P(E) = n(E) / n(S) = (10? x 2?) / 11? = 2 / 11

Q.: 6 boys and 6 girls sit in a row randomly, find the probability that all the '6' girls sit together.

Sol.: We have to arrange '6' boys and '6' girls in a row.

⇨ n(S) = 12?

Now, we have to arrange '6' girls in a way, such that all of them should sit together.

Regarding all the 6 girls as one person, we have to arrange 7 person in a row.

⇨ Total no. of ways = 7?

But 6 girls can be arranged among themselves in 6? ways.

⇨ n(E) = 7? x 6?

⇨ P(E) = n(E) / n(S) = (7? x 6?) / 12? = (6x5x4x3x2x1) / (12x11x10x9x8)

P(E) = 1 / 132

Q: If from a pack of '52' playing cards one card is drawn at random, what is the probability that it is either a kind or a queen?

Sol.: n(S) = Total number of ways of selecting 1 card out of 52 cards.

= $^{52}C_1$ = 52

n(E) = Total number of selections of a card, which is either a kind or a queen.

= $^{4}C_1$ + $^{4}C_1$ = 4 + 4 = 8

P(E) = n(E) / n(S) = 8 / 52 = 2 / 3

Q.: From a pack of 52 playing cards, three cards are drawn at random. Find the probability of drawing a king, a queen and a jack.

Sol.: Here n(S) = $^{52}C_3$ = 52? / (3? x 49?) = (52x51x50) / (3x2x1)

= 52x17x25

n(E) = $^{4}C_1$. $^{4}C_1$. $^{4}C_1$

= 4? / (1? x 3?) x 4? / (1? x 3?) = 4? / (1? x 3?)

n(E) = 4 x 4 x 4

⇨ P(E) = n(E) / n(S) = (4x4x4) / (52x17x25) = 16 / 5525

## High level Math example

Problems based on finding $P(E^1)$, by the use of $P(E^1) – 1 – P(E)$ :

Note : When an event has a lot of out comes, then we use this concept.

Ex.: What is the probability of getting a total of less than '12' in the throw of two dice?

Sol.: Here n(S) = 6x6x = 36

It is very difficult to find out all the cares, in which we can find the total less then '12'.

So let E = The event, that the sum of numbers is '12'.

Then E = { 6, 6}

n(E) = 1

⇨ P(E) = n(E) / n(S) = 1/36

Required probability, $P(E^1)$ = 1-P(E)

= 1 – 1/36

P(E1) = 35 /36

Ex.: There are '4' envelopes corresponding to '4' letters. If the letters are placed in the envelopes at random, what is the probability that all the letters are not placed in the right envelopes?

Sol.: We have to place '4' letters in 4 envelopes.

⇨ n(S) = 4!

Now:

Let E = The event, that all the 4 letters are placed in the corresponding envelopes.

So $E^1$ = The event that all the '4' letters are not placed in the right envelope.

Here n(E) = 1

P(E) = n(E) / n(S) = 1 / 4! = 1 / 24

Required probability, $P(E^1)$ = 1- P(E)

= 1 – (1/24)

$P(E^1) = 23 / 24$

Some information's about playing cards:

(1) A pack of 52 playing cards has 4 suits :

(a) Spades, (b) Hearts, (c) Diamonds, (d) Clubs.

(2) Spades and clubs are black and Hearts and Diamonds are red faced cards.

(3) The aces, kings, queens, and jacks are called face cards or honours – cards.

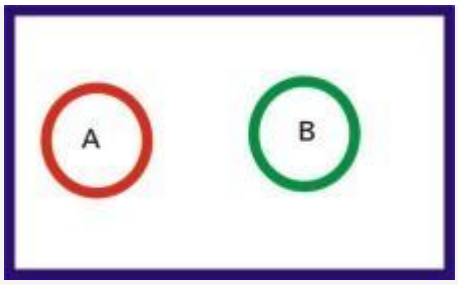
## Total Probability

Theorem – 1 : If 'A' and 'B' are mutually exclusive events then $P(A \cap B) = 0$ or P ( A and B) = 0

Proof : If 'A' and 'B' are mutually exclusive events then $A \cap B =$

⇨ $P(A \cap B) = P(\phi)$

$= n(\phi) / n(S)$ [ By definition of probability]



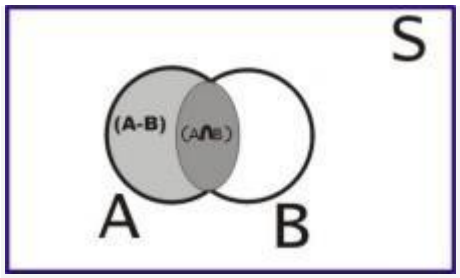

$= o / n(S)$ [Since the number of elements in a null – set is '0']

$P(A \cap B) = 0$

(2) Addition Theorem of Probability : If 'A' and 'B' by any two events, then the probability of occurrence of at least one of the events 'A' and 'B' is given by:

P(A or B) = P(A) + P(B) – P (A and B)

$P(A \cup B) = P(A) + P(B) – P (A \cap B)$




From set theory, we have :

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Dividing both sides by n(S) :

$n(A \cup B) / n(S) = n(A) / n(S) + n(B) / n(S) - n(A \cap B) / n(S)$

or $P(A \cup B) = p(A) + P(B) - P(A \cap B)$

Corollary : If 'A' and 'B' are mutually exclusive events,

Then $P(A \cap B) = 0$. [ As we have proved]

In this case :

⇨ $P(A \cup B) = p(A) + P(B)$

Addition theorem for '3' events 'A', 'B' and 'C' :

$P(A \cup B \cup C) = P(A) + P(B) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

Proof : $P(A \cup B \cup C) = P[(A \cup B) \cup C]$

$= P(A \cup b) + P(C) - P[(A \cup B) \cap C]$ [ By addition theorem for two events]

$= P(A \cup B) - P(C) - [P(A \cap C) + P(B \cap C) - P(A \cap C \cap B \cap C)]$

$= P(A) + P(B) - P(A \cap B) + P(C) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

Corollary : If 'A', 'B' and 'C' are mutually exclusive events, then $P(A \cap B) = 0$, $P(B \cap C) = 0$, $P(A \cap C) = 0$ and $P(A \cap B \cap C) = 0$.

In this case :

⇨ $P(A \cup B \cup C) = P(A) + P(B) + P(C)$

General Form of Addition Theorem of Probability:

$$P(A_1 \cup A_2 \cup \text{-----} \cup A_n) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j)$$

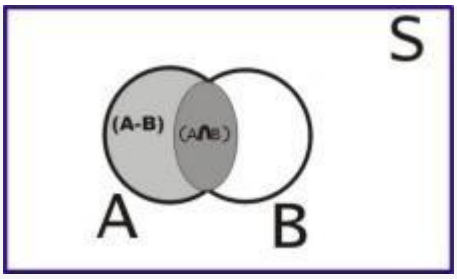
$$+ \sum_{i<j<k} P(A_i \cap A_j \cap A_k) \text{ ----- } (-1)^{n-1} P(A_1 \cap A_2 \cap \text{---- } \cap A_n)$$

Corollary : For any number of mutually exclusive events, $A_1, A_2, \text{------}, A_n$ :

$P(A_1 \cup A_2 \cup \text{---- } \cup A_n) = P(A_1) + P(A_2) + \text{------} + P(A_n)$

Theorem – 3 : For any two events ‘A’ and ‘B’

$P(A-B) = P(A) – P(A \cap B) = P(A \cap B^1)$




From the figure:

$(A-B) \cap (A \cap B) =$ ----------> (i)

and

$(A-B) \cup (A \cap B) = A$

$P[(A-B) \cup (A \cap B)] = P(A)$

or $P(A-B) + P(A \cap B) = P(A)$

[From (i) $(A-B) \cap (A \cap B) =$ i.e. These events are mutually exclusive]

⇨ $P(A-B) = P(A) – P(A \cap B)$

or $P(A \cap B) = P(A) – P(A \cap B)$

Similarly $P(A \cap B) = P(B) – P(A \cap B)$

Proof of $P(E) + P(E^1) = 1$, by the addition theorem of probability:

We know that :

$P(A \cup B) = P(A) + P(B) – P(A \cap B)$

Putting A = E and $B = E^1$

$P(E \cup E^1) = P(E) + P(E^1) – P(E \cap E^1)$ ----------> (1)

From set theory : $E \cup E^1 = S$

And $E \cap E^1 =$

From:

$P(S) = P(E) + P(E^1) – P()$

⇨ $1 = P(E) + P(E^1) – 0$

or $P(E) + P(E^1) = 1$

[P(S) = 1, P() = 0]

## EXAMPLES

Problems based on addition theorem of probability:

Working rule :

(i) A ∪ B denotes the event of occurrence of at least one of the event 'A' or 'B'

(ii) A ∩ B denotes the event of occurrence of both the events 'A' and 'B'.

(iii) P(A∪B) or P(A+B) denotes the probability of occurrence of at least one of the event 'A' or 'B'.

(iv) P(∩B) or P(AB) denotes the probability of occurrence of both the event 'A' and 'B'.

----------x----------x----------x----------x----------x-----

Ex.: The probability that a contractor will get a contract is '2/3' and the probability that he will get on other contract is 5/9 . If the probability of getting at least one contract is 4/5, what is the probability that he will get both the contracts ?

Sol.: Here P(A) = 2/3, P(B) = 5/9

P(A∪b) = 4/5, (P(A∩B) = ?

By addition theorem of Probability:

P(A∪B) = P(A) + P(B) - P(A∩B)

= 4/5 = 2/3 + 5/9 - P(A∩B)

or 4/5 = 11/9 – P(A∩B)

or P(A∩B) = 11/9 – 4/5 = (55-36) / 45

P(A∩B) = 19/45

Ex2.: Two cards are drawn at random. Find the probability that both the cards are of red colour or they are queen.

Sol.: Let S = Sample – space.

A = The event that the two cards drawn are red.

B = The event that the two cards drawn are queen.

⇨ A∩B = The event that the two cards drawn are queen of red colour.

⇨ $n(S) = {}^{52}C_2$, $n(A) = {}^{26}C_2$, $n(B) = {}^{4}C_2$

$n(A \cap B) = {}^{2}C_2$

⇨ $P(A) = n(A) / n(S) = {}^{26}C_2 / {}^{52}C_2$ , $P(B) = n(B) / n(S) = {}^{4}C_2 / {}^{52}C_2$

$P(A \cap B) = n(A \cap B) / n(S) = {}^{2}C_2 / {}^{52}C_2$

P(A∪B) = ?

We have $P(A \cup B) = P(A) + P(B) – P(A \cap B)$

$= {}^{26}C_2 / {}^{52}C_2 + {}^{4}C_2 / {}^{52}C_2 – {}^{2}C_2 / {}^{52}C_2$

$= ({}^{26}C_2 + {}^{4}C_2 – {}^{2}C_2) / {}^{52}C_2$

= (13X25+2X3-1) / (26X51)

P(A∪B) = 55/221

Ex.3: A bag contains ‘6’ white and ‘4’ red balls. Two balls are drawn at random. What is the chance, they will be of the same colour?

So.: Let S = Sample space

A = the event of drawing ‘2’ white balls.

B = the event of drawing ‘2’ red balls.

A∪B = The event of drawing 2 white balls or 2 red balls.

i.e. the event of drawing ‘2’ balls of same colour.

⇨ $n(S) = {}^{10}C_2 = 10! / (2! \times 8!) = 45$

$n(A) = {}^{6}C_2 = 6! / ((2! \times 4!) = (6 \times 5 ) / 2 = 15$

$n(B) = {}^{4}C_2 = 4! / (2! \times 2!) = (4x3) / 2 = 6$

P(A) = n(A) / n(S) = 15/45 = 1/3

P(B) = n(B) / n(S) = 6/45 = 2/15

⇨ P(A∪B) = P(A) + P(B)

= 1/3 + 2/15 = (5+2) / 15

P(A∪B) = 7/15

Ex.: For a post three persons ‘A’, ‘B’ and ‘C’ appear in the interview. The probability of ‘A’ being selected is twice that of ‘B’ and the probability of ‘B’ being selected is thrice that of ‘C’, what are the individual probability of A, B, C being selected?

Sol.: Let ‘$E_1$’, ‘$E_2$’, ‘$E_3$’ be the events of selections of A, B, and C respectively.

Let $P(E_3) = x$

⇨ $P(E_2) = 3.\ P(E_3) = 3x$

and $P(E_1) = 2P(E_2) = 2 \times 3x = 6x$

As there are only ‘3’ candidates ‘A’, ‘B’ and ‘C’ we have to select at least one of the candidates A or B or C, surely.

⇨ $P(E_1 \cup E_2 \cup E_3) = 1$

and $E_1$, $E_2$, $E_3$ are mutually exclusive.

⇨$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)$

$1 = 6x + 3x + x$

⇨ $10x - 1$ or $x = 1/10$

⇨ $P(E_3) = 1/10$, $P(E_2) = 3/10$ and $P(E_1) = 6/10 = 3/5$

**1. Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?**
Ans:
P(3 or 5) = P(3) + P(5) - P(3 and 5)

multiples of 3 = {3,6,9,12,15,18}
Therefore, P(3) = 6/20

multiples of 5 = {5,10,15,20}
Therefore, P(5) = 4/20

multiples of both 3 and 5 = {15}
Therefore, P(3 and 5) = 1/20

therefore, P(3 OR 5) = 6/20 + 4/20 - 1/20 = 9/20

**2. In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize?**
Ans:
10 prizes and 25 blanks
that means total = 35
favorable outcome/total = 10/35 =>
2/7

**3. One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card?**
Ans:
there are 12 face cards, 52 total cards
12/52 ===>
3/13

**4. A man and his wife appear in an interview for 2 vacancies in the same post. The probability of husband's selection is 1/7 and the probability of wife's selection is 1/5. What is the probability that only one of them selected?**
Ans:
This one was just a little bit tricky
probability that man will get the job and wife will also = 1/7 * 1/5 = 1/35
now probability that man will not get the job is 1-1/7 = 6/7
and probability that woman will not get the job is 1-1/5 = 4/5
so probability that they both will not get the job is 6/7 * 4/5 ==== > 24/35
now total probability is 1
so =====> 1- both will get job - no one will get job = only one got job
which gives
2/7

Alternatively,
probablility that man will not get the job is 1-1/7 = 6/7
and prob that woman will not get the job is 1-1/5 = 4/5
So probability = man gets and woman does not or woman get and man does not
= 1/7 * 4/5 + 1/5 *6/7

= 10/35 = 2/7

**7. A speaks truth in 75% cases and B in 80% of the cases. In what percentage of cases are they likely to contradict each other, narrating the same incident?**

**(a) 5% (b) 15% (c)35% (d)45%**

**8. Two dice are tossed. The probability that the total score is a prime number?**

**(a) 1/6 (b) 5/12 (c) 1/2 (d) 7/9**
Ans:
n(S) = 6*6 =36
Let E=Event that the sum is a prime number
E= {(1,1),(1,2)(1,4),(1,6),(2,1),(2,3),(2,5),(3,2),(3
,4),(4,1),(4,3),(5,2),(5,6),(6,1),(6,5)}
n(E) = 15
P(E) = n(E)/n(S) = 15/36 = 5/12

**9. In a class, 30% of the students offered English, 20% offered Hindi and 10% offered both. If a student is selected at random, what is the probability that he has offered English or Hindi?**

**(a) 2/5** (b) 3/4 (c)3/5 (d) 3/10
Ans:
the P(A OR B)= P(A)+P(B)-P(A AND B)
so it's 0,3 + 0,2 - 0,1 =0,4 = 2/5

**10. A box contains 20 electric bulbs, out of which 4 are defective. Two bulbs are chosen at random from this box. The probability that at least one of these is defective is :**

**(a) 4/9 (b) 7/19 (c) 12/19 (d)21/95**
Ans:
Probability of choosing 2 bulbs so that none of them are defective is 16C2/20C2
so probability of atleast one being defective is 1-16C2/20C2 = 7/19

**11. A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white?**
Ans: 4/7

**12. A box contains 5 green, 4 yellow and 3 white marbles. Three marbles are drawn at random. What is the probability that they are not of the same color?**
Ans;
Ways in which 3 marbles can be picked of same color is 5C3 + 4C3 + 3C3
and number ways of picking 3 marbles is 12C3
So probability is (5C3 + 4C3 + 3C3)/12C3 = 3/44
probability that they are not of the same color **= 1-3/44 = 41/44**

**13. Two cards are drawn together from a pack of 52 cards. The probability that one is a spade and one is a heart is?**

**14. From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings?**

**15. Two squares are chosen at random on a chessboard. What is the probability that they have a side in common?**
ans:
P(corner)=4/64
P(edge) =24/64
P(inner)=36/64

for a corner, you have two adjacent squares. P(corner ) = 2/63
for an edge, you have three adjacent squares. P(corner ) = 3/63
for an inner, you have four adjacent squares. P(corner ) = 4/63

Following the rule:
P(A) = P(A | x1) P(x1) + P(A | x2) P(x2) + ..... P(A | xn) P(xn)
P(A) = (4/64 * 2/63 + 24/64*3/63 + 36/64*4/63) = 1/(64*63) * (8+72+144) = 1/18

**17. From a group of 10 women and 5 men, 2 people are selected at random to form a committee. Find the probability that a) only men are selected and b) exactly 1 man and 1 woman is selected.**
Ans:
a) 2/21 b) 10/21
b) we have to select exactly one man and one women
so probab that we have a man selected first is 5/15
and prob that of the remaining u select a woman is 10/14
so total prob is 5/15 * 10/14 = 5/21
now u can also select the woman first and then a man
so that gives another 5/21
so total is 10/21

more simpler solution.

5C1 * 10 C1 / 15C2 = 10/21

**18. One student's name will be picked at random to win a CD player. There are 12 male seniors, 15 female seniors, 10 male juniors, 5 female juniors, 2 male sophomores, 4 female sophomores, 11 male freshmen and 12 female freshman. Find the probability that a) a senior or a junior is picked, b) a freshman or a female is picked, and c) a freshman is NOT picked. Round answers to three decimal places.**
Ans:
a) 42/71 b) 47/71 c) 48/71

**19. If two cards are drawn from a deck of cards, WITH REPLACEMENT, find the probability that the 1st card is a heart AND the 2nd card is an ace. Round answer to three decimal places.**
Ans:
1/52 or 0.011

**20. Find the probability of picking one card**

**Q1) a heart or club Q2) a heart and club**
Ans:
Q1) 1/2 Q2) 0

**21. TWO couples and a single person are to be seated on 5 chairs such that no couple is seated next to each other. What is the probability of the above??**
Ans:
Ways in which the first couple can sit together = 2*4! (1 couple is considered one unit)
Ways for second couple = 2*4!
These cases include an extra case of both couples sitting together
Ways in which both couple are seated together = 2*2*3! = 4! (2 couples considered as 2 units- so each couple can be arrange between themselves in 2 ways and the 3 units in 3! Ways)
Thus total ways in which at least one couple is seated together = 2*4! + 2*4! - 4! = 3*4!
Total ways to arrange the 5 ppl = 5!
Thus, prob of at least one couple seated together = 3*4! / 5! = 3/5
Thus prob of none seated together = 1 - 3/5 = 2/5

---

**22. Kurt, a painter, has 9 jars of paint:**
**4 are yellow**
**2 are red**
**rest are brown**
**Kurt will combine 3 jars of paint into a new container to make a new color, which he will name accordingly to the following conditions:**

**Brun Y if the paint contains 2 jars of brown paint and no yellow**
**Brun X if the paint contains 3 jars of brown paint**
**Jaune X if the paint contains at least 2 jars of yellow**
**Jaune Y if the paint contains exactly 1 jar of yellow**

**What is the probability that the new color will be Jaune**

**a) 5/42**
**b) 37/42**
**c) 1/21**
**d) 4/9**
**e) 5/9**
Ans:
1. This has at least 2 yellow meaning..

a> there can be all three Y => 4c3
OR
b> 2 Y and 1 out of 2 R and 3 B => 4c2 x 5c1

Total 34

2.This has exactly 1 Y and remaining 2 out of 5 = > 4c1 x 5c2

Total 40
Total possibilities = (9!/3!6!) = 84
Adding the two probabilities: probability = 74/84 = 37/42

---

**23. Four cards are drawn from a standard deck of playing cards without replacing a card after it is drawn. If the first 3 cards are hearts, what is the probability that the next card drawn is not a heart?**
Ans:
probability of drawing a heart at fourth attempt is 10/49
so not drawing will be 1-10/49 = 39/49

**24. A group of 7 boys and 5 girls went to the amusement park. Only 4 people were able to get on the last remaining car on the roller coaster. To the nearest hundredth, what is the probability that the 4 people consisted of 2 boys and 2 girls?**
Ans:
(7C2 * 5C2 )/ 12C4 = 42/99 ~ 0.42

**25. In a bag of cookies, there are 4 chocolate chip cookies, 8 sugar cookies, and 6 oatmeal cookies. Steve reaches into the bag without looking and takes out 2 cookies. What is the probability that they are both chocolate chip cookies?**
**Ans:**
4/18 * 3/17 = 12/306

**26. One integer will be randomly selected from the integers 11 to 60, inclusive. What is the probability that the selected integer will be a perfect square or a**
**perfect cube?**
Ans:
16,25,36,49 r perfect squares

27 is the only cube of 3 rest r bigger than 60

hence 5/50

or 0.1

**28. A model for a random spinner can be made by taking a uniform prob. space on the circumference of a circle with radius 1, so that the probability that the pointer of the spinner lands in an arc of length s is s/ (2*pi). Suppose the circle is divided into thirty-seven zones. What is the probability that the spinner stops in an even zone?**
Ans:
As there are in all 37 zones, 18 even and 19 odd zones.
P(even zone) = 18/37

**29. A college is composed of 70% men and 30% women. It is known that 40% of the men and 60% of the women smoke. What is the probability that a student seen smoking is a man?**
Ans:

Smoking men = 0.7*0.4 = 0.28
Smoking women = 0.3*0.6 = 0.18
P(Man|Smoking)=0.28/0.46=14/23

**30. A circular target of unit radius is divided into four annular zones with outer radii 1/4, 1/2, 3/4, and 1, respectively. Suppose ten shots are fired independently and at random into the target.**

**a) What is the prob. that at most three shots land in the zone bounded by the circles of radius 1/2 and 1.**
**b) If five shots land inside the disk of radius 1/2, find the probability that at least one is in the disk of radius 1/4.**
Ans:
a) P(1/2-1) = (1-0.5^2)/1=0.75.
[10C3*0.75^3*0.25^7+10C2*0.75^2*0.25^8+10C1*0.75*0.
25^9+0.25^10]~0.0035
I used the calculator to do the last one.

b) P(in the 1/4 | it falls in 1/2) = 0.25.
P(at least one in 1/4) = 1 - P(none in 1/4) = 1 - 0.75^5 = 0.7626.

**33. Three coins are tossed up in the air. What is the probability that two of them will land heads and one will land tails?**
Ans:
as...they r in total 8 possible outcome and among which 3 of them satisfies the condition.

So...the probability = **3/8**

**34. A letter is selected at random from the word TRAPEZOID**
**find the probability that the letter selected will have vertical or horizontal line symmetry.**
Ans:
9 letter
non vertically or horizontally symmetrics
RPZ
probability of selecting these 3/9=1/3
so the required answer is 1-1/3=2/3

**35. A softball team plays two games each weekend, one on Saturday and the other on Sunday. The probability of winning the game scheduled for next Saturday is 3/5 and the probability of winning the following game, scheduled for Sunday, is 4/7. What is the probability that the team will win at least one of the two games?**
Ans:
1- (2/5)*(3/7)
=1-6/35
=29/35

Alternatively,
Since A and B are independent events,
P(A or B) = P(A) +P(B) - P(A)P(B) = 3/5+4/7-3/5*4/7 = 29/35

**36. In a certain class, there are four students in the first row: three girls (Ann, Barbara, and Cathy), and one boy (David). The teacher will call on one of these students to solve a problem at the board. When the problem is completed, the teacher will call on one of the remaining students in the first row to do a second problem at the board. What is the probability that David will be one of the two students called?**
Ans:
1 - (3/4)*(2/3)
= 1/2

**37. Elton bought a pack of 16 baseball cards. He sorted them by position and noted that he had pictures of four pitchers, five outfielders, and seven infielders. Two cards are randomly chosen from the pack of 16 cards without replacement. Find the probability that:**
**a) at least one of the cards shows a pitcher**
**b) neither card shows a pitcher**
Ans:
a) 1-(12*11)/(16*15)
b) (12*11)/(16*15)

**38. What is the probability of seven people being randomly chosen from the general population, and only one of them having a birthday on a Thursday?**
Ans:
the assumption is that a randomly chosen person has a 1/7 chance of being born on Thursday (the same goes for the rest of the days).

**39. Sam wants to buy three doughnuts, and there are five varieties to choose from. He wants each doughnut to be a different variety. How many combinations are there?**
Ans: 56

**40. how many 3 digit positive integers are odd and do not contain the digit "5"?**
Ans:
Well now the options of nos. available to us are: 0,1,2,3,4,6,7,8,9.

We have a 3-digit no. _ _ _

For this to be odd we have 4 cases: ending with 1,3,7 & 9.

Case 1: no. ending with 1.

so the last blank contains the digit 1. We now have to fill the 1st two blanks. The first blank can be filled in 8 ways( we cant have 0 here as it'll then become a 2-digit no.).
The second blank can be filled in 9 ways now.( 'coz we can have any no. from the space as it is not mentioned that repetition of digits not allowed.)

Hence for case 1 total no. of possible ways are = 8* 9 = 72

Now for other cases( ending with 3,7,9) too u'll see we get 72 ways each.
Hence the Total ways for the given condn. = sum of all cases = 72 +72+72+72 = 288 ways.

Alternatively,
there is a good method for these kind of problems

draw 3 boxes since it requires 3 digits

|A|B|C|

A,B,C are the sets these set has special property A can contain 1,2,3,4,5,6,7,8,9

B anc C can contain 0,1,2,3,4,5,6,7,8,9

now problem reduces to finding subsets of these sets such that the constraints that the problem emphasizes

now for the above problem

A'={1,2,3,4,6,7,8,9}
B'={0,1,2,3,4,6,7,8,9}
C'={1,3,7,9}

the answer is multiplication of number of elemens in A',B',C' so
8*9*4

288

**41. Grace has 16 jellybeans in her pocket. She has 8 red ones, 4 green ones, and 4 blue ones. What is the minimum number of jellybeans she must take out of her pocket to ensure that she has one of each color?**
Ans:
we should think for the worst case scenario

RRRRRRRR (for the first 8 selection always selects red ones)
GGGG (for the following 4 selections selects greens)
B (there is no chance since only blues are in the pocket at the 13th selection)

**42. lets say we have a die we roll this die and note the numbers which shows up lets say this one is N and we write numbers from 1 to N to little pieces of paper and we draw a paper randomly what is the probability that the paper has 3 written on it**
Ans:
prob of getting 3 is 1/6

prob of getting 3 N times = 1/6^N
prob of getting 3 N-1 times = 1/6^(N-1)
....and so on

prob of choosing 3 when all are threes = 1
prob of choosing 3 when all but one are 3's = (N-1)/N
.....and so on

so choosing 3 from paper is combi of two events or
if you are a maths student youll know the total probability theorem

ill write the expression

P (3 ) = [ 1 * 1/6^N + (N-1) / N * 1/ 6 ^ (N-1) + ... + 1/N]

= N
SUMM {[ 1/ 6 ^ N ] * [ {(N - k ) / N }* 6^k]}
k=0

**43. A student arrives to bus stop between 7.55 and 8.05 every morning. And the bus leaves bus stop between 8.00 and 8.10 what is the probability that the student catches the bus**
Ans:

P(A)= prob of student arriving after 0800 = 0.5
P(B)= prob of bus arriving before 0805 = 0.5

therefore P(AB) = P(A)P(B) = 0.25 .....[ A and B independent]

**44. A 3-member rowing team is to be selected from 4 men and 5 women. How many different 3-member teams be formed subject to the requirement that each team have at least 1 woman in it?**
Ans:
all event is 9c3 and all men in group is 4c3 then at least woman in it is 9c3-4c3=84-4=80

Alternatively,
4C2 * 5C1 + 4C1 * 5C2 + 5C3 = 30 + 40 + 10 = 80

**45. There are 17 steps. A person is starting from the bottom climbing to the top. A person can climb, at a time, one or two steps. In how many ways can he climb the steps and reach the top**

**46. What is the probability of two people arriving at a restaurant within 11 minutes of each other between 1 PM and 5 PM? The answer needs to be rounded to 4 decimal places with no trailing zeroes. (e.g .15 needs to be represented as 0.15, 0.154567 needs to be represented as 0.1546)**

**47. A credit card number has 6 digits (between 1 to 9). The first two digits are 12 in that order, the third digit is** bigger **than 6, the forth is divisible by 3 and the fifth digit is 3** times **the** sixth**. How many different credit card numbers exist?**
Ans:

1st digit --> 1 choice {1}
2nd digit --> 1 choice {2}
3rd digit --> 3 choices {7,8,9}
4th digit --> 3 choices {3,6,9}
5th digit --> 1 choice. It depends on the 6th digit
6th digit --> 3 choices {1,2,3}

Number of CC numbers possible = 1*1*3*3*1*3 = 27

**48. 4 ETS big book and 5 barrons books are arranged in a row on a shelf at random. Find the probability that the books of the same publisher will be put together?**
Ans:
9!/(5!*4!) ways to do the whole arrangement
And 2 ways to do in the desired way
so prob = 2/( 9!/(5!*4!) ) = 1/63

**49. 4 verbal book and 5 quant books are arranged in a row on a shelf at random. Find the probability that the books on the same section will not be put together?**
Ans:
9! ways to do the whole arrangement
And 5!*4!*2 ways to do in the opposite way
so prob = 1 - 5!*4!* 2 / 9! = 62/63

**50. Two different integers are to be selected from a set of 10 different integers in which half of the integers are even and half are odd. how many of the 45 possible selection consists of one even and one odd integer.**
Ans:
Given set contains 5 even integers and 5 odd integers
So an even integer can be selected in 5 ways and an odd integer can be selected in 5 ways. Hence totally an even integer and an odd integer can be selected in 5x5= 25 ways.

Elaborating on the given question
combinations containing both even will be 5C2 = 10
combinations containing both odd will be 5C2 =10
combinations containing one odd and one even will be 25
total combinations =45

**51. A set of integers is created after discarding nonprime odd from 1 to 13. Two different integers are to be selected from this set of integers. How many of the 45 possible selections consist of one even and one odd integer.**
Ans:
Given set is 1,2,3,4,5,6,7,8,9,10,11,12,13
We have to discard non prime odd ie only 1 and 9 will be discarded and we will be left with 11 integers and hence the no. of combinations will be 55
Now the set is 2,3,4,5,6,7,8,10,11,12,13
even integers are 6 and odd integers are 5
Hence the no. of combinations containing one even and one odd will 6x5=30

Elaboration again
combinations containing both even will be 6C2=15
combinations containing both odd will be 5C2=10
combinations containing one even and one odd = 30
Hence total combinations = 55

**52. A certain deck of cards contains 2 blue cards, 2 red cards, 2 yellow cards, and 2 green cards. If two cards are randomly drawn from the deck, what is the probability that they will both are not blue?**
Ans:
For the first card draw...6 cards out of 8 are non blue...hence probability for non blue is 6/8

For the second card drawn...7 cards remain which have 2 as blue cards and 5 as non blue...Again probability for non blue will be 5/7

Hence (6/8)*(5/7) = 15/28

Or,
Prob of not getting blue = 6C2/8C2 = 15/28

**53. Find the probability of winning the California lottery. The possible numbers are numbers 1 - 51. Six numbers are selected and all six must match to win the grand prize.**
Ans:
i think the probability is 1/51C6.... since all 6 should match....there is only 1 way of getting all 6 correct....so my ans.

Or,

For the California lottery question:
the desired combination is : 1
the total number of possible combinations: 51 raised to power 6
(because each of the six positions has 51 possibilities.. so total no. of possible combinations is 51*51*51*51*51*51..we cannot use concept of "combinations" because then we assume that once a number is used in one of the positions it cant be repeated.. however such a condition is not mentioned...i am talking of a possible combinations of the form 111111 or 122112 etc..)

So the probability is **1/(51^6)**

**54. If the chance of a coin toss landing on heads is 1/2, then the probability of getting at least three heads after four tosses is ?**
Ans:
The probability of getting exactly 3 heads would be 4C3 * .5^3 * .5^1 = 1/4
plus the probability of getting exactly one head is 4C0 * .5^1 * .5^3 = 1/16
so the total prob = 1/4 + 1/16 = 5/16.

**55. A guy throws a pair of dice. If the sum of the numbers on the dice is even, then the person tosses a coin. What is the probability of him getting a head?**
Ans:
1/2*1/2 =1/4

**56. A guy throws a pair of dice. If he doesn't get the sum as even number, he throws the dice again and again until he gets it. And then he tosses the coin. What is the probability of him getting a head?**
Ans:

prob even = 1/2
prob odd = 1/2

prob head = 1/2
prob tail = 1/2

independent events!

answer: ½

---

**How many odd numbers between 50 and 450 can be written with the elements of the set {0,1,2,3,4,5}?**
**Ans:**

__....__...__

0,1,2,3,4,5

2 digit number > 50
5 _?
we can put here 1,3,5 hence 3 numbers

3 digit number < 400
__..__..__
we can place 1,2,3 in leftmost position
0,1,2,3,4,5 in middle position
1,3,5 in rightmost position
hence total 3*6*3=54 possible numbers

3 digit number >= 400 and <=450
4..__..__
we can put 0,1,2,3,4 in middle position
and 1,3,5 in rightmost position
hence here we get 5*3=15 possible numbers

so ans = 3 + 54 + 15 = 72

---

**A certain stock exchange designates each stock with a one-, two- or three-letter cod, where each letter is selected from the 26 letters of the alphabet. If the letters may be repeated and if the same letters used in a different order constitute a different code, how many different stocks is it possible to uniquely designate with these codes?**
Ans:
If order doesn't matter and you have 1,2 and 3 letter combinations then you have 26 + 26^2 + 26^3 i.e. 1 letter options, 2 letter options, and 3 letter options.

If two of the four expressions $x + y$, $x + 5y$, $x - y$, and $5x - y$ are chosen at random, what is the probability that their product will be of the form of $x^2 - (by)^2$, where $b$ is an integer?

- ◉ $\frac{1}{2}$
- ○ $\frac{1}{3}$
- ○ $\frac{1}{4}$
- ○ $\frac{1}{5}$
- ○ $\frac{1}{6}$

Ans:
4C2=6 ( total possibilities) only (x+y)(x-y) i.e one combination gives the given form.
Hence 1/6

OR,

Probability of choosing x+y , x-y in two successive picks:
1/4 * 1/3 *2 (since you can choose x+y first, then x-y or vice versa) = 1/6

---

**A chemical plant has an emergency alarm system. When an emergency situation exists, the alarm sounds with probability 0.98. When an emergency situation does not exist, the alarm system sounds with probability 0.01. A real emergency situation is a rare event occurring with probability 0.002. Given that the alarm has just sounded, what is the probability that a real emergency situation exists?**

Ans:
Bayes' Theorem: P(A|B) = P(B|A)*P(A) / P(B)
Here we have P(A) = .002 (probability of a happening without dependence of b)
P(B) = .98*.002 + .01*.(1-.002) = .01194 (probability of b happening without dependence of a)
P(B/A) = 0.98 (Probability of b happening given a)
so we have P(A/B) = .98*.002 / .01194 = .164

---

**A rectangular table seats 4 people on each of two sides, with every person directly facing another person across the table. If eight people choose their seats at random, what is probability that any two of them directly face other?**

Ans:
Does not matter how you seat the first person (A).

Now for another person (say B) to be seated opposite to this first person(A), there is only one place.

Total possible places (empty chairs) = 7
probability = 1/7

OR,

Total number = 8!

If two are facing each other, there are 6! configurations to seat the rest.
You can place the couple in 8 different configurations.

8*6!/8! = 1/7

---

**Probability of an employee to get infected by a disease X is 20%. What is the probability that out of randomly selected four employees, 2 employees will suffer from disease X?**

Ans:

4c2*(0.2)^2*(0.8)^2 = 96 / 625.

---

**If a 3-digits integer is selected at random from the integers 100 to 199, inclusive, what is the probability that the first digit and the last digit of the integer are each equal to one more than the middle digit?**

**A) 2/225 B) 1/111 C) 1/110 D) 1/100 E) 1/50**

Ans:

1/100 (only possible 101)

---

**S is a set of all prime numbers less than 10.If a number is selected from the set at random and then another number, not necessarily different, what is the probability that the sum of the two randomly selected numbers is odd ?**

Ans:

Set of Primes < 10 = {2,3,5,7}

Number of ways of selecting 2 numbers with repetitions = 4 * 4 = 16

We need to select 2 exactly once to get an odd sum (either 2X or X2) = 3+3 = 6

---

**A certain club has 10 members, including Harry. One of the 10 members is to be chosen at random to be the president, one of the remaining 9 members is to be chosen at random to be the secretary, and one of the remaining 8 members is to be chosen at random to be the treasurer. What is the probability that Harry will be either the member chose to be the secretary or the member chosen to be the treasurer??**

Ans:

<u>case1</u>

don't pick him as pres. p=9/10

pick him as secr. p=1/9

p1 = 9/10*1/9 = 1/10 (1)

<u>case2</u>

don't pick him as pres. p=9/10

don't pick him as secr. p=8/9

pick him as tr. p=1/8

p2=9/10*8/9*1/8 = 1/10 (2)

so p=p1+p2 = 2/10=1/5

OR,

Total number of ways to choose people = 10C1*9C1*8C1

Harry can be either sec or tr. So 2 choice. Fixing a position for Harry, we can choose the rest of the people in 9C1*8C1 ways

So reqd prob = 2*9C1*8C1/10 C1*9C1*8C1 = 1/5

---

**An urn contains 10 balls colored either black or red. When selecting two balls from the urn at random the probability that you select a ball of each color is 8/15. Assuming that the urn contains more black balls than red balls, what is the probability of obtaining at least one black ball when selecting two balls from the urn?**

Ans:

P(selecting exactly 1 black & one red) = rC1*bC1/10C2 = 8/15
rb = 24
r(10-r) = 24
r = 4 (r<b)
P(at least 1 black) = 1-P(2 red) = 1- 4/10*3/9 = 1-12/90 = 39/45

---

**5 identical balls are distributed at random into 4 boxes marked A,B,C,D. Find the probability that these boxes contain respectively 1,2,2,0 balls.**

Ans:
I think the answer is C=1/56. Imagine that you have these 5 undistinguishable balls in one line. To find the number of all possible combinations you need to know how to divide them between 4 boxes. balls are all the same so you only care how many ball are there in each of the boxes. To find out this figure lets say we add 3 extra balls or whatever else which serve as dividers and show us how many balls are there in each of the boxes . B for ball, D for divider. Eg bDbbDbDb or DDbbbbbD, in the first case it means 1 inA, 2 in B, 1 in C and 1 in D, in the second all are in C. So the figure is C 3 out of 8 (how to find 3 places for dividers out of 5+3=8 places). And it equals to 56. The 1/2/2/0 combination we are after is one of 56. so here we go with 1/56.

---

**A box contains ten assemblies of which two are defective. A sample of three assemblies is selected at random. What is the probability that the two defective parts will be selected?**

Ans:
2C2*8C1/10C3 = 1/15

OR,

There are 2 possibilities
(a) NonDefective,Defective,Defective : (8/10) * (2/9) * (1/8)
(b) Defective,NonDefective,Defective : (2/10) * (8/9) * (1/8)
(c) Defective,Defective,NonDefective : (2/10) * (1/9) * (8/8)

so total probaility is (a) + (b) + (c) = > (16+16+16)/(8*9*10)
=> (3*16)/(8*9*10) = 1/15

---

**In the x-y plane, a rectangular region has the following coordinates: O (0,0), M (0,4), P (6,4), N (6,0). What is the probability that the sum of the integer coordinates of a particular point is less than 4 within the rectangular region?**

Ans:
for the denominator (6,0) there are 7 integers (since we start at 0)
and similarly (0,4) we have 5 integers so total 7*5=35
for the numerator another way to look at is if you draw the line y=x
then values are symmetric about it so suffices to look at values say to the left of it
(0,1),(0,2),(0,3) (1,2) so 4+4=8 + 2 values that lie on the line y=x
(0,0) and (1,1) = 10
and p=10/35

---

**A family has 3 children. What is the possibility that of the 3 children, exactly 2 of them are boys?**

Ans:
3c2*(1/2)^2 * (1/2)^1 = 3/8

OR,

There are 8 possible outcomes, 2*2*2 = 8, only 3 of them could have exactly 2 boys.
Ans = 3/8

---

**A student has to select 3 subjects out of 6 subjects A, B, C, D, E and F. If he has already chosen E, what is the probability that he will choose B also?**
Ans:
Already chosen E. So remaining are 5.

Out of 4 he need to chose 2.

Number of ways to chose B & one among (A,C,D,F) are 1*4C1

Total number ways to chose 2 out of 5 are 5c2 = 10.

4 / 10 = 0.4

OR,

P(B/E) = P(E & B)/P(E)

This says that the probability of happening B when E happens is the probability when both happens divided by when E alone happens

1) Find P(E & B)
no of ways to always choose both E & B from 6 subject / no of ways to choose 3 subjects from 6
= 4C1/6C3 = 4/20

2) P(E)
no of ways to always choose E / no of ways to choose 3 subjects from 6
5C2/6C3 = 10/20

P(B/E) = P(E & B)/P(E) = 4/20 * 20/10 = 4/10 = .4

OR,

1/5 + (4/5)*(1/4) = 2/5 = .4

1/5 = Probability of picking B on first try

(4/5)*(1/4) = Probability of picking B on second try

---

**Urn A contain 6 white and 4 black balls. Urn B contains 2 white and 2 black balls. From urn A two balls are selected at random and placed in urn B. From urn B two balls are then selected at random. What is the probability that exactly one of these two balls is white?**
Ans:
from **6w 4b**
p(2w) = (6/10)*(5/9) = 5/15 (1)
p(1b1w) = 2*(6/10)*(4/9) = 8/15 (2)
p(2b) = (4/10)*(3/9) = 2/15 (3)
just to check that everything is OK if we sum then we have 1.
now from **2w 2b**

if we add 2 w then we have 4w 2b we want
p(1b1w) = 2*(4/6)*(2/5) = 8/15 (4)
if we add 1 w and 1 b then we have 3w 3b we want
p(1b1w) = 2*(3/6)*(3/5) = 9/15 (5)
if we add 2 b then we have 2w 4b we want
p(1b1w) = 2*(2/6)*(4/5) = 8/15 (6)
so (5/15)*(8/15) + (8/15)*(9/15)+(2/15)*(8/15) = 128/225

---

**Suppose 5 % of Channel inheritants are Cricket fans. Determine the approximate probability that a sample of 100 inheritants will contain at least 3 cricket fans.**
Ans:

let cf stand for "cricket fan"
P(cf) = 0.05 so p(not cf)=0.95
p(#cf >= 3)
= 1 - p(#cf=0) - p(#cf=1) - p(#cf=2)
= 1 - (0.95)^100 - (0.05)(0.95)^99 - (0.05)^2(0.9)

---

**How many randomly assembled people do u need to have a better than 50% prob. that at least 1 of them was born in a leap year?**
Ans:

Probability that a person, taken at random, born in a leap year: 1/4

Let x be the number of assembled people:

We need that x(1/4)>0.5 ==> x>2 Therefore we need at least 3 people

---

**A coin is biased to an extent that head occurs two times as frequently as tail. if this coin is tossed 5 times then what is the probability that head and tail come alternately?**
**a) 77/81**
**b)1/81**
**c)1/16**
**d)4/81**
Ans:

number of heads : number of tails is 2 : 1
so P(head) = 2/3 and P(tail) = 1/3

favorable outcomes are HTHTH or THTHT

ie (2/3)^3 * (1/3)^2 + (1/3)^3 * (2/3)^2 = 4/81

---

**a fair deck of 52 playing cards contains 4 suits: diamonds, spades, hearts and clubs. Each suit contains 13 cards.**

**Column A**
**Without replacing any cards drawn, the probability of randomly drawing 2 diamonds in a row.**

**Column B**
**Without replacing any cards drawn, the probability of randomly drawing a heart, a club and a spade, but not necessarily in that order.**
Ans:

column A:
(13/52) * (12/51) = 0.0588
column B:
(39/52) * (26/51) * (13/50) = 0.099

coulmn B is bigger.
on column B they are asking for the probability of getting either:
a heart or a club or a spade therefore 39/52
next card can only be of two diff suits:
26/51
third card can only be of one suit:
13/50

---

**If n is an integer from 1 to 96, what is the probability for n*(n+1)*(n+2) being divisible by 8?**

**a) 25%**
**b) 50%**
**c) 62.5%**
**d) 72.5%**
**e) 75%**

Ans:

All the even numbers substituted in n*(n+1)*(n+2) are divisible by 8. There are 48 even numbers from 1 to 96.
And the odd numbers that produce the multiples of 8, for ex. 7, when substituted gives a product 7*8*9. There are 12 such odd numbers.
So the probability = (48+12)/96
= 62.5 %

---

**Two cards are drawn from a pack of well-shuffled cards. Find the probability that one is a CLUB and other is an ACE.**
Ans:

u hv two cases

1) club and a non-club ace
2) club and a club ace

in case (1)
num ways = C(13,1) . C(3,1) = 39

in case (2)
num ways = C(12,1) . C(1,1) = 12

Total = 51 .... (A)

Total num ways to draw two cards = C(52,2) = 26.51 .... (B)

reqd prob = (a) / (b) = 1/26

---

**A bag of 10 marbles contains 3 red marbles and 7 blue marbles. If two marbles are selected at random, what is the probability that at least one**

**marble is blue?**
**A. 21/50**
**B. 3/13**
**C. 47/50**
**D. 14/15**
**E. 1/5**
Ans:
Ok so atleast one blue = 1 - (probability of no blue)
=1 - 1/15 = 14/15

---

**A bag contains six marbles: two red, two blue, and two green. If two marbles are drawn at random, what is the probability that they are the same color?**
Ans:
Probability of drawing anything = 1 but if anyone color marble is chosen then there will br only 1 choice of that color left among the 5 marbles. Therefore ans = 1/5

---

**how do I find the probability of choosing a number which is a factor of 2 and 8 out of the first 1000 consecutive numbers?**
Ans:
every 8th number is a multiple of both 2 and 8 so probability that u will get a multiple of 8 is 1 out of every 8 numbers so 1/8
Any multiple of 8 is a multiple of 2, so the 2 part seems irrelevant. 1000/8 = 125. So there are 125 multiples of 8 from 1-1000. 125/1000 = 1/8

---

**I have 4 types of marbles:**
**Red: 3**
**Blue: 5**
**Green: 4**
**Cyan: 7**

**If 8 marbles are picked at random, what is the probability of getting exactly 2 of each color?**
Ans:

3c2*5c2*4c2*7c2/19c8

---

**A box contains 100 balls, numbered from 1 to 100. If three balls are selected at random and with replacement from the box, what is the probability that the sum of the three numbers on the balls selected from the box will be odd?**
Ans:
(0.5*0.5*0.5) + [3*( 0.5 * 0.5* 0.5)]
= 4/8
= 1/2

---

**There are 10 marble balls, including 5 green, 3 blue, and 2 red, in a container. One ball is picked out then put back, then another ball is selected. What is the probability that a green ball and a blue ball will be selected?**
Ans:
(1st pick blue AND then green) OR (1st pick green ball and the blue ball)
= 2*(3/10)*(5/10)

---

**In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize?**
Ans:

10 prizes and 25 blanks

that means total = 35

fav. outcome/total = 10/35 =>

2/7

---

**A jar contains 4 marbles: 2 Red and 2 White.**
**2 marbles are chosen at random.**

**Column A: The probability that the marbles chosen are the same color.**
**Column B: The probability that the marbles chosen are different colors.**
Ans:
In both case when chose the first marble there are 4 ways out of 4 marbles..
But in case of chosing the 2nd ball its depends on the colour..
in case of same colour there are only one ways out of 3 marbles.. so in case of SAME colour the probability will be..4/4 * 1/3 => 1/3
in case of Different colour there are only 2 ways out of 3 marbles.. so in case of SAME colour the probability will be..4/4 *2/3 =>2/3

So column **B** is greater.....

OR,
Col A
2/4c2 = 2/6 = 1/3

Col B
2c1*2c1/4c2 = 2/3

Or if taken together
I think even when you draw the marbles both at the same time the probability of having two of a different color is higher. Altogether you have six options:

W1W2
R1R2
W1R1
W1R2
W2R1
W2R2

and thus the probability of having two of different colors is 4/6 while the probability of the same color is 2/6.

---

**If 4 digit numbers greater than 5000 are randomly formed from the digits 0,1,3,5 and 7. What is the probability of forming a number divisible by 5 when the repetition of digits is not allowed?**
Ans:
total nos. possible:
5abc, 7abc = 2 * 4p3 = 2 * 24 = 48 total possibilities.
nos. possible that satisfy the req of divisible by 5 are:
5ab0, 7ab0, 7ab5 :
so, 3p2 + 3p2 + 3p2 = 18
thus prabability should be 18/48 = 3/8

---

**In an entrance test that is graded on the basis of two examinations, the probability of a randomly chosen student passing the first examination is 0.8 and the probability of passing the second examination is 0.7. The probability of passing at least one of them is 0.95. What is the probability of passing both?**
Ans:
For the second question:
P(A and B) = P(A) + P(B) - P(A or B) = 0.8 + 0.7 - 0.95 = 0.55

---

**P(A or B) = .6 and P(A) = .2**
**Find P(B) given that events A and B are independent.**
Ans:
Isnt P(A and B) = 0 only for mutually exclusive?

For example, look at the probability of the following two events:

Event A: It is Monday
Event B: It is raining

These two events are independent but P(A and B) is not 0. It can rain on a Monday

x = 0.4 + 0.2x => 0.8x = 0.4 => x = 0.5 (P(B) =x)

---

**Throw: ----- Outcome:**
**-- 1 ---------- Lose $3.00**
**-- 2 ---------- Lose $2.00**
**-- 3 ---------- Lose $1.00**
**-- 4 ---------- No Effect**
**-- 5 ---------- Win $1.00**
**-- 6 ---------- Win $5.00**

**The results of throwing a single die in a certain gambling game are shown in the table above. What is the probability that a player will have won at least $5.00 after two throws?**

**A) 1/36, B) 1/12, C) 1/9, D) 5/36, E) 1/6**
Ans:

well there are 36 options (6*6) and there are 5 ways you can get at least 5:
6 & 5 =6
5 & 6 =6
6 & 4 = 5
4 & 6 =5
6 & 6 =10
so probability is 5/36

---

**Thirteen cards are drawn from pack of 52 cards. The probability that out of 13 cards, 8 belong to the same suit is ?**
Ans:

i think when it is given specifically that there shuld be 8 cards and not atleast 8 cards we can take 4C1*13C8*39C5/52C13

---

**I have 6 coins of different denominations in my pocket. If i take them out of my pocket one after another, what is the probability of taking out the 6**

**coins in descending order of their denominations?**

Ans: 6 coins can be taken from the pocket one after the other in 6! ways out which there will be only one way of taking out the coins in descending order of their denominations. Hence the probability is 1/720.

---

**In a function if 20 speakers give their speech, what is the probability that a speaker A will speak before speaker B?**

Ans:

for any combination of 20 man A and B can themselves have only two choices. Either A is before B or B is before A. So, its 1/2.

---

**Two couples and one single person are seated at random in a row of five chairs. What is the probability that neither of the couples sits together in adjacent chairs?**

Ans:

The probabilty equal (1 - probability of one couple sitting together - probability of the other couple sitting together + the probability of both couple sitting together)
=1 - (2x4!)/5! -(2x4!)/5! + (4x3!)/5! = 2/5

---

**A man speaks the truth 3 out of 4 times. He throws a die and reports it to be a 6. What is the probability of it being a 6?**

Ans:

P(S|R)=(P(R|S)*P(S) ) / P(R)

So,
P(R|S)*P(S)=(1/6*3/4)

P(R) is interesting. It is the probability that the person spoke the truth and it was 6 and the person lied and said it was 6and it was something else

so that would be (1/6*3/4) +the person lied and said it was 6 and it was something else
The probability that it was something else is (1/6)*5.
For every number, 1-5, the person can select from 5 other numbers to lie. so , the probabilty that the person lied and said it was 6 is
1/6*5*1/5 *1/4=1/6*1/4 (1/4 is the prob. of lying)

That brings us back to
(1/6*3/4) / {1/6*3/4)+(1/6*1/4)}

Solving this comes to 3/4.

OR,

Step 1: Let's assume that the man throws the die 120 times, in which he gets a 1, a 2, a 3, a 4, a 5 and a 6 20 times each.

He will report 120 times.

Step 2: Let's call the times he reports a 6 is M, the times it's really a 6 out of M times is N.

Now the probability we must calculate is:

x=N/M.

Step 3: (Calculating M, N)

In 20 times he gets a 1, he will report it to be a 1 itself 15 times (=20*(3/4)). In 5 times remained he will not report a 1 again, therefore the probability of him speaking each of a 2 to a 6 is 1/5 => he will report a 2 to a 6, each 1 time (=5*(1/5)).

So in those 20 times he gets a 1, he will report it to be a 6 1 time.

Similarly for 20 times he gets a 2, a 3, a 4, a 5, he will report it to be a 6 1 time.

We already have 5 times he speaks a 6 out of 100 times. In those 5 times there is 0 time it **is** really a 6. (1)

In the last 20 times he gets a 6, he will speak the truth 75% => the number of times he report a 6 itself is 15 times (=20*(75%)). In those 15 times there are 15 times it **is** really a 6. (2)

From (1) and (2) we have

M= 5+15 = 20 (times)
N= 0+15 =15 (times)

Step 3: The answer is

x = N/M =3/4.

---

**Bill has a small deck of 12 playing cards made up of only 2 suits of 6 cards each. Each of the 6 cards within a suit has a different value from 1 to 6; thus, there are 2 cards in the deck that have the same value. Bill likes to play a game in which he shuffles the deck, turns over 4 cards, and looks for pairs of cards that have the same value. What is the chance that Bill finds at least one pair of cards that have the same value?**

Ans:
first card 12/12 = 1
second card take away duplicate (-1 pair) choose from 10/11
third card take away 2nd duplicate (-2 pairs) choose from 8/10
fourth card take away 3rd duplicate (-3 pairs) choose from 6/9

OR,

prob( no pair in the card) = no of ways of choosing 4 different valued cards / no of ways of choosing 4 cards from deck of 12 cards.

no of ways of choosing 4 cards from deck of 12 cards= 12C4

no of ways of choosing 4 different cards from value 1 to 6= 6C4
but each card can be of different color so 2^4 ways for each combination of 4 different cards. total no of ways = 16*6C4

prob= 16*6c4 /12c4= 16/33

required probability = 1- 16/33= 17/33

---

**In a room filled with 7 people, 4 people have exactly 1 friend in the room and 3 people have exactly 2 friends in the room (Assuming that friendship is a mutual relationship, i.e. if John is Peter's friend, Peter is John's friend). If two individuals are selected from the room at random, what is the probability that those two individuals are NOT friends?**

Ans:

4/7*5/6 is the probability of selecting a person with only one friend (4/7) and then selecting anyone but the friend (5/6).

3/7*4/6 is the probability of selecting a person with two friend (3/7) and then selecting anyone but the friends (4/6).

(4/7)(5/6) + (3/7)(4/6) = 32/42 = 16/21.

OR,

There're 21 mutually exclusive friendship relationships in total

$$7!/(2!\ 5!) = 21$$

We are told that 4 people have exactly 1 friend. This would account for 2 "friendship" relationships (e.g. AB and CD). We are also told that 3 people have exactly 2 friends. This would account for another 3 "friendship" relationships (e.g. EF, EG, and FG). Thus, there are 5 total "friendship" relationships in the group.

The probability that any 2 individuals in the group are friends is 5/21. The probability that any 2 individuals in the group are NOT friends = 1 – 5/21 = 16/21.

---

**A certain roller coaster has 3 cars and a passenger is equally likely to ride in any of the 3 cars each time that passenger rides the roller coaster. If a certain passenger is to ride the roller coaster 3 times, what is the probability that the passenger will ride in each of the cars?**

Ans:

p = favourable/possible

possible =

1st time any of the 3 so 3
2nd time any of the 3 so 3
3rd time any of the 3 so 3

total possible = 3*3*3

favourable =

1st time any of the 3 so 3
2nd time remaining 2 so 2
3rd time remaining 1 so 1

total favourable = 3*2*1

p = f/t
3*2*1/3*3*3

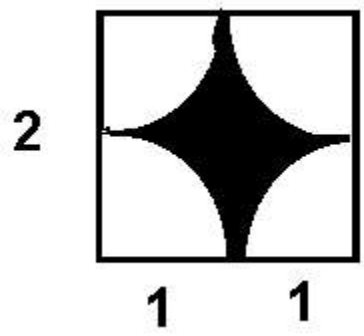
6/27
2/9

OR,

say person goes in 2 the first round
in the second round he should go in either 1st or 3rd with p = 2/3
in the third round goes in the one that's left with p=1/3
so p = 2/9

---

**one point inside a square of side 2 cm is taken randomly. whats the probability of it being away from all corners by atleast 1 cm?**
Ans:




Well.. the problem seems to be simple if observed careful..

Area of Square = 4
Area of Unshaded = 4 - 3.14 = 0.86

Sample space = 4.. and P(A) = 0.86

i suspect that Probabability of point away from all corners by more than 1 cm is 0.86/4 = 0.215

---

**From the word ANTALYA meaningless words are to be formed. What is the** probability **that the word will begin with N and will end with Y?**
Ans:
Actually, there are at least 2 or 3 other ways that the problem can be solved (all require more than 2 minutes). However logic says: The **probability** that N is in the first slot is 1/7 and the **probability** that Y comes last (GIVEN that N is first) is 1/6. Thus, p=1/42.

The median of five numbers is 15. The mode is 6. The mean is 12. What are the five numbers?
Ans:
6, 6, 15, 16, 17

---

**A membership of 620 people shows that 31 of them have first and last name begining with same letter. If a person is selected at random then whats the probability that the person choosen doesnt has his first and last name begining with the same letter.**
**0.05, 0.25, 0.50 , 0.75, 0.95**

Ans:
1-(31/620) = 1-(1/20) = 1-0.05 = 0.95

---

**A guy throws a pair of dice. If he doesn't get the sum as even number, he throws the dice again and again untill he gets it. And then he tosses the coin. What is the probability of him getting a head?**

Ans:
prob even = 1/2
prob odd = 1/2

prob head = 1/2
prob tail = 1/2

independent events!

answer: ½

---

**S is a set of all prime numbers less than 10.If a number is selected from the set at random and then another number, not necessarily different, what is the probability that the sum of the two randomly selected numbers is odd ?**

Ans:
Set of Primes < 10 = {2,3,5,7}
Number of ways of selecting 2 numbers with repetitions = 4 * 4 = 16
We need to select 2 exactly once to get an odd sum (either 2X or X2) = 3+3 = 6

Answer: 6/16 = 3/8

OR,

case1 first number 2 with p=1/4 the other number is odd p=3/4
case2 reverse order
p=1/4*3/4 + 3/4*1/4 = 3/8

---

**How many randomly assembled people do u need to have a better than 50% prob. that at least 1 of them was born in a leap year?**

Ans:
Prob. of a randomly selected person to have NOT been born in a leap yr = 3/4
Take 2 people, probability that none of them was born in a leap = 3/4*3/4 = 9/16.
The probability at least one born in leap = 1- 9/16 = 7/16 < 0.5
Take 3 people, probability that none born in leap year = 3/4*3/4*3/4 = 27/64.
The probability that at least one born = 1 - 27/64 = 37/64 > 0.5
Thus min 3 people are needed.

---

**A man has invited 7 for the party. 3 are his close friends. 4 others. He has to give two prize them. what is the probability that the 2 of them will be his close friends?**

Ans:
3C2/7C2 = **1/7**.

---

**5 girls and 3 boys are arranged randomly in a row. Find the probability that:**
**A) there is one boy on each end.**
**B) There is one girl on each end.**

Ans:
For the first scenario:
A) there is one boy on each end.

The first seat can be filled in 3C1 (3 boys 1 seat) ways = 3

the last seat can be filled in 2C1 (2 boys 1 seat) ways = 2
the six seats in the middle can be filled in 6! (1 boy and 5 girls) ways
Total possible outcome = 8!
Probability= (3C1 * 2C1 * 6!)/ 8! = 3/28

For the second scenario:
A) there is one girl on each end.

The first seat can be filled in 5C1 (5 girls 1 seat) ways = 5
the last seat can be filled in 4C1 (2 girls 1 seat) ways = 4
the six seats in the middle can be filled in 6! (3 boys and 3 girls) ways
Total possible outcome = 8!
Probability= (5C1 * 4C1 * 6!)/ 8! = 5/14

---

**There are 8 students. 4 of them are men and 4 of them are women. If 4 students are selected from the 8 students. What is the probability that the number of men is equal to that of women?**
**A.18/35 B16/35 C.14/35 D.13/35 E.12/35**
Soln: there has to be equal no of men & women so out of 4 people selected there has to be 2M & 2W.
Total ways of selecting 4 out of 8 is 8C4
total ways of selecting 2 men out of 4 is 4C2
total ways of selecting 2 women out of 4 is 4C2

so probability is (4C2*4C2)/ 8C2 = 18/35

---

**A business school club, Friends of Foam, is throwing a party at a local bar. Of the business school students at the bar, 40% are first year students and 60% are second year students. Of the first year students, 40% are drinking beer, 40% are drinking mixed drinks, and 20% are drinking both. Of the second year students, 30% are drinking beer, 30% are drinking mixed drinks, and 20% are drinking both. A business school student is chosen at random. If the student is drinking beer, what is the probability that he or she is also drinking mixed drinks?**
**A. 2/5**
**B. 4/7**
**C. 10/17**
**D. 7/24**
**E. 7/10**
**Soln:** The probability of an event A occurring is the number of outcomes that result in A divided by the total number of possible outcomes.

The total number of possible outcomes is the total percent of students drinking beer.

40% of the students are first year students. 40% of those students are drinking beer. Thus, the first years drinking beer make up (40% * 40%) or 16% of the total number of students.

60% of the students are second year students. 30% of those students are drinking beer. Thus, the second years drinking beer make up (60% * 30%) or 18% of the total number of students.

(16% + 18%) or 34% of the group is drinking beer.

The outcomes that result in A is the total percent of students drinking beer and mixed drinks.

40% of the students are first year students. 20% of those students are drinking both beer and mixed drinks. Thus, the first years drinking both beer and mixed drinks make up (40% * 20%) or 8% of the total number of students.

60% of the students are second year students. 20% of those students are drinking both beer and mixed drinks. Thus, the second years drinking both beer and mixed drinks make up (60% * 20%) or 12% of the total number of students.

(8% + 12%) or 20% of the group is drinking both beer and mixed drinks.

If a student is chosen at random is drinking beer, the probability that they are also drinking mixed drinks is (20/34) or 10/17.

---

**Find the probability that a 4 person committee chosen at random from a group consisting of 6 men, 7 women, and 5 children contains**
**A) exactly 1 woman**
**B) at least 1 woman**
**C) at most 1 woman**
**Soln:**
A) 7C1* 11C3/ 18C4
B) 1 - (11C4/18C4)
C) (11C4/18C4) + (7C1*11C3/18C4)

---

**A serial set consists of N bulbs. The serial set lights up only if all the N bulbs are in working condition. Even if one of the bulbs fails then the entire set fails. The probability of a bulb failing is x. What is the probability of the serial set failing?**
Ans:
Probability of x to fail. Probability of a bulb not failing = 1-x
probability that none of the N bulbs fail, hence serial set not failing = (1-x)^N
probability of serial set failing = 1-(1-x)^N

---

**As a part of a game, four people each much secretly chose an integer between 1 and 4 inclusive. What is the approximate likelihood that all four people will chose different numbers?**
Soln:
The probability that the first person will pick unique number is 1 (obviously) then the probability for the second is 3/4 since one number is already picked by the first, then similarly the probabilities for the 3rd and 4th are 1/2 and 1/4 respectively. Their product 3/4*1/2*1/4 = 3/32

---

**From a group of 3 boys and 3 girls, 4 children are to be randomly selected. What is the probability that equal numbers of boys and girls will be selected?**
**A. 1/10**
**B. 4/9**
**C. 1/2**
**D. 3/5**
**E. 2/3**
Ans:
Total number of ways of selecting 4 children = 6C4 = 15

with equal boys and girls. => 2 boys and 2 girls. => 3C2 * 3C2 = 9.

Hence p = 9/15 = 3/5

---

**A bag contains 3 red, 4 black and 2 white balls. What is the probability of drawing a red and a white ball in two successive draws, each ball being put back after it is drawn?**
**(A) 2/27**
**(B) 1/9**
**(C) 1/3**
**(D) 4/27**
**(E) 2/9**

**Ans:** Case I: Red ball first and then white ball

P1 = 3/9*2/9= 2/27

Case 2: White ball first and then red ball

P2 = 2/9*3/9 = 2/27

Therefore total probability: p1 + p2 = 4/27

---

**A and B alternately toss a coin. The first one to turn up a head wins. if no more than five tosses each are allowed for a single game.**

**1- Find the probability that the person who tosses first will win the game?**

**2- What are the odds against A's losing if she goes first?**

Ans**:**

look at the conditions; it says that the first person who tosses a head wins.

Let's say A tosses first.

what is the probability that he wins

H + TTH + TTTTH + TTTTTTH + TTTTTTTTH

i.e. either the first toss is head,
or the first time A tosses the coin he gets a tail and B also gets a tail , n in the second throw A gets a head.....

This continues for a max till 5 throws, because the game is for 5 throws only.
So,

1. 1/2 + (1/2)^3 + (1/2)^5 + (1/2)^7 + (1/2)^9

2. (1/2)^2 + (1/2)^4 + (1/2)^6 + (1/2)^8 + (1/2)^10

---

**My name is AJEET. But my son accidentally types the name by interchanging a pair of letters in my name. What is the probability that despite this interchange, the name remains unchanged?**
Ans:

There are actually 20 ways to interchange the letters, namely, the first letter could be one of 5, and the other letter could be one of 4 left. So total pairs by product rule = 20.

Now, there are two cases when it wouldn't change the name. First, keeping them all the same. Second, interchanging the two EEs together. Thus 2 options would leave the name intact.

Prob = 2/20 = 0.1, or 10%.

---

**Three letters are selected at random from the word SCHOOL. Find the probability that the selection**
**a) does not contain the letter O**
**b) contains both the letters O.**

Ans:

we have 3 cases
xxx --> 3 letters excluding O so 4C3 = 4
Oxx -->take one O and chose from 4 other letters 2, so 4C2 = 6
OOx -->take two O's and choose from 4 other letters 1, so 4C1 = 4
so total # of cases is 4+6+4 = 14 and combining statements we get required probabilities.
One could say here. Isn't the denominator 6C3=20 since we choose 3 letters out of 6? Answer is no since when we speak about combinations we speak about selecting out of DIFFERENT objects (all objects different)

So total no of combinations without restrictions is 6c3 = 20. This contains picks with 2 Os (no repetition here), picks with only O1 and picks with only O2. So, the duplication is betn 'only O1' & 'only O2' ..

P(only O1) = _ _ O. This can be done in 4C2 ways = 6. So, subtract this from 20.
So, no of ways to pick 3 from 6 is 14.
P(no O) = 4/14 (4C3)
P(both O) = 4/14 (4C1)

---

**In a certain school, 40% of the students are male and 20% of the male are members of a committee. If 10% of the committee members are from senior grade, what is the probability that a student selected from the total students randomly is a senior student?**

Ans:
lets take: tot. number of students=100
Males would be 40
members of a committee= 8
senior grade=0.8

P=0.8/100=0.008

---

**when tossed , a certain coin has equal probability of landing on either side. If the coin is tossed 3 times .what is the probability that it will land on the same side each time ?**

Ans:
we can have HHH or TTT so (1/2)^3+(1/2)^3=1/4

---

**In a certain city 6% of teenagers are married, 25% of married teenagers have children, and 15% of unmarried teenagers have children. If a teenager has a child, what is the probability that the teenager is not married?**

**a. 0.156**
**b. 0.200**
**c. 0.500**
**d. 0.904**
**e. 0.940**
Ans:
If there are 100 teenagers in the city 6 are married. So 94 are unmarried.

6x0.25=1.5 have children (married teenager)
94x0.15=14.1 have children (unmarried teenager)

Now the question is if the teenager has a child, what is the prob that she is Unmarried.

Prob =
= Unmarried teenagers with child/Total teenagers with child
= 14.1/(1.5+14. 1)
= 14.1/15.6
= 0.904

---

**In jar A there are 3 white balls and 2 green ones, in jar B there is one white ball and three green ones. A jar is randomly picked, what is the probability of picking up a white ball out of jar A?**
Ans:
The probability of picking the first jar is ½, the probability of picking up a white ball out of jar A Is 3/(3+2) = 3/5. The probability of both events is 1/2 x 3/5 = 3/10.

---

**Out of a box that contains 4 black and 6 white mice, three are randomly chosen. What is the probability that all three will be black?**
Ans:
The probability for the first one to be black is: 4/(4+6) = 2/5. The probability for the second one to be black is: 3/(3+6) = 1/3. The probability for the third one to be black is: 2/(2+6) = 1/4. The probability for all three events is (2/5) x (1/3) x (1/4) = 1/30.

---

**The probability of pulling a black ball out of a glass jar is 1/X. The probability of pulling a black ball out of a glass jar and breaking the jar is 1/Y. What is the probability of breaking the jar? a) 1/(XY). b) X/Y. c) Y/X. d) 1/(X+Y). e) 1/(X-Y).**
Ans:
Let Z be the probability of breaking the jar, therefore the probability of both events happening is Z x (1/X) = (1/Y). Z = X/Y.

---

**Danny, Doris and Dolly flipped a coin 5 times and each time the coin landed on "heads". Dolly bet that on the sixth time the coin will land on "tails", what is the probability that she's right?**
Ans:
The probability of the coin is independent on its previous outcomes and therefore the probability for "head" or "tail" is always ½.

---

**In a blue jar there are red, white and green balls. The probability of drawing a red ball is 1/5. The probability of drawing a red ball, returning it, and then drawing a white ball is 1/10. What is the probability of drawing a white ball?**
Ans:

Indicate A as the probability of drawing a white ball from the jar. The probability of drawing a red ball is 1/5. The probability of drawing both events is 1/10 so, 1/5 x A = 1/10. Therefore A = ½.

---

**Out of a classroom of 6 boys and 4 girls the teacher picks a president for the student board, a vice president and a secretary. What is the probability that only girls will be elected?**

Ans:

The basic principle of this question is that one person can't be elected to more than one part, therefore when picking a person for a job the "inventory" of remaining people is growing smaller. The probability of picking a girl for the first job is 4/10 = 2/5. The probability of picking a girl for the second job is (4-1)/(10-1) = 3/9. The probability of picking a girl for the third job is (3-1)/(9-1) = 1/4. The probability of all three events happening is: 2/5 x 3/9 x ¼ = 1/30.

---

**Two dice are rolled. What is the probability the sum will be greater than 10?**

Ans:

When rolling two dice, there are 36 possible pairs of results (6 x 6). A sum greater than 10 can only be achieved with the following combinations: (6,6), (5,6), (6,5). Therefore the probability is 3/36 = 1/12.

---

**The probability of having a girl is identical to the probability of having a boy. In a family with three children, what is the probability that all the children are of the same gender?**

Ans:

(1/2)^3 + (1/2)^3 = 1/4

---

**On one side of a coin there is the number 0 and on the other side the number 1. What is the probability that the sum of three coin tosses will be 2?**

Ans:

The coin is tossed three times therefore (2 x 2 x 2). We are interested only (0,1,1), (1,0,1), (1,1,0). The probability requested is 3/8.

---

**In a jar there are balls in different colors: blue, red, green and yellow. The probability of drawing a blue ball is 1/8. The probability of drawing a red ball is 1/5. The probability of drawing a green ball is 1/10. If a jar cannot contain more than 50 balls, how many yellow balls are in the Jar?**
**a) 23. b) 20. c) 24. d) 17. e) 25.**

Ans: LCM (8,5,10) = 40.

If 1/8 is the probability of drawing a blue ball then there are 40/8 = 5 blue balls in the jar. And with the same principle there are 8 red balls and 4 green ones. 40 – 5 – 8 – 4 = 23 balls (yellow is the only color left).

---

**In a jar there are 3 red balls and 2 blue balls. What is the probability of drawing at least one red ball when drawing two consecutive balls randomly?**
**a) 9/10 b) 16/20 c) 2/5 d) 3/5 e) ½**

Ans:

Since we want to draw at least one red ball we have four different possibilities: 1. Drawing blue-blue. 2. Drawing blue-red. 3. Drawing red-blue. 4. Drawing red-red. There are two ways to solve this question: One minus the probability of getting no red ball (blue-blue): 1-2/5 x ¼ = 1-2/20 = 18/20 = 9/10/ Or summing up all three good options: Red-blue --> 3/5 x 2/4 = 6/20. Blue-red --> 2/5 x ¾ = 6/20. Red-red --> 3/5 x 2/4 = 6/20. Together = 18/20 = 9/10.

---

**John wrote a phone number on a note that was later lost. John can remember that the number had 7 digits, the digit 1 appeared in the last**

**three places and 0 did not appear at all. What is the probability that the phone number contains at least two prime digits?**

Ans:

Since 1 appears exactly three times, we can solve for the other four digits only. For every digit we can choose out of 8 digits only (without 1 and 0). Since we have 4 prime digits (2, 3, 5, 7) and 4 non-prime digits (4, 6, 8, 9), the probability of choosing a prime digit is ½. We need at least two prime digits: One minus (the probability of having no prime digits + having one prime digit): There are 4 options of one prime digit, each with a probability of (1/2)^4. There is only one option of no prime digit with a probability of (1/2)^4. So: [1- ((1/2)^4+(1/2)^4*4)] = 11/16.

---

**What is the probability for a family with three children to have a boy and two girls (assuming the probability of having a boy or a girl is equal)?**

Ans:

There are three different arrangements of a boy and two girls:(boy, girl, girl), (girl, boy, girl), (girl, girl, boy). Each has a probability of (1/2)3. The total is 3*(1/2)3=3/8.

---

**What is the probability of getting a sum of 12 when rolling 3 dice simultaneously?**

Ans:

(1.5.6) -> 3!
(2,4,6) -> 3!
(3,3,6) -> 3!/2!
(3,4,5) -> 3!
(4,4,4) -> 1
(5,5,2) -> 3!/2!

Summing up -> 3*3!+2*3!/2!+1 = 25

Probability = 25/6^3 = 25/216

---

**A drawer holds 4 red hats and 4 blue hats. What is the probability of getting exactly three red hats or exactly three blue hats when taking out 4 hats randomly out of the drawer and returning each hat before taking out the next one?**

Ans:

Getting three red out of 4 that are taken out has 4 options (4!/(3!*1!)) each option has a probability of (1/2)4 since drawing a red or blue has a 50% chance. 4*1/16= ¼ to get three red hats. The same goes for three blue hats so ¼+¼ =1/2.

---

**In a department store prize box, 40% of the notes give the winner a dreamy vacation; the other notes are blank. What is the approximate probability that 3 out of 5 people that draw the notes one after the other, and immediately return their note into the box get a dreamy vacation?**

Ans:

The chance of winning is 0.4 and it stays that way for all people since they return their note. The number of different options to choose 3 winners out of 5 is 5!/(3!*2!) = 10. Each option has a chance of 0.4*0.4*0.4*0.6*0.6 = 0.02304 * 10 = 0.2304.

---

**A buyer buys 3 different items out of the newly introduced 10 different items. If two items were to be selected at random, what is the probability that the buyer does not have both the chosen items?**

Ans:

Number of ways 2 can be selected without taking the three in question. : 7C2
Number of ways 2 can be selected out of 10 = 10C2 Prob. = 7C2/10C2 = 21/45 = 7/15

---

**A certain junior class has 1,000 students and a certain senior class has 800 students. Among these students, there are 60 sibling pairs, each consisting of 1 junior and 1 senior. If 1 student is to be selected at random from each class, what is the probability that the 2 students selected will be a sibling pair?**
**A) 3/40,000**
**B)1/3,600**
**C)9/2,000**
**D)1/60**
**E)1/15**
Ans:
The first member you picked has only ONE sibling, not 60.
60/1000 * 1/800 = 60/800,000 = 3/40,000

---

**There are three secretaries who work for four departments. If each of the four departments has one report to be typed out, and the reports are randomly assigned to a secretary, what is the probability that all three secretaries are assigned at least one report?**
Ans
Total no of ways: 3^4
1st sec can get a report in 4c1 ways = 4
2nd sec .....3c1 way = 3
3rd sec.....2c1 way = 2
last can be distributed to any of the three = 3
So probability is = 4*3*2*3/ 3^4 = 8/9

---

**From (1, 2, 3, 4, 5, 6), one number is picked out and replaced and one number is picked out again. If the sum of the 2 numbers is 8, what is the probability that the 2 numbers included the number 5?**
Ans:
It is already given that sum is 8. So total number of events is 5 i.e. (2,6) (6,2) (3,5) (5,3) (4,4) Events in which 5 is included 2. So probability = 2/5.

---

**Set *S* is the set of all prime integers between 0 and 20. If three numbers are chosen randomly from set *S* and each number can be chosen only once, what is the positive difference between the probability that the product of these three numbers is a number less than 31 and the probability that the sum of these three numbers is odd?**
Ans:
8 prime nos. One of them is even. So, the only way to have an odd sum is to have all three numbers odd.

P(Odd) = 7C3/8c3.
P(Product less than 31) = 1/8c3
=> 34/56 = 17/28

---

**A and B throw a die alternately till one of them gets a '6' and wins the game. Find their respective probabilities of winning if A starts first.**
Ans:
A takes the odd terms -- 1, 3, 5, 7,...
B takes the even terms -- 2, 4, 6, 8,...
so it's ABABABABAB..... infinity

Let AW = A wins, BW = B wins

Assuming we know P(AW), the calculation of P(BW) would be identical (don't have to worry about the end points b/c both sequences theoretically have infinite terms) except it needs to be multiplied by an extra 5/6 factor resulting from the fact the entire sequence is conditioned on the fact that A didn't roll a 6 the first time.
Know P(AW) + P(BW) = 1 <-- either AW or BW, so they're complementary events
P(AW) + (5/6)P(AW) = 1 => P(AW) = 6/11
P(BW) = 1 - 6/11 = 5/11

the tedious (easier?) way -
P(AW) = P(A rolls 6 on 1st) + P(A rolls 6 on 3rd | no 6 yet) + P(A rolls 6 on 5th | no 6 yet)+.....
= (1/6) + (5/6)^2*(1/6) + (5/6)^4*(1/6) + (5/6)^6*(1/6) +.....
= 1/6 [ 1 + (25/36)^1 + (25/36)^2 + (25/36)^3 +...]
= 1/6 * (1 / (1 - 25/36)) = 1/6 * 36/11 = 6/11
P(BW) = 1 - 6/11 = 5/11

---

**urn A contains 6 red and 4 black balls and urn B contains 4 red and 6 black balls.one ball is drawn at random from urn A and placed in urn B and then one ball is drawn at random from urn B and placed in urn A . if one ball is now drawn from urn A ,the probability that it is red is?**

Ans:
Let A and B be the urns,

RED and BLUE are balls

now following events are possible:

A -Red
B -Red

6/10 x 5/11 x 6/10

A -Red
B- Blue

6/10 x 6/11 x 5/10

A -Blue
B -Red

4/10 x 4/11 x 7/10

A -Blue
B -Blue

4/10 x 7/11 x 6/10

on calculating the probabilities and summing them up - its 32/55

---

**A box contains 100 balls, numbered from 1 to 100. If three balls are selected at random and with replacement from the box, what is the probability that the sum of the three numbers on the balls selected from the**

**box will be odd?**
**A) 1/4**
**B) 3/8**
**C) 1/2**
**D) 5/8**
**E) ¾**
Ans:
We may draw odd or even number with the same probability.
We can have 8 possible combinations to draw 3 numbers which are o or e.
Only 4 combinations sastisfy the condition:
o+o+o
e+e+o
o+e+e
e+o+e
Thus, the probability is 4/8 = ½

---

**In an insurance company, each policy has a paper record and an electric record. For those policies having incorrect paper record, 60% also having incorrect electric record; For those policies having incorrect electric record, 75% also having incorrect electric record. 3% of all policies have both incorrect paper and incorrect electric records. If we randomly pick out one policy, what's the probability that the one having both correct paper and correct electric records?**
**(A)0.80 (B)0.94 (C)0.75 (D)0.88 (E)0.92**
Ans:
Let x = incorrect paper record
y = incorrect electric record

Incorrect paper record also having incorrect electric record (common set)=0.6x
Incorrect electric record also having incorrect paper record (common set)=0.75y

so, 0.6x=0.75y
x=5/4 y

Now, let T= total policy
Common set=0.03T
So,
0.03T=0.6x=0.75y
x=T/20
y=T/25
So total number of Incorrect paper record and Incorrect electric record
T/20+T/25-0.03T
=3T/50
So total number of correct paper record and correct electric record

=T-3T/50 = 47T/50

So, probability = 47/50=0.94

OR,

Let B = Incorrect both Records
IP = Incorrect Paper Records
IE = Incorrect Electronic Records

C = Correct

Given B = 3%
Out of IP, 60% are IE, meaning ... (.6)*IP = B --> (.03)
Therefore, IP = (.03)/(.6) = (.05)

Similarly, out of IE, 75% are IP, meaning ... (.75)*IE = B --> (.03)
Therefore, IE = (.03)/(.75) = (.04)

Using Sets ...
IP U IE = IP + IE - B
= (.05) + (.04) - (.03) = (.06)

C = 1 - (IP U IE) = 1 - .06 = .94

OR,

P(E and P)=0.03 ||| P(E | P)=0.6 ||| P(P | E)= 0.75 ||| P(P)=0.03/0.6=0.05 ||| P(E)=0.03/0.75=0.04
P(E')=1-P(E)=0.96
P(P' and E') = P(P' | E') * P(E') = [1-P(P | E')]*P(E') = [1-P(E' | P)*P(P)/P(E')]*P(E')=[1-0.4*0.05/0.96]*0.96=0.94

---

**If 2 numbers a and b are chosen at random from the set {1,2,3.....9,10} with replacement. What is the probability that the roots of the quadratic equation x^2 + ax + b = 0 has real roots?**
**A) 1/2**
**B) 31/50**
**C) 17/41**
**D) 9/100**
**E) 7/11**
Ans:
For real roots the discriminant of a quadratic equation should be greater than or equal to zero.
so .. a^2-4*1*b >= 0
a^2 >= 4b
(a/2)^2 >= b ---1

Following pairs satisfy the inequality 1.

a = 2 b = 1
a = 3 b = 1,2
a = 4 b = 1,2,3,4
a = 5 b = 1,2,3,4,5,6
a = 6 b = 1,2,3,4,5,6,7,8,9
a = 7 b = 1,2,3,4,5,6,7,8,9,10
a = 8 b = 1,2,3,4,5,6,7,8,9,10
a = 9 b = 1,2,3,4,5,6,7,8,9,10
a = 10 b = 1,2,3,4,5,6,7,8,9,10

so out of 100 (10*10), a total of 62 ordered pairs fulfill the condition.
hence P(E) = 62/100 or 31/50.

---

**10% of the people in a certain population use an illegal drug. A drug test yields the correct result 90% of the time, whether the person uses drugs or not. A random person is forced to take the drug test and the result is positive. What is the probability he uses drugs?**
**a)1/2 b)1/3 c) 1/9 d) 1/8**
Ans:
Assume 100 people (10 D ; 90 Non D)
...of the 10 D:
# of correct results (i.e. actual number of D) = 10 x 90% = 9
# of incorrect results(i.e. actual number of Non D) = 10 x 10% = 1

...of the 90 Non D:
# of correct results (i.e. actual number of Non D) = 90 x 90% = 81
# of incorrect results (i.e. actual number of D) = 90 x 10% = 9

Prob that a random person uses a drug
= correct positive/ (correct positive + incorrect positive)
= 9/(9+9) = 1/2

---

**There are ants at each vertex of a triangle and they all simultaneously crawl along a side of the triangle to the next vertex. The probability that no two ants will encounter one another is 2/8, since the only two cases in which no encounter occurs is when all the ants go left, i.e., clockwise -- LLL -- or all go right, i.e., counterclockwise -- RRR. In the six other cases -- RRL, RLR, RLL, LLR, LRL, and LRR -- an encounter occurs.**

**If there is an ant at each vertex of a cube and the ants all simultaneously move along one edge of the cube to the next vertex, each ant choosing its path randomly, what is the probability that no two ants will encounter one another, either on route or at the next vertex?**
Ans:
Each ant can move in three ways, so the eight ants can move in 3^8 ways. For example, the reference at position (0,0,0) could move to (1,0,0), (0,1,0), or (0,0,1). If it moves along one of the two edges bordering the left face, it will force the other three ants who start at the other three vertices of that face to move along the edges to the next vertices. The other four ants must stay on the edges bordering the right face, so they could rotate in either of the two directions. Thus there are four ways the ants could move so that the one at (0,0,0) could stay on the left face. Similarly, there are four ways the ants could move so that the reference ant could stay on the bottom face and four ways they could move so that the reference ant stays on the back face. TOTAL=12

So I get 12/(3^8) as the required probability.

---

**An access card code has 10 digits. Gretchen does not remember the last two digits, but she knows they are from the set 3, 4, 6, 8 or 9.**
**What is the probability to guess the code in at most 2 attempts?**
**(A) 1/625**
**(B) 2/625**
**(C) 4/625**
**(D) 25/625**
**(E) 50/625**
Ans:

P(one attempt)=1/25
p(exactly two attempts)=P( not first attempt)*P(second attempt/not first attempt)=(24/25)*(1/24)=1/25
P(at most two attempts)=2/25 (E)

It is probably easier to see it as a random selection , without replacement, of 2 of the 25 possible endings, one of which corresponds to the code. As the probability that any one ending corresponds to the code is 1/25, the desired probability is 2/25

---

**Some toys include large, middle, and small model with red, yellow, green, or blue color. If numbers of all model-color combinations are the same, for example, number of red large toys is equal to number of green little toys. A boy wants a red-large toy. If his mother select one for him at random, what is the probability that at least one of the color and model will satisfy the boy?**

Ans:
Total no of outcomes = 3*4=12
Desired outcomes= 3 out comes with Red (includes Large+Red also) + Large models with other 3 colours = 6

Ans: 6/12=1/2.

---

**n is a 3 digit integer chosen at random. k is the integer that results when the hundreds and units digits of n are switched. For example, if n=143, k=341 and if n=870, k=78. To the nearest percentage point, what is the probability that |n-k|<600?**

Ans:
Let the 3 digit no be : 100x + 10y + z.
|n-k| = 99.|(x-z)| < 600
Therefore, |x-z| <= 6
So, its only dependent on the first and last digit. There are only 3 combinations for which the diff between first and last digit will be greater than 6. ie. (1,8), (1,9) and (2,9)
Total combinations of first and last digit = 9.10 = 90
Total combinations which will have diff between first and last < 6 = 90 - 3 = 87

Therefore, probability that |n-k| < 600 = 87/90.

---

**A 7 digit Telephone Number has the digit 0 exactly 3 times and the number 1 is not used at all. What is the probability that the phone number contains one or more prime digits?**
**A) 1/24**
**B) 1/16**
**C) 1/2**
**D) 15/16**
**E) 23/24**
Ans:
A simple answer choice isn't much help. The poster likely has the correct answer. Four of the seven digits are not 0's or 1`s. Of the remaining 8 choices for each digit {2,3,4,5,6,7,8,9}, 4 of them are prime numbers. Thus the probability that any one of these four undetermined numbers in the telephone number is prime is 1/2. The probability that at least one is prime= 1- probability that they are all non-prime i.e 1-(1/2)^4=15/16

---

**Involved a figure:- a square MNOP is plotted on x-y plane, vertex are (0,0) (0,4) (4,0) and (4,4) the points aren't shown as (0,0) and all, but the graph**

**is divided in parts so for sake of explaining i have put the vertex. There is a triangle in this square with vertex (0,0) (3,3) and (4,0). The question is:- what is the probability that if a point is randomly selected, will lie within the triangular region.. Answer choices.. (1/8, 1/5, 1/6)**

Ans:
area of triangle
---------- = 3/8 = prob(event)
area of square

area of triangle = 4*3/2 = 6
area of square = 4* 4 = 16

so the prob(e) = 6/16 = 3/8

---

**A group of 6 is chosen from 8 men and 5 women so as to contain at least 2 men and 3 women. How many different groups can be formed if two of the men refuse to serve together?**
**a) 3510**
**b) 2620**
**c) 1404**
**d) 700**
**e) 635**

Ans:
There are two possibilities,
1) group of 3men and 3 women
2) group of 2 men and 4 women

now for 1) total cases are:

6c3*5c3 + 6c2*2c1*5c3

6c3 is the number of ways to pick 3 men from 6 (excluding the 2 men)
5c3 is the ways to pick 3 women from 5

6c2*2c1 is the case when 2 men are picked from remaining 6 men and 1 is picked from those 2.

Similarly in the second case when there are 2 men and 4 women:
6c2*5c4 + 6c1*2c1*5c4

adding 1 and 2 we get 635.

---

**A fair coin is tossed 5 times. What is the probability of getting at least three heads on consecutive tosses?**

Ans:
HHHHH
THHHH
HHHHT
HHHTT
TTHHH
THHHT
HTHHH
HHHTH

So there are 8 desired outcomes. There are a total of 32 possible outcomes (5 slots (tosses), 2 options for each toss (H or T), so 2^5=32). Hence the correct answer is Probability of at least 3 consecutive heads = 8/32 = 1/4.

---

**There are 200 houses. Two different operators are working independently. What is the probability that they will call the same number?**

Ans:

We want the probability of both calling the same number. For that, we don't need to calculate for both.

The first call's probability is 1 and not 1/200.

The second operator's probability of calling that particular number i.e the number which the first operator called is 1/200

So the answer is 1/200

---

**A guy throws a pair of dice. If the sum of the numbers on the dice is even, then the person tosses a coin. if he doesn't get the sum as even number, he throws the dice again and again until he gets it. What is the probability of him getting a head?**

Ans:

lets say he threw the dice and got even no. and then tossed the coin..

the probability will be 1/2*1/2

but if he doesn't get even ( probability 1/2) in the first attempt then he throws the dice again and lets say gets an even no. then the probability will be 1/2*1/2*1/2

similarly if he gets even in 3rd attempt P =1/2*1/2*1/*1/2*1/2

so the prob will be

1/2*1/2+1/2*1/2*1/2+1/2*1/2*1/2*1/2+.....
which is an infinite GP with first term as 1/4 and ratio as 1/2

hence sum = 1/4/(1-1/2)=2/4=1/2

so the answer should be ½

---

**Given a circle, find the probability that a chord chosen at random be longer than the side of an inscribed equilateral triangle.**

Ans:

Just take a point over the circle. Let's suppose this point A is the one point of the chord. Now if the other point is within the 1/3 distance of the perimeter of the circle of A, then the chord will be less than the side of the triangle. So...the chord to be greater than the side of the triangle...we have to take the other point B in the (1 - 1/3 - 1/3) = 1/3 portion of the perimeter.

So...ultimately the probability stands out to be 1/3.

---

**2 cards are drawn successively without replacing. What's the probability that first one will be spade and the second one will be a king?**

Ans:

there can be two alternative scenerio to get the desired combination.

**First one:** The first card drawn is a spade but not a king and the second one is a king.

The probability is: 12/52* 4/51

**Second one:** The first card drawn is the king of spade and the second one is a king.

probability : 1/52 * 3/51

So..total probability is : 12/52 * 4/51 + 1/52 * 3/51
= **1/52.**

OR,

let, F = the first card drawn is a spade.

E = the second card drawn is a king.

we have to determine, P( EF ).

from conditional probability: P( E | F ) = P( EF )/ P( F ).

=> P( EF ) = P( E | F ) * P( F ).

now, P( E | F ) is the probability of drawing a king given that the first one drawn is a spade = 1/4.
and, P( F ) is the probability of drawing a spade = 1/13

So.. P( EF ) = 1/4 * 1/13 = **1/52.**

It's ok upto this, i got the correct result except that there is a subtle problem( I'm not sure about it though). **How can we determine P( E|F ) to be 1/4 for sure?** We could do that only if F = the event that the first card drawn is a spade except a king. Then, surely P( E | F ) = 1/4.

Now, i tried it in a alternative way.

let...F = the event that the first card drawn is a spade except a king.
E = the second card drawn is a king.

thus i calculated P( E | F ).
Again, i let..
F = the event that the first card drawn is a king of spade.
E = the event that the second one is a king.

Then, calculated P( E | F ).

**So, the result is the summation of these two probabilities, which is, indeed, 1/52.**

---

**A bag contains a counter, known to be either white or black. A white counter is put in, the bag is shaken, and a counter is drawn out, which proves to be white. What is now the chance of drawing a white counter?**
*Ans:*
Let us denote the initial counter (if it is black) by B or (if it is white) by W1. The second white counter is W2.

There is actually 3 possibilities here.

possibility 1: B was in the bag. W2 was put in and then W2 was pulled out of the bag.
possibility 2: W1 was in the bag. W2 was put in and then W2 was pulled out of the bag.
possibility 3: W1 was in the bag. W2 was put in and then W1 was pulled out of the bag.

Now...out of the three possibilities, two of them make sure that we have a white bag left in the bag.
So, the answer is **2/3.**

**OR,**

Let, W1= event to get white first time
W2= event to get white second time

we need to calculate, P(W2|W1)= P(W2W1)/P(W1)

P(W1)= probability to choose the second ball first time+ probability to choose the first ball first time and get it white
= 1/2 +1/2*1/2
=3/4

P(W2W1) = Probability to choose white ball both times
= 1/2

So, P(W2|W1) = (1/2)/(3/4) = 2/3

---

**Stanton Marketing conducted a taste preference test among married and single persons for Super Cola Company. Among single people, 11% of the population questioned preferred Super Cola to all other brands. For married couples the data was:**

| | **Wife Prefers** | **Wife doesn't prefer** |
|---|---|---|
| **Husband prefers** | **0.08** | **0.06** |
| **Husband doesn't prefer** | **0.07** | **0.79** |

**Using this study as a basis for a probability measure, find the Probability that is two single people are questioned:**
**a)**

1. **They both prefer Super Cola**
2. **They both do not prefer Super Cola**
3. **Only the first person questioned would prefer Super Cola**
4. **Only one would prefer Super Cola**

**b) What is the Probability that for a married couple, at least one of them prefer Super Cola?**
**c) What is the probability that for a married couple, none prefer Super Cola?**

**d) Given that a husband prefers SC, what is the Probability that his wife also prefers it?**
**e) If the wife prefers SC, what is the probability that her husband would not prefer it?**
**f) Is preference for SC independent of the gender? Why?**
Ans:
Answers - Section a
a.1) 0.11^2
a.2) (1-0.11)^2 = 0.89^2
a.3) 0.11*0.89
a.4) (0.11*0.89)*2

Explanation - Section a
Possibly the easiest way to look at this would be to first consider a population of 100 single folks.
Number who like SC = 11
Number who don't like SC = 89
a.1) (11/100)*(10/99)
a.2) (89/100)*(88/99)
a.3) (11/100)*(89/99)
a.4) 2*(11*89)/(100*99) --> Need to multiply by 2 as you can have an "aye" first and a "nay" second or a "nay" first and an "aye" second

You'll notice that the answers in the explanation (with the population of 100) don't exactly match the earlier answers. The implicit assumption is that the population is >> 100

b) P(at least 1 member of a couple likes SC) = 1 - P(Neither member likes SC) = 1-0.79 = 0.21

c) P(Neither member likes SC) = 0.79

d) Given that a husband likes SC, P(Wife also likes SC)
= P(Both members like SC)/[P(Husband likes SC)] = 0.08/(0.08+0.06) = 0.08/0.14 = 4/7

e) Given that a wife likes SC, P(Husband dislikes SC) = P(Wife likes SC but husband doesn't)/[P(Wife likes SC)] = 0.07/(0.08+0.07) = 7/15

f) I assume we're looking at ... erm ... conventional families here. Assuming also that we're to look only at married men and women.
P(Men like SC) = P(Husbands like SC) = 0.14
P(women like SC) = P(wifes like SC) = 0.15
Clearly, Preference is dependent on gender

---

**the probability that 3 people chosen at random were born in same month**
**the probability that 2 people chosen at random were born in same day of the week.**
ans:
1. example:

favorable : 12x1x1
all: 12x12x12

favorable/all =1/144

example 2:
favorable: 7x1
all: 7x7
favorable/all =1/7

---

**P (A or B)=0.60, P (A)=0.20**
**(a) Find P (B) given that events A and B are mutually exclusive.**
**(b) Find P (B) given that events A and B are**
Ans:
(b) P(A or B) = P(A) + P(B) - P(A)*P(B)
P(B) = 0.6 - 0.2 / 0.8 = 0.4/0.8 = 0.5
(a) P(A or B) = P(A)+P(B)
P(B) = 0.4

---

**Lin and Mark each attempt independently to decode a message. If the probability that Lin will decode the message is 0.80, and the probability that Mark will decode the message is 0.70, find the probability that**
**(a) both will decode the message**
**(b) at least one of them will decode the message----xplain..**
**(c) neither of them will decode the message**
Ans:
(a) 0.8*0.7 = 0.56
(b) Probability (At least one of them) = 1 – Probability (none) = 1 - 0.2*0.3 = 1-0.06 = 0.94
(c) Probability (neither of them) = 0.2*0.3 = 0.06

---

**John wrote a phone number on a note that was later lost. John can remember that the number had 7 digits, the digit 1 appeared exactly 3 times and 0 did not appear at all. What is the probability that the phone number contains at least two prime digits?**
**a)15/16**
**b)11/16**
**c)11/12**
**d)½**
**e)5/8**
Ans:
P(atleast 2 prime) means the remaining numbers can have 2, 3 or 4 primes.
Probability of a prime = 4/8 = 1/2
P(composite) = 1/2
P(atleast 2 prime)= (1/2)^2(1/2)^2 (4c2) + (1/2)^3(1/2)^1 (4C3)+(1/2)^4 (4c4) = 11/16.

P(2 primes) is (1/2)^2(1/2)^2 (4c2)
p(3 primes) is (1/2)^3(1/2)^1 (4C3)
p(4 primes) is (1/2)^4 (4c4)

---

**A no. is drawn at random from factors of 441 .What is probability that the no. selected is divisible by 3**
Ans:
441 = 21 * 21 = 3^2 * 7^2

total factors = 3 * 3 = 9

out of these not divisible by 3 are 1, 7, 49

so divisible by 3 are 6

hence the reqd probability = 6/9 = 2/3

---

**A multiple-choice test consists of five choices per question. You think you know the answer for 75% of the questions and for the other 25% you guess at random. When you think you know the answer, you are right only 80% of the time. Find the probability of getting an arbitrary question right.**

Ans:

75% of the time you are 80% right, so

(.75)(.8) = .6

And 25% of the time you randomly guess on a 5 choice answer (20% chance of getting it right)

(.25)(.2) = .05

So: .6 + .05 = .65

---

**There are 3 points a(5,0), b(0,2) and c(0,0). And point p(x,y) is a point in triangle abc, what is the probability that y<x?**

Ans:

Draw the whole thing and draw the line y=x. Assume that the point of intersection between y=x and the triangle is a from both x axis and y axis. Trangle area = 5. Small upper triangle = a. Lower triangle area = 2.5 a.

5=3.5a==> a=10/7.

Prob = 2.5a/5=(2.5*5/3.5)/5=5/7

---

**A bag contains 3 blue marbles and 3 yellow marbles. If a marble is drawn with replacement, what is the minimum number of draws required so that there is at least a 90% probability of drawing a yellow marble?**

Ans:

Assume n draws. Since each has the same prob=1/2, hence, Prob that all marbles are blue = (1/2)^n. At least one yellow =

1-(1/2)^n > 0.9

0.1>(1/2)^n

Hence, n=4

OR,

p=1/2.

p+p^2+p^3+...+p^n >= 9/10

sum of n terms in a GP where First term =1/2 and ratio =1/2

=> p(1-p^n)/(1-p) >= 9/10

=>1-p^n >= 9/10

=> p^n <=1/10

=> 2^n >=10

=> n=4.

---

**In a high school debating team consisting of 2 freshmen, 2 sophomores, 2 juniors, and 2 seniors, two students are selected to represent the school at**

**the state debating championship. The rules stipulate that the representatives must be from different grades, but otherwise the 2 representatives are to be chosen by lottery. What is the probability that the students selected will consist of one freshman and one sophomore?**
Ans:
Total number of ways of selecting 2 reps: 8C2 = 28
Out of this 4 combinations are not allowed (different grades)

So total number of valid combinations = 24

Probability of 1 F and 1 S

(2C1*2C1) / 24

= 1/6

---

**In a certain game, a one-inch square piece is placed in the lower left corner of an eight-by-eight grid made up of one-inch squares. If the piece can move one grid up or to the right, what is the probability that the center of the piece will be exactly 4*sqrt(2) inches away from where it started after 8 moves?**
**a. 24/256**
**b. 64/256**
**c. 70/256**
**d. 128/256**
**e. 176/256**
Ans:
Total number of possible moves = 2^8 = 256 (as you can move either up or right for every move -> 2 options)

Favorable moves:
If the centre of the piece is exactly 4*sqrt(2) inches away from where it started => it is like a hypotenuse in a right angled triangle with sides 4 & 4. Since it started at (1,1), it is currently at (5,5).
Compare it to a lattice bridge of 5 x 5 squares=> the number of possible moves to reach 5,5 with moves up or to the right = (2n-2)!/(n-1)!(n-1)! = 8!/4!4! = 70

Prob = 70/256

---

**John purchased a lottery ticket and is watching the lottery results on television. For his ticket, John selected 7 numbers from 1 to 12 without replacement. In this lottery, 7 numbers from 1 to 12 are selected randomly without replacement. The first 5 numbers selected match John's lottery ticket. If John's ticket matches the next two numbers, he will win $500. What is the probability that John will win $500?**
**A. 1/15**
**B. 1/21**
**C. 1/42**
**D. 1/210**
**E. 1/10**
Ans:
The order may not matter in some of the lotteries(like Powerball where the numbers need not be in the same order), then the desired prob = 1/7c2 = 1/21—

---

**A certain club has 10 members, including Harry. One of the 10 members is to be chosen at random to be the president, one of the remaining 9 members is to be chosen at random to be the secretary, and one of the remaining 8 members is to be chosen at random to be the treasurer. What is the probability that Harry will be either the member chosen to be the secretary or the member chosen to be the treasurer?**

Ans:

The number of ways of filling the president and treasurer positions without Harry is 9P2, and the number of ways of filling the president and secretary positions without Harry is 9P2. The total number of ways of filling the 3 positions out of 10 people is 10P3.

Probability = (9P2 + 9P2)/10P3 = (72 + 72)/720 = 1/5.
OR,

There are two ways I do this:

First

Probability that harry is not selected for president = 9/10
Probablity that Hary is selected for secretary = 1/ 9
Probablity that Hary is not selected for secretary = 8/ 9
Probablity that Hary is selected for treasurer = 1/ 8

The probability that Harry is selected for either Secretary or Treasurer = 9/10* 1/9 + 9/10 * 8/9 * 1/8 = 1/10 + 1/10 = 1/5

Second

The sequence of selection is used to trick us. But the fact is that every member on the board has equal chance of becoming a president or secretary or treasurer and it is 1/10.

Therefore the probability that Harry (or anybody else) to become a Secretary or Treasurer is 1/10 + 1/10 = 1/5

OR,

P(Harry to be selected as Treasury or Secretary) = 1 - (P(Harry not selected for any positions) + P(Harry selected as President))
= 1- (9p3/10p3 + 9p2*10p3)
= 1 - 8/10 = 2/10 = 1/5

---

**A box contains 2 tennis, 3 baseball and 4 squash balls. 3 balls are drawn in succession with replacement. Find the probability that all 3 are of the same type.**

Ans:

p(all 3 of the same type) = p(all 3 tennis) + p(all 3 baseball) + p(all 3 squash) = (2/9*2/9*2/9) + (3/9*3/9*3/9) + (4/9*4/9*4/9) = 99/729 = 11/81

---

**P(A)=1/3 ; P(B) = 1/2 ; P(A intersection B) = 1/4. Find P(A' union B' )**

Ans:

P(A' union B' ) = P(A intersection B)' = 1 - 1/4 = ¾

---

**If the integers m and n are chosen at random between 1 and 100, then the probability that a number of the form 7^m + 7^n is divisible by 5 equals**
**(a) 1/4**
**(b) 1/7**
**(c) 1/8**
**(d) 1/16**
**(e) 1/49**
Ans:
The number 7^m ends with 1 or 3 or 7 or 9. Same thing with 7^n. Since both m and n are randomly selected, there are 4*4 = 16 possibilities, as far as end digit is concerned.

Out of them only combinations of m and n with last digit pair of (1,9), (3,7),(7,3),(9,1) will make the sum end with zero, and divisible by 5.

The overall probability that 7^m + 7^n is divisible by 5 is = 4/16 = ¼

---

**Triplets Adam, Bruce, and Charlie enter a triathlon. If there are 9 competitors in the triathlon, and medals are awarded for first, second, and third place, what is the probability that at least two of the triplets will win a medal?**
Ans:
(1) 2 of them get medals ==> (3c2 * 6c1)/9c3 ==> 3/14
(2) 3 of them get medals ==> 3c3/9c3 ==> 1/84

3/14 + 1/84 ==> 19/84

the question is based on either 2 or all 3 triplets winning an award. Order does not matter, hence combination

Decided to solve it another way (order is important):

Total # of ways = 9P3 = 504
Two medals = 3P2*6P1*3!/2! = 108
Three medals = 3P3 = 6
P(>=2) = (6+108)/504 = 114/504 = 19/84

---

**A bag contains 2 red beads, 2 blue beads, and 2 green beads. Sara randomly draws a bead from the bag, and then Victor randomly draws a bead from the bag. What is the probability that Sara will draw a red bead and Victor will draw a blue bead?**

**sol. 1.**
**2/6*2/5=2/15**

**sol. 2.**
**2c1*2c1/6c2 = 4/15**

**Where is the mistake? and why?**
Ans:
Solution 1 is right.

In solution 2, you have assumed that the order does not matter. ie., a red and blue

bead are picked and it doesn't matter how they are picked. So you end up with twice the # of favorable ways: red and then blue; blue and then red. For what is asked, you have to divide the result by 2; and doing so yields 2/15.

---

**A drawer holds 4 red hats and 4 blue hats. What is the probability of getting exactly three red hats or exactly three blue hats when taking out 4 hats randomly out of the drawer?**

Ans:
If you replace the hats:

Number of different ways to select (3R and 1 B) or (1R and 3B) = (4 * 4 *4) * 4 + (4 * 4 *4) * 4 = 2* 4^4
The total ways to select 4 hats is = 8 *8 *8 *8 = 8^4

The probability is 2 * 4^4 / 8^4 = 2/16 = 1/8

If you don't replace:

Number of different ways to select (3R and 1 B) or (1R and 3B) = (4c3*4c1) * 2 = 32
The total ways to select 4 hats is = 8c4

probability is = 32 / 70 = 16/35

---

**A bag contains six marbles: two red, two blue, and two green. If two marbles are drawn at random, what is the probability that they are the same color?**

Ans:
3* 2c2/6c2 = 3/15 = 1/5

---

**The probability that event A will happen is 0.5 and the event B will happen is 0.4. What is the range of the probability that neither A nor B will happen?**

Ans:
maximize P(A or B)
P(A or B) = P(A) + P(B) - P(A&B)
for max prob => P(A&B) = 0 ( A and B are mutually exclusive)
=> 0.9
neither = 0.1

minimize p(A or B) = p(A) + p(B) - P(A&B)
p(A&B) = p(A)*P(B) for independent events
= 0.5*0.4 = 0.2
p(A or B) = 0.7
neither = 0.3

Range = 0.3-0.1 = 0.2

---

**Two cards are drawn at random from a standard deck of 52 cards, without replacement. What is the probability of drawing a 7 and a king in that order?**

**whether the answer should be:**

**a. (4/52) * (4/51)**

**or**

**b. (4C1*4C1)/ (52C2)**

**which one is wrong and why?**
Ans:
In the fisrt option, cards are taken out in sequence and the first one is 7 and the second one is King. So we are looking for sequence 7-K. In this choice, the order is important.
Therefore the probability is (4C1 * 4C1) / 52P2 = 4 * 4 / 52 * 51

The second choice is where two cards are taken and we expect one card to be 7 and the other to be King. Here two sequences are possible 7-K and K-7. Since each (7K , K7) have the same probability, this option has twice probability when compared to the first one. Since in this option order is not important, the probability is (4C1*4C1) / 52C2 = 4 * 4 / (52 * 51 / 2).

---

**9 people, including 3 couples, are to be seated in a row of 9 chairs.**
**What is the probability that**
**a. None of the Couples are sitting together**
**b. Only one couple is sitting together**
**c. All the couples are sitting together**
Ans:
c) 6!*2!*2!*2!

b) The number of arrangements that a given couple is together, irrespective of the status of other couples, is 8! * 2!. Here the focus is on a given couple. It doesn't mean that there are some arrangements in this where not only couple1 is together but couple 2, 3 may also be together. We need to eliminate those arrangements.

Let f(a) is the number of arrangements that couple 'a' is together

Then f(1) = f(2) = f(3) = 8! * 2!

The number of arrangements that two couples are together is

f(1&2) = f(2&3) = f(3&1) = 7! * 2! * 2!

The number of arrangements that all couples are together is
f(1&2&3) = 6! * 2! * 2! * 2!

The number of arrangements that only one couple is together is = f(1) + f(2) + f(3) – 2*f(1&2) –2*f(2&3) –2*f(3&1) + 3* f(1&2&3) = 3* 8! * 2! – 6*7! * 2! * 2! + 3* 6! * 2! * 2! * 2!

Therefore, the probability that only one couple is together = 6! * 192 / 9! = 192/504 = 8/21

a) 6!*2!*2!*2! is the number of arrangements that all three couples are together. But there are other arrangements that either one couple is together or two couples are together that need to be eliminated.

The number of unique arrangements that at least one couple is together are = f(1) + f(2) + f(3) – f(1&2) – f(2&3) – f(3&1) + f(1&2&3) = 3* 8! * 2! – 3*7!*2!*2! + 6! * 21 * 2! * 2! = 260 * 6!

The number of arrangements that no couple is together are 9! – 260*6! = 244 * 6!

The probability is 244*6!/9!

---

**An unbiased die is tossed 4 times. The probability that the minimum face value is NOT less than two and the maximum face value is NOT greater than 5 is?**

Ans:

The probability that the minimum face value is NOT lesss than two and the maxm face value is NOT greater than 5 = The probability that we get 2,3,4 & 5 = 4/6 = 2/3 = p = probability of success.

If we assume the die is tossed 4 times, then the distribution is (p+q)^4.

=>(4C0)*(p^4)*(q^0) = (2/3)^4 = 16/81

---

**There are n letters and n envelopes. The probability that At LEAST one letter is in the wrong envelope is?**

Ans:

1-1/n!

n! - is the number of possible permutations. Only one of it has all letters in the right envelopes=> all the other have at least one letter misplaced.

---

**The prob that a man speaks the truth is ¾. He reports that the face value of a die when it is tossed is 6. The probability that he is speaking the truth is?**

Ans:

P (truthful | saying_6) = P (truthful & saying_6) / P(saying_6)

= (3/4*1/6) / (3/4*1/6 + 1/4*5/6) = 1/8 / 1/3 = 3/8

---

**Suppose we know there are 3 tails and 2 heads, what is the probability that the order is exactly HTHTT?**

Ans:

1/10 since we know that there are 3 tails and 2 heads - which can be ordered in 10 ways - and [HTHTT] is 1 way of ordering them.

---

**A single card is drawn from a standard deck of cards. Find the following probabilities:**

**a) P(Jack|face card)**

**b) P(Face card|Jack)**

**c) P(two|not face card)**

**d) P(queen|black)**

Ans:

a) P(Jack|face card) = 1/3 --> P(Jack) given that the card drawn is a "Face Card"

b) P(Face card|Jack) = 1

c) P(two|not face card) = 4/40 = 1/10

d) P(queen|black) = 2/26 = 1/13

---

**For the experiment "toss a coin and spin a spinner with three equal sectors labelled A,B and C."**

**a) List the sample space S.**

**b) List the event E,"toss a head and spin an A or B."**

**C) Find P(E)**

**d) List E'**

**e) Find P(E')**

Ans:

4a) [HA HB HC TA TB TC] ---> 2C1*3C1
4b) [HA HB]
4c) 1/3
4d) HC TA TB TC
4e) 2/3

---

**A box contains 100 balls numbered from 1 to 100 if three balls are selected at random and with replacement from the box, what Is the probability that the sum of the three numbers on the balls selected from the box will be odd.**
**A ¼**
**B 3/8**
**C ½**
**D 5/8**
**E ¾**
Ans:
to get 3 numbers sum to be odd, the following should be the pattern
o e e (3 such possibilities)
o o o (1 possibility)

selecting odd or even number from 1 - 100 number = 1/2
1/2 * 1/2 * 1/2 * 4 = ½

---

**4 letters are randomly selected from the word TRAMPLE. What is the probability that the 4 letters can be arranged to form the word TRAM?**
Ans:

P = 1/7c4 = 1/35

---

**There are two identical bottles. One bottle contains 2 green balls and 1 red ball. The other contains 2 red balls. A bottle is selected at random and a single ball is drawn. What is the probability that the ball is red?**
Ans:
you can choose a red ball by choosing the first bottle OR the second one. The prob for choosing a bottle is 1/2. Say we choose the first one.
choosing a red ball from that one is 1/3
so the total prob for the first bottle is (1/2)(1/3)
suppose I choose the second bottle. Prob for doin that is 1/2 as well. Choosing a red one from the second one is done in 2/2 ways. Total prob is (1/2)(2/2)
Add them up, 1/6 + 1/2 is 2/3.

---

**Two canoe riders must be selected from each of two groups of campers. One group consists of three men and one woman, and the other group consists of two women and one man. What is the probability that two men and two women will be selected?**
**(A) 1/6 (B) 1/4 (C) 2/7 (D) 1/3 (E) ½**
Ans:
Group 1: (3M,1W) ; G2 = {2W,1M}

G1: p(MM)=3/4*2/3 or p(MW)=1/4*3/3=1/4
G2: p(WW)=2/3*1/2=1/3 or p(MW)=1/3

Prob(MM,WW) = p(MM)*p(WW)+p(MW)*p(MW)=1/2*1/3+1/4*1/3=2/12+1/12=1/4

---

**There are 5 different boxes B1, B2, B3, B4, B5 and 5 different hats H1, H2, H3, H4 and H5. The hats are to be distributed among the different boxes. Each box can accommodate all the hats. What's the probability that B1 has**

**either H1 or H2?**
Ans:
if a bag can contain all 5 hats, then;
(2*5^4-5^3)/5^5=9/25

---

**John is trying to remember a 3 digit ID number. He knows that one of the last 2 digits in the number 138 is wrong, but he's not sure which. He also knows that all the digits in the ID number are distinct. If he were to start trying all the combinations that fit these conditions, what is the probability he would get the right combination on the first try?**
Ans:
Consider the number 138. One of the last two digits is wrong. Either 3 or 8 could be wrong. If 3 is wrong we have 7 options to fill this place-0,2,4,5,6,7,9.
similarly if 8 is wrong we have 7 options to fill this place-0,2,4,5,6,7,9. In all we have 14 options. He can choose any one in the first take and only one of them is correct. Therefore the probability is 1/14.

---

**A drawer holds 4 red hats and 4 blue hats. What is the probability of getting exactly three red hats or exactly three blue hats when taking out 4 hats randomly out of the drawer?**
**(a) 1 / 8**
**(b) ¼**
**(c) ½**
**(d) 3 / 8**
**(e) 7 / 12**
Ans:
4c3*4c1/8c4 + 4c1 * 4c3/8c4

---

**If a single throw of two dice, find the probabilty that neither a double nor a total of 9 appear ???**
Ans:
total number of permutations of oucomes = 6 * 6 = 36
outcomes with same number on both dice = 6
outcomes with sum of 9 = 4 ((4,5),(5,4),(3,6),(6,3))
probability that neither of events A or B occur = 1 - probability that either of them occur.
Ans. = 1 - (10/36) = 13/18.

---

**A card is drawn from a deck of 52 cards , find the probability of getting a king or a heart or a red card ???**
Ans:
the number of cards which are red in colour ( include all hearts) or kings are 26 + 2 ( two kings are blacks as well )= 28 .
Ans. 28/52.

OR,

P (king or heart or red)

= p(k) + p(h) + p(r) - p(k and h) - p(h and r) - p(r and k) + p(k and h and r)

= 4/52 + 13/52 + 26/52 - 1/52 - 13/52 - 2/52 + 1/52

= 28/52 = 7/13

---

**Four cars are drawn at a time from a pack of 52 playing cards. FIND THE PROBABILITY of getting four cards of the same suit ???**
Ans:
(4 * 13C4)/52C4

---

**A deck of cards contains 6 cards numbered from 1 to 6. If three cards are randomly selected from the deck, what is the probability that the numbers on the cards are drawn in order (example: first draw: 3, second draw: 4, third draw: 5)?**
**A. 1/60 B. 1/216 C. 1/30 D. 1/24 E. 1/6**
Ans:
4/6p3=1/30

---

**6 men and 4 women are waiting for an interview for a job. 4 of the men and 2 of the women are in room A, the rest in room B. Just before the interview starts, one of the candidates moves from room A to room B. The first candidate for interview is now chosen at random from room A. What is the probability that it'll be a man?**
Ans:
Room A - 4m 2w
Scenario 1 - a man is moved from A to B
Scenario 2 - a woman is moved from A to B

Now probability of selecting a man from A = P(scenario 1) * P(selecting a man from A) + P(scenario 2) * P(selecting a man from A)
= 4/6 * 3/5 + 2/6 * 4/5 = 2/3

---

**Of two men, it is said that one sometimes tells the truth while the other only tells the truth two times out of three. If one is chosen at random and asked a question, the probability of a truthful answer is 17/24. How often does the first man tells the truth?**
Ans:
The probability of selecting a guy is 1/2
The probability (selecting a guy at random and answer is truthful) =
1/2(p1)+1/2(2/3)=17/24
p1+2/3= 17/12
p1= 3/4
The first man tells the truth 3 times out of four.

---

**A point is selected at random inside a circle.Find the probability that the point is closer to the circumference of the circle than to its center.**
Ans:
Dividing the circle of radius r, into two parts with a smaller circle of radius r/2 and the remaining portion between the smaller circle and the larger one.
The smaller circle has an area of Pi *(r/2)^2 and the remaining region with an area Pi*r^2 - Pi *(r/2)^2.
The point that leis in this region is more closer to the circumference than the center.
So, P = (Pi*r^2 - Pi *(r/2)^2)/ (Pi*r^2) =3/4

---

**LECTURE 2**
**CONDITIONAL PROBABILITY, INDEPENDENCE, BAYES' RULE**

# 1 Conditional probability

The probability model is concerned with evaluating the likeliness of events. However often when evaluating the likeliness of events, researchers need to incorporate some prior information. Suppose one is interested in the measure of likeliness of an event $A$ when it is known that another event $B$ has occurred. In this situation it seems unwise to use $P(A)$ since that measure ignores the information on the event $B$. The revised measure that takes into account the additional information (the $B$ has occurred) is called the conditional probability of $A$ given $B$.

**Definition 1.** Suppose that events $A$ and $B$ are defined on the same probability space, and the event $B$ is such that $P(B) > 0$. The *conditional probability* of A given that $B$ has occurred is given by $P(A|B) = P(A \cap B)/P(B)$.

To understand the intuition behind the concept of conditional probability, note that if $B$ has occurred then we know that the selected $\omega$ belongs to $B$. Therefore, when evaluating $A$, one should ignore all $\omega$'s that are not in $B$. Thus, a revised measure of the likeliness of $A$ should depend on $P(A \cap B)$. Intuitively this is equivalent to reducing the effective sample space from $\Omega$ to $B \subset \Omega$. However, according to the second axiom of probability, the total probability measure must be equal to one. Hence, we re-scale $P(A \cap B)$ by $P(B)$ to obtain $P(\Omega|B) = P(B|B) = 1$.

**Example.** Consider rolling a dice example: $\Omega = \{1, 2, \ldots, 6\}$. Let $A = \{6\}$ and $B = \{2, 4, 6\}$. In this example, the unconditional probability of rolling a six is $P(A) = 1/6$. Now suppose you know that a player rolled an even number, however, you do not know what number exactly. In that case, you can update the probability of rolling a six to $P(A|B) = P(A \cap B)/P(B) = P(\{6\})/P(\{2, 4, 6\}) = 1/3$.

**Theorem.** *The conditional probability function is a probability function, i.e. it satisfies the axioms of probability.*

*Proof.* Let $B$ be such that $P(B) > 0$. Since $P(\cdot)$ is a probability function and satisfies the axioms of probability, $P(A \cap B) \geq 0$. It follows that $P(A|B) = P(A \cap B)/P(B) \geq 0$. Hence, the first axiom holds.

The second axiom has be already shown to hold above. To verify the third axiom, suppose

that $A_1, A_2, \ldots$ are mutually exclusive. Then, for $i \neq j$

$$\begin{aligned}(A_i \cap B) \cap (A_j \cap B) &= A_i \cap A_j \cap B \\ &= \emptyset \cap B \\ &= \emptyset,\end{aligned}$$

where the first equality holds by the associative property of the union. Thus, the events $(A_i \cap B)$ are mutually exclusive for $i = 1, 2, \ldots$. Next,

$$\begin{aligned}P(A_1 \cup A_2 \cup \ldots | B) &= P((A_1 \cup A_2 \cup \ldots) \cap B)/P(B) \\ &= P((A_1 \cap B) \cup (A_2 \cap B) \cup \ldots)/P(B) \\ &= (P(A_1 \cap B) + P(A_2 \cap B) + \ldots)/P(B) \\ &= P(A_1 \cap B)/P(B) + P(A_2 \cap B)/P(B) + \ldots \\ &= P(A_1|B) + P(A_2|B) + \ldots,\end{aligned}$$

where the second equality holds do to the distributive property, the third equality holds since the events $(A_i \cap B)$ are mutually exclusive, and the last equality holds by the definition of conditional probability. □

Note that once it has been established that conditional probability satisfies the axioms of probability, other properties such as those discussed in Theorem 7 in Lecture 1 follow immediately.

The definition of conditional probability implies that

$$P(A \cap B) = P(A|B)P(B).$$

In many cases, the computation of conditional probability is very natural and straightforward, which makes it much easier to calculate the probability of an intersection using the conditional probability approach rather than the counting technique.

**Example.** (Based on Wonnacott & Wonnacott) Suppose that there are 10 domestic and 10 international students in a class. Two students are selected randomly one after another. What is the probability that they are both international? Let $I_1$ be the event that the first selected student is international. Let $I_2$ denote the event that the second selected student is international. We are interested in $P(I_1 \cap I_2)$. Assuming that the selection procedure is purely random, $P(I_1) = 1/2$. Next, consider $P(I_2|I_1)$. Since we condition on $I_1$, we assume that the first student was international. Thus, after the first round, there are 10 domestic and 9 international students left. Hence, $P(I_2|I_1) = 9/19$, and therefore, $P(I_1 \cap I_2) = 9/38$.

The same answer can be obtained by the counting technique. We can use the permutations

formula or ignore the order of selection and use the combinations formula. Both approaches will give the same answer. The number of possible permutations of choosing 2 students out of 20 is

$$P(20,2) = \frac{20!}{18!} = 20 \times 19.$$

The number of possible permutations with 2 international students (out of 10 international students) is

$$P(10,2) = \frac{10!}{8!} = 10 \times 9.$$

Hence, the probability that the two selected students are international is

$$\frac{10 \times 9}{20 \times 19} = \frac{9}{38}.$$

## 2 Independence

Independence is one of the central concepts in probability and statistics. It will be used repeatedly in this course and its continuations. Intuitively, the independence of events $A$ and $B$ means that the information about $B$ is useless for predicting the occurrence of $A$:

**Definition 2.** Suppose that $P(B) > 0$. Two events $A$ and $B$ are *independent* if

$$P(A|B) = P(A).$$

This definition of independence together with the formula for conditional probability in Definition 1, imply:

$$\begin{aligned} P(A) &= P(A|B) \\ &= P(A \cap B)/P(B) \end{aligned}$$

or

$$P(A \cap B) = P(A)P(B). \tag{1}$$

Suppose that $P(A) > 0$ and $P(B) > 0$. Then (1) implies that $P(A|B) = P(A)$ and $P(B|A) = P(B)$. Thus, equation (1) can be used that as the definition of independence between $A$ and $B$.

**Definition 3.** Two events $A$ and $B$ are *independent* if $P(A \cap B) = P(A)P(B)$.

The advantage of Definition (2) is that it clearly shows the intuition behind the concept of independence: conditioning on $B$ does not change the probability of occurrence of $A$. However,

it can only be applied with events $B$ that satisfy $P(B) > 0$. On the other hand, Definition (3) applies to all events including events including events that occur with probability zero. For events that occur with non-zero probabilities, the two definitions are equivalent as they connected by the definition of conditional probability (Definition 1).

**Example.** Consider two events: {to be a hockey player} and {to live in Canada}. Are they independent?

**Example.** Can disjoint events be independent? The intuition suggests not: if $A$ and $B$ are disjoint, they have no common outcomes, and therefore if $B$ occurred we can immediately deduce that $A$ did not occur. Hence, the information contained in $B$ can be used to revise the probability of occurrence of $A$. Nevertheless, disjoint events can be independent if one also considers events that occur with probability zero. We have $P(A \cap B) = P(\emptyset) = 0$. If $P(A) = 0$ or $P(B) = 0$, then $P(A)P(B) = 0$ and, therefore, $A$ and $B$ satisfy the criterion for independence (according to Definition 3). Thus, mutually exclusive events can be independent.

**Example.** Can an event be independent from itself?

Next, we will discuss certain properties and implications of independence. First, note that independence is a property of the probability function $P(\cdot)$ rather than events themselves. For the same sample space and a collection of events one can define many probability functions. Thus, it is possible that $A$ and $B$ are independent according to $P(\cdot)$, but not independent according to some other probability function $Q$: i.e $P(A \cap B) = P(A)P(B)$ but $Q(A \cap B) \neq Q(A)Q(B)$.

**Theorem.** *Suppose that $A$ and $B$ are independent (with respect to some probability function $P$). Then the following statements are true (with respect to the same probability function $P$):*

*(a) $A^c$ and $B$ are independent.*

*(b) $A^c$ and $B^c$ are independent.*

*(c) $A$ and $B^c$ are independent.*

*Proof.* To show (a), we need to evaluate $P(A^c \cap B)$. Recall from Lecture 1 that $B = (A \cap B) \cup (A^c \cap B)$ or $P(B) = P(A \cap B) + P(A^c \cap B)$. After re-arranging the terms, we obtain:

$$\begin{aligned} P(A^c \cap B) &= P(B) - P(A \cap B) \\ &= P(B) - P(A)P(B) \\ &= P(B)P(1 - P(A)) \\ &= P(A^c)P(B), \end{aligned}$$

where the equality in the second line holds by independence of $A$ and $B$, and the last equality holds because $P(A^c) = 1 - P(A)$.

Parts (b) and (c) follow immediately from (a). □

The concept can be extended to more than two events. The following definition applies in the case of three events.

**Definition 4.** Events $A_1$, $A_2$, and $A_3$ are *mutually independent* if:

(a) $P(A_1 \cap A_2) = P(A_1)P(A_2)$, $P(A_1 \cap A_3) = P(A_1)P(A_3)$, $P(A_2 \cap A_3) = P(A_2)P(A_3)$, and

(b) $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$.

Thus, in the case of three events, they all must be *pairwise independent* (condition (a)), but also the probability of the intersection of the three events should be equal to the product of the individual probabilities.

The concept can be generalized to an arbitrary number of events:

**Definition 5.** Events $A_1, A_2, A_3, \ldots$ are *mutually independent* if $P(A_{i_1} \cap A_{i_2} \cap \ldots A_{i_j}) = P(A_{i_1})P(A_{i_2}) \ldots P(A_{i_j})$ for any *finite* collection of indices $\{i_1, i_2, \ldots, i_j\}$, $j = 2, 3, \ldots$.

Sometimes experiments consist of multiple stages or trials. We say that the trials are independent if events from different trials are mutually independent. This occurs when the outcome of one trial does not help with predicting the outcomes in other trials

**Definition 6.** Suppose that an experiment consists of $n$ trials. Let $\mathcal{A}_i$ denote the collection of events associated with trial $i = 1, 2, \ldots, n$. The trials are said to be independent if $P(A_1 \cap A_2 \cap \ldots \cap A_n) = P(A_1)P(A_2) \ldots P(A_n)$ for any combination of $A_1 \in \mathcal{A}_1$, $A_2 \in \mathcal{A}_2$, $\ldots$,$A_n \in \mathcal{A}_n$.

**Example. (Hockey example revisited)** Consider the hockey example from Lecture 1. Recall that for a complete lineup one has to select 3 forwards out of 12, 2 defencemen out of 6, and 1 goalie out of 2. Suppose that selection of forwards does not affect selection of defencemen or the goalie, and etc. In other words, suppose that the three stages, selection of forwards, of defencemen, and of the goalie are independent. Then the probability that Higgins (forward) and Bieksa (defencemen) will be in the same lineup is given by

$$\frac{3}{12} \times \frac{2}{6} = \frac{1}{12} \approx 0.083333.$$

The probability that Higgins and Bieksa will both play L on the same shift is

$$\frac{1}{12} \times \frac{1}{6} \approx 0.013889.$$

## 3 Bayes' rule

Baye's rule allows one to deduce $P(B|A)$ given $P(A|B)$ and the probabilities of $A$ and $B$. The next theorem presents the simplest version of the rule.

**Theorem 7.** *Let $A$ and $B$ be two events such that $P(A) > 0$ and $P(B) > 0$. Then*

$$P(B|A) = P(A|B)\frac{P(B)}{P(A)}.$$

*Proof.* $P(A|B)P(B)/P(A) = P(A \cap B)/P(A) = P(B|A)$, where the first equality is by the definition of conditional probability. □

Next, we will present a more general version of the result.

**Definition 8.** A collection of events $\{B_1, B_2, \ldots, B_n\}$ is called a *partition* of the sample space $\Omega$ if:

(a) The events $B$'s are mutually exclusive, i.e. $P(B_i \cap B_j) = \emptyset$ for all $i \neq j$.

(b) $\cup_{i=1}^{n} B_i = \Omega$.

The following result is know as *the law of total probability.*

**Theorem 9.** *Let $A$ be an event on $\Omega$, and let $\{B_1, B_2, \ldots, B_n\}$ be a partition of $\Omega$ such that $P(B_i) > 0$ for all $i = 1, 2, \ldots, n$. Then*

$$P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(A|B_i)P(B_i).$$

*Proof.* Since $B$'s partition $\Omega$,

$$A = (A \cap B_1) \cup (A \cap B_2) \cup \ldots \cup (A \cap B_n) = \cup_{i=1}^{n}(A \cap B_i).$$

Since $B$'s are mutually exclusive, the events $(A \cap B_i)$ and $(A \cap B_j)$ are also mutually exclusive for $i \neq j$. Hence,

$$\begin{aligned} P(A) &= P(\cup_{i=1}^{n}(A \cap B_i)) \\ &= \sum_{i=1}^{n} P(A \cap B_i). \end{aligned}$$

Lastly, by the definition of conditional probability,

$$P(A \cap B_i) = P(A|B_i)P(B_i).$$

□

Now we can state a more general version of Bayes' theorem.

**Theorem 10. (Bayes' Theorem)** *Let $A$ be an event on $\Omega$, and let $\{B_1, B_2, \ldots, B_n\}$ be a partition of $\Omega$ such that $P(B_i) > 0$ for all $i = 1, 2, \ldots, n$. Suppose further that $P(A) > 0$.*

*Then*

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^{n} P(A|B_j)P(B_j)}$$

*for all* $i = 1, 2, \ldots, n$.

*Proof.* By Theorem 9, the denominator is equal to $P(A)$. By the definition of conditional probability, the numerator is equal to $P(B_i \cap A)$. The result follows from the definition of conditional probability. □

**Example.** Suppose that 30% of Canadians play hockey. Suppose further that 10% of people in the USA and 1% of people in Mexico play hockey, and that for the rest of North American countries and territories the probability of playing hockey is zero. If a randomly selected North American person plays hockey, what is the probability that he is from Canada? What is the probability that he is from the US? Note that Canada has 6% of the population in North America, the USA has 57%, and Mexico has 21%.

Answer: Let $A$ denote hockey, and $B_1$, $B_2$, and $B_3$ denote Canada, USA, and Mexico respectively. Let $B_4, \ldots, B_{41}$ denote the rest of North American countries and territories. We have

$$\begin{aligned} P(A|B_1) &= 0.30, \\ P(A|B_2) &= 0.10, \\ P(A|B_3) &= 0.01, \end{aligned}$$

and $P(A|B_j) = 0$ for $j = 3, \ldots, 41$. Applying Bayes' Theorem, the probability that a randomly selected North American hockey player is from Canada is

$$P(B_1|A) = \frac{0.30 \times 0.06}{0.30 \times 0.06 + 0.10 \times 0.57 + 0.01 \times 0.21} = \frac{0.018}{0.0771} \approx 0.23.$$

The probability that he is from the US is

$$P(B_2|A) = \frac{0.10 \times 0.57}{0.30 \times 0.06 + 0.10 \times 0.57 + 0.01 \times 0.21} = \frac{0.057}{0.0771} \approx 0.74.$$

Question -1 : A factory has two machines I and II. Machine I produces 40% of items of the output and Machine II produces 60% of the items. Further 4% of items produced by Machine I are defective and 5% produced by Machine II are defective. An item is drawn at random. If the drawn item is defective, find the probability that it was produced by Machine II.

Solution:

Let $A_1$ be the event that the items are produced by Machine-I, $A_2$ be the event that items are produced by Machine-II. Let B be the event of drawing a defective item. Now we are asked to find the conditional probability P ($A_2$ / B). Since $A_1$ , $A_2$ are mutually exclusive and exhaustive events, by Bayes' theorem,

We have,

$P(A_1) = 0.40$ , $P(B / A_1) = 0.04$

$P(A_2) = 0.60$, $P(B / A_2) = 0.05$

$$P(A_2 / B) = \frac{P(A_2)\,P(B / A_2)}{P(A_1)\,P(B / A_1) + P(A_2)\,P(B / A_2)}$$

$$P(A_2 / B) = \frac{(0.60)(0.05)}{(0.40)(0.04) + (0.60)(0.05)} = \frac{15}{23} .$$

Question -2:

A construction company employs 2 executive engineers. Engineer-1 does the work for 60% of jobs of the company. Engineer-2 does the work for 40% of jobs of the company. It is known from the past experience that the probability of an error when engineer-1 does the work is 0.03, whereas the probability of an error in the work of engineer-2 is 0.04. Suppose a serious error occurs in the work, which engineer would you guess did the work?

## Solution

Let $A_1$ and $A_2$ be the events of job done by engineer-1 and engineer-2 of the company respectively. Let B be the event that the error occurs in the work.

We have to find the conditional probability

$P(A_1 / B)$ and $P(A_2 / B)$ to compare their errors in their work.

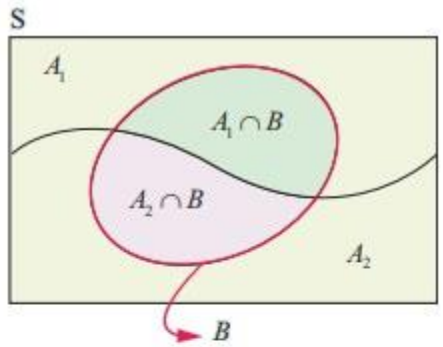
From the given information, we have

$P(A_1) = 0.60$, $P(B / A_1) = 0.03$

$P(A_2) = 0.40$, $P(B / A_2) = 0.04$

$A_1$ and $A_2$ are mutually exclusive and exhaustive events.

Applying Bayes' theorem,




$$P(A_1 / B) = \frac{P(A_1)\,P(B / A_1)}{P(A_1)\,P(B / A_1) + P(A_2)\,P(B / A_2)}$$

$$= \frac{(0.60)(0.03)}{(0.60)(0.03) + (0.40)(0.04)}$$

$$P(A_1 / B) = \frac{9}{17}.$$

$$P(A_2 / B) = \frac{P(A_2)\,P(B / A_2)}{P(A_1)\,P(B / A_1) + P(A_2)\,P(B / A_2)}$$

$$P(A_2 / B) = \frac{(0.40)(0.04)}{(0.60)(0.03) + (0.40)(0.04)}$$

$$P(A_2 / B) = \frac{8}{17}.$$

Since $P(A1/B) > P(A2/B)$, the chance of error done by engineer-1 is greater than the chance of error done by engineer-2. Therefore one may guess that the serious error would have been be done by engineer-1.

## Question- 3 :

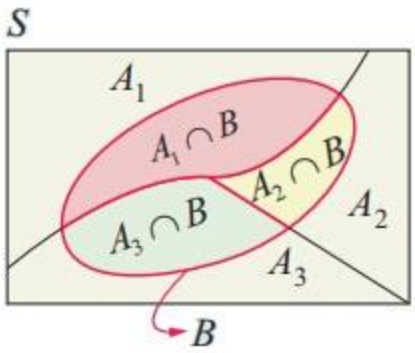
The chances of *X*, *Y* and *Z* becoming managers of a certain company are 4 : 2 : 3. The probabilities that bonus scheme will be introduced if *X*, *Y* and *Z* become managers are 0.3, 0.5 and 0.4 respectively. If the bonus scheme has been introduced, what is the probability that *Z* was appointed as the manager?

**Solution**

Let $A_1$, $A_2$ and $A_3$ be the events of *X*, *Y* and *Z* becoming managers of the company respectively. Let *B* be the event that the bonus scheme will be introduced.

We have to find the conditional probability $P(A_3/B)$.

Since $A_1$, $A_2$ and $A_3$ are mutually exclusive and exhaustive events, applying Bayes' theorem




We have

$$P(A_3 / B) = \frac{P(A_3)\,P(B / A_3)}{P(A_1)\,P(B / A_1) + P(A_2)\,P(B / A_2) + P(A_3)\,P(B / A_3)}$$

$$P(A_1) = \frac{4}{9}, \quad P(B / A_1) = 0.3$$

$$P(A_2) = \frac{2}{9}, \quad P(B / A_2) = 0.5$$

$$P(A_3) = \frac{3}{9}, \quad P(B / A_3) = 0.4$$

$$P(A_3 / B) = \frac{P(A_3)\,P(B / A_3)}{P(A_1)\,P(B / A_1) + P(A_2)\,P(B / A_2) + P(A_3)\,P(B / A_3)}$$

$$P(A_3 / B) = \frac{\left(\frac{3}{9}\right)(0.4)}{\left(\frac{4}{9}\right)(0.3) + \left(\frac{2}{9}\right)(0.5) + \frac{3}{9}(0.4)}$$

$$= \frac{12}{34} = \frac{6}{17}.$$

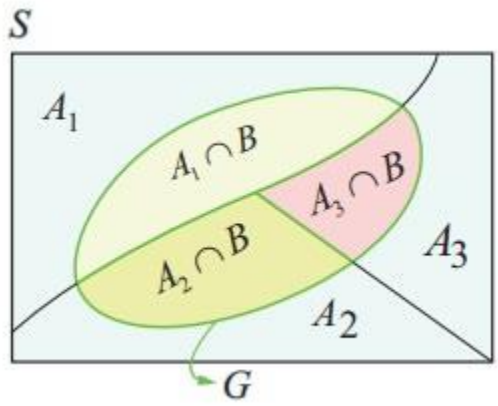
Question -4 :

A consulting firm rents car from three agencies such that 50% from agency *L*, 30% from agency *M* and 20% from agency *N*. If 90% of the cars from *L*, 70% of cars

from *M* and 60% of the cars from *N* are in good conditions (i) what is the probability that the firm will get a car in good condition? (ii) if a car is in good condition, what is probability that it has come from agency *N*?

## Solution

Let $A_1$ , $A_2$ , and $A_3$ be the events that the cars are rented from the agencies X, Y and Z respectively.




Let *G* be the event of getting a car in good condition.

We have to find

(i) the total probability of event *G* that is, *P*(*G*)

(ii) find the conditional probability $A_3$ given G that is, $P\ (A_3 / G)$ We have

$P\ (A_1) = 0.50$, $P\ (G / A_1) = 0.90$

$P\ (A_2) = 0.30$, $P\ (G / A_2) = 0.70$

$P\ (A_3) = 0.20$, $P\ (G / A_3) = 0.60$.

(i) Since $A_1$ , $A_2$ and $A_3$ are mutually exclusive and exhaustive events and *G* is an event in S, then the total probability of event *G* is *P*(G).

$$P\ (G) = P\ (A_1)\ P\ (G / A_1) + P\ (A_2)\ P\ (G / A_2) + P\ (A_3)\ P\ (G / A_3)$$

$$P(G) = (0.50)(0.90) + (0.30)(0.70) + (0.20)(0.60)$$

$$P(G) = 0.78.$$

(ii) The conditional probability $A_3$ given *G* is $P\ (A_3 / G)$

By Bayes' theorem,

$$P(A_3/G) = \frac{P(A_3)\,P(G/A_3)}{P(A_1)\,P(G/A_1) + P(A_2)\,P(G/A_2) + P(A_3)\,P(G/A_3)}$$

$$P(A_3/G) = \frac{(0.20)(0.60)}{(0.50)(0.90) + (0.30)(0.70) + (0.20)\,(0.60)}$$

$$= \frac{2}{13}.$$

## Question -5:

A rare genetic disease is discovered. Although only one in a million people carry it, you consider getting screened. You are told that the genetic test is extremely good; it is 100% sensitive (it is always correct if you have the disease) and 99.99% specific (it gives a false positive result only 0.01% of the time). Having recently learned Bayes' theorem, you decide not to take the test. Why?

(From Durbin et.al. "Biological Sequence Analysis", Cambridge University Press, 1998)

Solution:

Bayes' Theorem states that for events X and Y:

P(X|Y)=P(Y|X)*P(X)/P(Y).

We want to know the probability of being healthy(X) given the positive test(PT) results(Y).

According to the Bayes' Theorem,

P(healthy|PT)=P(PT|healthy)*P(healthy)/P(PT).

From the problem we know that

P(healthy)=1-0.000001=0.999999

and getting a false positive

P(PT|healthy)=0.0001.

The only unknown in the formula above is the probability of having a positive test P(PT). It can be calculated using the definition of *marginal* probability

$$P(Y)=P(Y|Z_1)*P(Z_1)+...+P(Y|Z_n)*P(Z_n),$$

where $Z_i$, i=1...n are *all* possible events. In our case there are only two possible events: "being healthy" and "being sick". Therefore

$$P(PT)=P(PT|healthy)*P(healthy)+P(PT|sick)*P(sick).$$

From the problem we know that

$$P(PT|sick)=1.0$$

(test is always correct in presence of the disease) and

$$P(sick)=0.000001.$$

Substituting the numbers into the formula we get

$$P(PT)=0.0001*0.999999+1.0*0.000001=0.000101.$$

Finally,

$$P(healthy|PT)=0.0001*0.999999/0.000101=0.990098,$$

that is very close to 1.

So, the probability of still being healthy given that the results of the test turned positive is above 99%. That is a good reason for not taking the test.

Question -6:

Of the one million squirrels on MIT's campus most are good-natured. But one hundred of them are pure evil! An enterprising student in Course 6 develops an "Evil Squirrel Alarm" which she offers to sell to MIT for a passing grade. MIT decides to test the reliability of the alarm by conducting trials.

- When presented with an evil squirrel, the alarm goes off 99% of the time.
- When presented with a good-natured squirrel, the alarm goes off 1% of the time.

(a) If a squirrel sets off the alarm, what is the probability that it is evil?

(b) Should MIT co-opt the patent rights and employ the system?

Solution:

One solution

(This is a base rate fallacy problem)

We are given:

$$P(\text{nice}) = 0.9999, \qquad P(\text{evil}) = 0.0001 \text{ (base rate)}$$

$$P(\text{alarm} \mid \text{nice}) = 0.01, \quad P(\text{alarm} \mid \text{evil}) = 0.99$$

$$P(\text{evil} \mid \text{alarm}) = \frac{P(\text{alarm} \mid \text{evil})P(\text{evil})}{P(\text{alarm})}$$

$$= \frac{P(\text{alarm} \mid \text{evil})P(\text{evil})}{P(\text{alarm} \mid \text{evil})P(\text{evil}) + P(\text{alarm} \mid \text{nice})P(\text{nice})}$$

$$= \frac{(0.99)(0.0001)}{(0.99)(0.0001) + (0.01)(0.9999)}$$

$$\approx 0.01$$

## Evil Squirrels Solution

**answer:** (a) This is the same solution as in the slides above, but in a more compact notation. Let $E$ be the event that a squirrel is evil. Let $A$ be the event that the alarm goes off. By Bayes' Theorem, we have:

$$\begin{aligned} P(E \mid A) &= \frac{P(A \mid E)P(E)}{P(A \mid E)P(E) + P(A \mid E^c)P(E^c)} \\ &= \frac{.99\frac{100}{1000000}}{.99\frac{100}{1000000} + .01\frac{999900}{1000000}} \\ &\approx .01. \end{aligned}$$

(b) No. The alarm would be more trouble than its worth, since for every true positive there are about 99 false positives.

Question -7:

## Table Question: Dice Game

1. The Randomizer holds the 6-sided die in one fist and the 8-sided die in the other.
2. The Roller selects one of the Randomizer's fists and covertly takes the die.
3. The Roller rolls the die in secret and reports the result to the table.

Given the reported number, what is the probability that the 6-sided die was chosen? (Find the probability for each possible reported number.)

**answer:** If the number rolled is 1-6 then $P(\text{six-sided}) = 4/7$.
If the number rolled is 7 or 8 then $P(\text{six-sided}) = 0$.

*Explanation on next page*

Solution:

## Dice Solution

This is a Bayes' formula problem. For concreteness let's suppose the roll was a 4. What we want to compute is $P(\text{6-sided}|\text{roll 4})$. But, what is easy to compute is $P(\text{roll 4}|\text{6-sided})$.
Bayes' formula says

$$P(\text{6-sided}|\text{roll 4}) = \frac{P(\text{roll 4}|\text{6-sided})P(\text{6-sided})}{P(4)}$$
$$= \frac{(1/6)(1/2)}{(1/6)(1/2)+(1/8)(1/2)} = 4/7.$$

The denominator is computed using the law of total probability:

$$P(4) = P(4|\text{6-sided})P(\text{6-sided}) + P(4|\text{8-sided})P(\text{8-sided}) = \frac{1}{6}\cdot\frac{1}{2}+\frac{1}{8}\cdot\frac{1}{2}.$$

Note that any roll of 1,2,...6 would give the same result. A roll of 7 (or 8) would give clearly give probability 0. This is seen in Bayes' formula because the term $P(\text{roll 7}|\text{6-sided}) = 0$.

Question -8 :

EXAMPLE

If a single card is drawn from a standard deck of playing cards, the probability that the card is a king is 4/52, since there are 4 kings in a standard deck of 52 cards. Rewording this, if King is the event "this card is a king," the prior probability $P(\text{King}) = \frac{4}{52} = \frac{1}{13}$.

If evidence is provided (for instance, someone looks at the card) that the single card is a face card, then the posterior probability $P(\text{King} \mid \text{Face})$ can be calculated using Bayes' theorem:

$$P(\text{King} \mid \text{Face}) = \frac{P(\text{Face} \mid \text{King})}{P(\text{Face})} P(\text{King}).$$

Since every King is also a face card, $P(\text{Face} \mid \text{King}) = 1$. Since there are 3 face cards in each suit (Jack, Queen, King) , the probability of a face card is $P(\text{Face}) = \frac{3}{13}$. Combining these gives a likelihood ratio of $\frac{1}{\frac{3}{13}} = \frac{13}{3}$.

Using Bayes' theorem gives $P(\text{King} \mid \text{Face}) = \frac{13}{3}\frac{1}{13} = \frac{1}{3}$. □

Question-9:

EXAMPLE

A family has two children. Given that one of the children is a boy, what is the probability that both children are boys?

---

We assume that the probability of a child being a boy or girl is $\frac{1}{2}$. We solve this using Bayes' theorem. We let $B$ be the event that the family has one child who is a boy. We let $A$ be the event that both children are boys. We want to find $P(A \mid B) = \frac{P(B|A) \times P(A)}{P(B)}$. We can easily see that $P(B \mid A) = 1$. We also note that $P(A) = \frac{1}{4}$ and $P(B) = \frac{3}{4}$. So $P(A \mid B) = \frac{1 \times \frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$. □

Question -10:

EXAMPLE

A family has two children. Given that one of the children is a boy, and that he was born on a Tuesday, what is the probability that both children are boys?

---

Your first instinct to this question might be to answer $\frac{1}{3}$, since this is obviously the same question as the previous one. Knowing the day of the week a child is born on can't possibly give you additional information, right?

Let's assume that the probability of being born on a particular day of the week is $\frac{1}{7}$ and is independent of whether the child is a boy or a girl. We let $B$ be the event that the family has one child who is a boy born on Tuesday and $A$ be the event that both children are boys, and apply Bayes' Theorem. We notice right away that $P(B \mid A)$ is no longer equal to one. Given that there are 7 days of the week, there are 49 possible combinations for the days of the week the two boys were born on, and 13 of these have a boy who was born on a Tuesday, so $P(B \mid A) = \frac{13}{49}$. $P(A)$ remains unchanged at $\frac{1}{4}$. To calculate $P(B)$, we note that there are $14^2 = 196$ possible ways to select the gender and the day of the week the child was born on. Of these, there are $13^2 = 169$ ways which do not have a boy born on Tuesday, and $196 - 169 = 27$ which do, so $P(B) = \frac{27}{196}$. This gives is that $P(A \mid B) = \frac{\frac{13}{49} \times \frac{1}{4}}{\frac{27}{196}} = \frac{13}{27}$. □

**Note:** This answer is certainly not $\frac{1}{3}$, and is actually much closer to $\frac{1}{2}$.

Question-11:

- Pam put in 15 paintings, 4% of her works have won First Prize.
- Pia put in 5 paintings, 6% of her works have won First Prize.
- Pablo put in 10 paintings, 3% of his works have won First Prize.

What is the chance that Pam will win First Prize?

$$P(\text{Pam}|\text{First}) = \frac{P(\text{Pam})P(\text{First}|\text{Pam})}{P(\text{Pam})P(\text{First}|\text{Pam}) + P(\text{Pia})P(\text{First}|\text{Pia}) + P(\text{Pablo})P(\text{First}|\text{Pablo})}$$

Put in the values:

$$P(\text{Pam}|\text{First}) = \frac{(15/30) \times 4\%}{(15/30) \times 4\% + (5/30) \times 6\% + (10/30) \times 3\%}$$

Multiply all by 30 (makes calculation easier):

$$P(\text{Pam}|\text{First}) = \frac{15 \times 4\%}{15 \times 4\% + 5 \times 6\% + 10 \times 3\%}$$

$$= \frac{0.6}{0.6 + 0.3 + 0.3}$$

$$= 50\%$$

A good chance!

Pam isn't the most successful artist, but she did put in lots of entries.

Question- 12:

**Example 1:Bag I contains 4 white and 6 black balls while another Bag II contains 4 white and 3 black balls. One ball is drawn at random from one of the bags and it is found to be black. Find the probability that it was drawn from Bag I.**

Solution:Let $E_1$ be the event of choosing the bag I, $E_2$ the event of choosing the bag II and A be the event of drawing a black ball.

Then, $P(E_1) = P(E_2) = \frac{1}{2}$

Also, $P(A|E_1) = P$(drawing a black ball from Bag I) = $\frac{6}{10} = \frac{3}{5}$

$P(A|E_2) = P$(drawing a black ball from Bag II) = $\frac{3}{7}$

By using Bayes' theorem, the probability of drawing a black ball from bag I out of two bags,

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1)+P(E_2)P(A|E_2)}$$

$$= \frac{\frac{1}{2} \times \frac{3}{5}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{3}{5}} = \frac{7}{12}$$

Question – 13:

**Example 2:A man is known to speak truth 2 out of 3 times. He throws a die andreports that number obtained is a four. Find the probability that the number obtained is actually a four.**

Solution:Let $A$ be the event that the man reports that number four is obtained.

Let $E_1$ be the event that four is obtained and $E_2$ be its complementary event.

Then, $P(E_1)$ = Probability that four occurs = $\frac{1}{6}$

$P(E_2)$ = Probability that four does not occurs = $1 - P(E_1) = 1 - \frac{1}{6} = \frac{5}{6}$

Also, $P(A|E_1)$ = Probability that man reports four and it is actually a four = $\frac{2}{3}$

$P(A|E_2)$ = Probability that man reports four and it is not a four = $\frac{1}{3}$

By using Bayes' theorem, probability that number obtained is actually a four,

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)} = \frac{\frac{1}{6} \times \frac{2}{3}}{\frac{1}{6} \times \frac{2}{3} + \frac{5}{6} \times \frac{1}{3}} = \frac{2}{7}$$

Definition: A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses $a_n$ in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$, where $n_0$ is a nonnegative integer. A sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

Simply, A recurrence relation is an equation that recursively defines a sequence where the next term is a function of the previous terms (Expressing Fn as some combination of Fi with i<n).

**Example** − Fibonacci series − $F_n=F_{n-1}+F_{n-2}$

, Tower of Hanoi − $F_n=2F_{n-1}+1$

## Linear Recurrence Relations

A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots\ldots + c_k a_{n-k}$$

where $C_1, C_2, \ldots, C_k$ are real numbers, and $C_k \neq 0$.

These are some examples of linear recurrence equations –

| Recurrence relations | Initial values | Solutions |
|---|---|---|
| $F_n = F_{n-1} + F_{n-2}$ | $a_1 = a_2 = 1$ | Fibonacci number |
| $F_n = F_{n-1} + F_{n-2}$ | $a_1 = 1, a_2 = 3$ | Lucas Number |
| $F_n = F_{n-2} + F_{n-3}$ | $a_1 = a_2 = a_3 = 1$ | Padovan sequence |
| $F_n = 2F_{n-1} + F_{n-2}$ | $a_1 = 0, a_2 = 1$ | Pell number |

## Solving Linear Homogeneous Recurrence Relations with Constant Coefficients

The basic approach for solving linear homogeneous recurrence relations is to look for solutions of the form $a_n = r^n$, where r is a constant. Note that $a_n = r^n$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \ldots + c_k a_{n-k}$ if and only if

$$\mathbf{r^n = c_1 r^{n-1} + c_2 r^{n-2} + \ldots\ldots + c_k r^{n-k}}$$

When both sides of this equation are divided by $r^{n-k}$ and the right-hand side is subtracted from the left, we obtain the equation

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \ldots\ldots - c_{k-1} r - c_k = 0.$$

Consequently, the sequence {an} with $a_n = r^n$ is a solution if and only if r is a solution of this last equation, which is called the **characteristic equation** of the recurrence relation. The solutions of this equation are called the **characteristic roots** of the recurrence relation.

### How to solve linear recurrence relation

Suppose, a two ordered linear recurrence relation is – $F_n = AF_{n-1} + BF_{n-2}$ where A and B are real numbers.

The characteristic equation for the above recurrence relation is −

$$x^2 - Ax - B = 0$$

Three cases may occur while finding the roots −

**Case 1** − If this equation factors as $(x - x_1)(x - x_1) = 0$

and it produces two distinct real roots $x_1$ and $x_2$, then $F_n = ax_1^n + bx_2^n$ is the solution. [Here, a and b are constants]

**Case 2** − If this equation factors as $(x - x_1)^2 = 0$

and it produces single real root $x_1$, then $Fn = ax_1^n + bnx_1^n$ is the solution.

**Problem 1**

Solve the recurrence relation $Fn = 5F_{n-1} - 6F_{n-2}$ Where $F_0 = 1$ and $F_1 = 4$

**Solution**

The characteristic equation of the recurrence relation is −

$$x^2 - 5x + 6 = 0,$$

So, $(x-3)(x-2) = 0$

Hence, the roots are −

$x_1 = 3$ and $x_2 = 2$

The roots are real and distinct. So, this is in the form of case 1

Hence, the solution is −

$$F_n = ax_1^n + bx_2^n$$

Here, $Fn = a3^n + b2^n$ (As $x_1 = 3$ and $x_2 = 2$)

Therefore,

$1 = F_0 = a3^0 + b2^0 = a + b$

$4 = F_1 = a3^1 + b2^1 = 3a + 2b$

Solving these two equations, we get a=2 and b=−1

Hence, the final solution is −

$$F_n=2.3^n+(-1).2^n=2.3^n-2^n$$

**Problem 2**

Solve the recurrence relation − $Fn=10F_{n-1}-25F_{n-2}$ where $F_0=3$ and $F_1=17$

**Solution**

The characteristic equation of the recurrence relation is −

$x^2-10x-25=0$

So $(x-5)^2=0$

Hence, there is single real root $x_1=5$

As there is single real valued root, this is in the form of case 2

Hence, the solution is −

$F_n= ax_1^n + bnx_1^n$

$3=F_0=a.5^0+b.0.5^0=a$

$17=F_1=a.5^1+b.1.5^1=5a+5b$

Solving these two equations, we get a=3

and b=2/5

Hence, the final solution is − $Fn=3.5^n+(2/5).n.2^n$

Problem 3:

What is the solution of the recurrence relation

$$a_n = a_{n-1} + 2a_{n-2}$$

$$\text{with } a_0 = 2 \text{ and } a_1 = 7?$$

*Solution:*

The characteristic equation of the recurrence relation is $r^2 - r - 2 = O$. Its roots are $r = 2$ and $r = -1$. Hence, the sequence $\{a_n\}$ is a solution to the recurrence relation if and only if

$$a_n = \alpha_1 2^n + \alpha_2 (-1)^n,$$

for some constants $\alpha_1$ and $\alpha_2$. From the initial conditions, it follows that

$a_0 = 2 = \alpha_1 + \alpha_2$,

$a_1 = 7 = \alpha_1 \cdot 2 + \alpha_2 \cdot (-1)$.

Solving these two equations shows that $\alpha_1 = 3$ and $\alpha_2 = -1$. Hence, the solution to the recurrence relation and initial conditions is the sequence $\{a_n\}$ with

$$a_n = 3 \cdot 2^n - (-1)^n.$$

Problem 4:

Find the solution to the recurrence relation $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with the initial conditions $a_0 = 2$, $a_1 = 5$, and $a_2 = 15$.

*Solution:*
The characteristic polynomial of this recurrence relation is $r^3 - 6r^2 + 11r - 6$.
The characteristic roots are $r = 1$, $r = 2$, and $r = 3$, because $r^3 - 6r^2 + 11r - 6 = (r - 1)(r - 2)(r - 3)$. Hence, the solutions to this recurrence relation are of the form

$a_n = \alpha_1 1^n + \alpha_2 2^n + \alpha_3 3^n$

To find the constants $\alpha_1$, $\alpha_2$, and $\alpha_3$, use the initial conditions. This gives
$a_0 = 2 = \alpha_1 + \alpha_2 + \alpha_3$,
$a_1 = 5 = \alpha_1 + \alpha_2 . 2 + \alpha_3 . 3$,
$a_2 = 15 = \alpha_1 + \alpha_2 \cdot 4 + \alpha_3 \cdot 9$.
When these three simultaneous equations are solved for $\alpha_1$, $\alpha_2$, and $\alpha_3$, we find that $\alpha_1 = 1$, $\alpha_2 = -1$, and $\alpha_3 = 2$. Hence, the unique solution to this recurrence relation and the given initial conditions is the sequence $\{a_n\}$ with

$$a_n = 1 - 2^n + 2 \cdot 3^n.$$

**Non-Homogeneous Recurrence Relation and Particular Solutions**

A recurrence relation is called non-homogeneous if it is in the form

$F_n = AF_{n-1} + BF_{n-2} + f(n)$ where $f(n) \neq 0$

Its associated homogeneous recurrence relation is $F_n = AF_{n-1} + BF_{n-2}$

The solution $(a_n)$ of a non-homogeneous recurrence relation has two parts.

First part is the solution ($a_h$) of the associated homogeneous recurrence relation and the second part is the particular solution ($a_t$)

.

$$a_n=a_h+a_t$$

Solution to the first part is done using the procedures discussed in the previous section.

To find the particular solution, we find an appropriate trial solution.

Let $f(n)=cx^n$ ; let $x^2=Ax+B$ be the characteristic equation of the associated homogeneous recurrence relation and let $x_1$ and $x_2$ be its roots.

- If $x\neq x_1$ and $x\neq x_2$, then $a_t=Ax^n$
- If $x= x_1$, $x\neq x_2$, then $a_t=Anx^n$
- If $x= x_1 = x_2$, then $at=An^2x^n$

## Example

Let a non-homogeneous recurrence relation be $F_n=AF_{n-1}+BF_{n-2}+f(n)$ with characteristic roots $x_1$ =2 and $x_2$=5. Trial solutions for different possible values of $f(n)$ are as follows –

| f(n) | Trial solutions |
|---|---|
| 4 | A |
| $5.2^n$ | $An2^n$ |
| $8.5^n$ | $An5^n$ |
| $4^n$ | $A4^n$ |
| $2n^2+3n+1$ | $An^2+Bn+C$ |

**Problem 5:**

Solve the recurrence relation $Fn=3Fn-1+10Fn-2+7*5^n$ where $F0=4$ and $F1=3$

**Solution**

This is a linear non-homogeneous relation, where the associated homogeneous equation is $Fn=3Fn-1+10Fn-2$ and $f(n)=7*5^n$

The characteristic equation of its associated homogeneous relation is −

$$x^2-3x-10=0$$

Or, $(x-5)(x+2)=0$

Or, $x_1=5$ and $x_2=-2$

Hence $a_h=a.5^n+b.(-2)^n$, where a and b are constants.

Since $f(n)=7.5^n$, i.e. of the form $c.x^n$, a reasonable trial solution of at will be $Anx^n$

$a_t=Anx^n=An5^n$

After putting the solution in the recurrence relation, we get −

$An5^n=3A(n-1)5^{n-1}+10A(n-2)5^{n-2}+7.5^n$

Dividing both sides by $5^{n-2}$, we get

$An5^2=3A(n-1)5+10A(n-2)5^0+7.5^2$

Or, $25An=15An-15A+10An-20A+175$

Or, $35A=175$

Or, $A=5$

So, $Fn=An5^n=5n5^n=n5^{n+1}$

The solution of the recurrence relation can be written as −

$Fn=a_h+at$

$=a.5^n+b.(-2)^n+n5^{n+1}$

Putting values of $F_0=4$ and $F_1=3$, in the above equation, we get $a=-2$ and $b=6$

Hence, the solution is −

$F_n=n5^{n+1}+6.(-2)^n-2.5^n$

*For reference:*


- https://www.tutorialspoint.com/discrete_mathematics/discrete_mathematics_recurrence_relation.htm
- https://brilliant.org/wiki/recurrence-relations/
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